

KALASIN STARCH PLANT - JET COOKER PROJECT

PLC SYSTEM CONTROL CABINET CA1 & I/O WIRING SCHEMATICS

Rockwell Automation ControlLogix 1756 Architecture | 1:1 Metric Standard

PROJECT SPECIFICATION & STAKEHOLDERS

Client / Owner:	INGREDION (THAILAND) CO., LTD.
Plant Location:	Kalasin Starch Plant, Thailand
Project Name:	Jet Cooker Automation & Control System
Project Code:	R5THS00138-JC / xPrj-2603001
System Integrator:	AEC INDUSTRIAL ENGINEERING CO., LTD.
Lead Controller:	Allen-Bradley ControlLogix 1756-L950TPSXT
I/O Architecture:	5x 1756 Chassis (C1..C5) + Remote I/O
Main Panel Designation:	CA1 (Main Control Cabinet) / 4-Bay Suite
Total I/O Points:	1,211 Documented Channels
Drawing Standard:	AutoCAD AC1015 / ISO 3098 Metric Standard
Document Status:	APPROVED FOR CONSTRUCTION (Rev 3.6)

DRAWING PACKAGE REVISION RECORD

REV	DATE	DESCRIPTION	BY	APPR
Rev 1.0	2026-06-15	Initial Engineering Concept Draft	AEC	ENG
Rev 2.0	2026-07-10	Chassis & Slot Allocation Baseline	AEC	LEAD
Rev 3.0	2026-08-01	Updated with Client Tagging Dev35	AEC	INGR
Rev 3.5	2026-08-25	Aligned with P&ID and JB Destinations	AEC	PE
Rev 3.6	2026-09-08	Approved I/O List -6 Master Schematics	AEC	MGR

PREPARED BY

AEC INDUSTRIAL ENG.

Instrumentation Dept.

CHECKED BY

LEAD AUTOMATION ENG.

System Architecture

VERIFIED BY

PROJECT ELECTRICAL PE

Licensed Professional Eng.

APPROVED BY

INGREDION PROJECT MGR.

Client Automation Lead

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECT

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

COVER SHEET & PROJECT DIRECTORY
MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756

DWG NO: KAL-JC-WIR-00001
REV: 3.6 DATE: 2026-09-08

DRAWN: AEC-ENG
CHECKED: LEAD-PE

STATUS: APPROVED
SCALE: N.T.S.

PROJECT NO:
R5THS00138-JC

SHEET NO:
01 / 40

DOCUMENT INDEX - COMPLETE DRAWING LIST (KALASIN JET COOKER)

SHT	DRAWING NUMBER	SHEET TITLE & MODULE DESCRIPTION	REV	STATUS
01	KAL-JC-WIR-00001	COVER SHEET & PROJECT DIRECTORY	3.6	APPROVED
02	KAL-JC-WIR-00002	DRAWING INDEX & SHEET DIRECTORY	3.6	APPROVED
03	KAL-JC-SCD-10001	SYSTEM CONFIGURATION DIAGRAM (SCD)	3.6	APPROVED
04	KAL-JC-WIR-20001	CHASSIS C1 SLOT 3 - 1756-IB32 (DI)	3.6	APPROVED
05	KAL-JC-WIR-20002	CHASSIS C1 SLOT 4 - 1756-IB32 (DI)	3.6	APPROVED
06	KAL-JC-WIR-20003	CHASSIS C1 SLOT 5 - 1756-IB32 (DI)	3.6	APPROVED
07	KAL-JC-WIR-20004	CHASSIS C1 SLOT 6 - 1756-IB32 (DI)	3.6	APPROVED
08	KAL-JC-WIR-20005	CHASSIS C1 SLOT 7 - 1756-OB32 (DO)	3.6	APPROVED
09	KAL-JC-WIR-20006	CHASSIS C1 SLOT 8 - 1756-OB32 (DO)	3.6	APPROVED
10	KAL-JC-WIR-20007	CHASSIS C1 SLOT 9 - 1756-IF16 (AI)	3.6	APPROVED
11	KAL-JC-WIR-20008	CHASSIS C1 SLOT 10 - 1756-IF16 (AI)	3.6	APPROVED
12	KAL-JC-WIR-20009	CHASSIS C2 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
13	KAL-JC-WIR-20010	CHASSIS C2 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
14	KAL-JC-WIR-20011	CHASSIS C2 SLOT 3 - 1756-IB32 (DI)	3.6	APPROVED
15	KAL-JC-WIR-20012	CHASSIS C2 SLOT 4 - 1756-IB32 (DI)	3.6	APPROVED
16	KAL-JC-WIR-20013	CHASSIS C2 SLOT 5 - 1756-IB32 (DI)	3.6	APPROVED
17	KAL-JC-WIR-20014	CHASSIS C2 SLOT 6 - 1756-OB32 (DO)	3.6	APPROVED
18	KAL-JC-WIR-20015	CHASSIS C2 SLOT 7 - 1756-OB32 (DO)	3.6	APPROVED
19	KAL-JC-WIR-20016	CHASSIS C2 SLOT 8 - 1756-OB32 (DO)	3.6	APPROVED
20	KAL-JC-WIR-20017	CHASSIS C2 SLOT 9 - 1756-IF16 (AI)	3.6	APPROVED
21	KAL-JC-WIR-20018	CHASSIS C2 SLOT 10 - 1756-IF16 (AI)	3.6	APPROVED

SHT	DRAWING NUMBER	SHEET TITLE & MODULE DESCRIPTION	REV	STATUS
22	KAL-JC-WIR-20019	CHASSIS C2 SLOT 11 - 1756-IF16 (AI)	3.6	APPROVED
23	KAL-JC-WIR-20020	CHASSIS C2 SLOT 12 - 1756-IF16 (AI)	3.6	APPROVED
24	KAL-JC-WIR-20021	CHASSIS C2 SLOT 13 - 1756-IF16 (AI)	3.6	APPROVED
25	KAL-JC-WIR-20022	CHASSIS C3 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
26	KAL-JC-WIR-20023	CHASSIS C3 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
27	KAL-JC-WIR-20024	CHASSIS C3 SLOT 3 - 1756-OB32 (DO)	3.6	APPROVED
28	KAL-JC-WIR-20025	CHASSIS C3 SLOT 4 - 1756-IF16 (AI)	3.6	APPROVED
29	KAL-JC-WIR-20026	CHASSIS C3 SLOT 5 - 1756-IF16 (AI)	3.6	APPROVED
30	KAL-JC-WIR-20027	CHASSIS C3 SLOT 13 - 1756-OF8 (AO)	3.6	APPROVED
31	KAL-JC-WIR-20028	CHASSIS C4 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
32	KAL-JC-WIR-20029	CHASSIS C4 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
33	KAL-JC-WIR-20030	CHASSIS C4 SLOT 3 - 1756-IB32 (DI)	3.6	APPROVED
34	KAL-JC-WIR-20031	CHASSIS C4 SLOT 4 - 1756-OB32 (DO)	3.6	APPROVED
35	KAL-JC-WIR-20032	CHASSIS C4 SLOT 5 - 1756-IF16 (AI)	3.6	APPROVED
36	KAL-JC-WIR-20033	CHASSIS C4 SLOT 6 - 1756-IF16 (AI)	3.6	APPROVED
37	KAL-JC-WIR-20034	CHASSIS C5 SLOT 1 - 1756-IB32 (DI)	3.6	APPROVED
38	KAL-JC-WIR-20035	CHASSIS C5 SLOT 2 - 1756-IB32 (DI)	3.6	APPROVED
39	KAL-JC-WIR-20036	CHASSIS C5 SLOT 3 - 1756-OB32 (DO)	3.6	APPROVED
40	KAL-JC-WIR-20037	CHASSIS C5 SLOT 4 - 1756-IF16 (AI)	3.6	APPROVED

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
DRAWING INDEX & SHEET DIRECTORY MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-00002 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 02 / 40

KALASIN JET COOKER AUTOMATION ARCHITECTURE - CONTROLLOGIX DLR & RIO

High-Availability Redundant Fiber Optic Device Level Ring (DLR) & Stratix Switch Network

SUPERVISORY LEVEL (SERVER ROOM / MCC)

PRIMARY SCADA SERVER (DELL PowerEdge)
FactoryTalk View SE Server / Redundant Historian

ENGINEERING WORKSTATION & HMI CLIENT
Studio 5000 Logix Designer V35 / RSLinx Classic

LOCAL HMI: PanelView Plus 7 15-inch (Door M1)

CORE INDUSTRIAL SWITCH (STRATIX 5700 / 1783-BMS10CGN)

Port 1-2: Fiber Optic Uplink to Plant Backbone Network

Port 3-4: Supervisory Control Network (Servers & EWS)

Port 5-6: Device Level Ring (DLR) Ring Supervisor

Port 7-8: 1783-ETAP Tap Modules & PanelView HMI

Subnet: 192.168.1.0/24 | Default Gateway: 192.168.1.1

MAIN CONTROLLER CHASSIS C1 (1756-A13 13-SLOT) - PANEL M1

PS	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13
1756-PA	1756-L9	RESERVE	1756-IB	1756-IB	1756-IB	1756-IB	1756-OB	1756-OB	1756-IF	1756-IF	1756-EN	1756-N2	1756-N2
75	50TPSXT		32	32	32	32	32	32	16	16	2TR		
120/24	Contro	Spare	32DI 2	32DI 2	32DI 2	32DI 2	32DO 2	32DO 2	16AI D	16AI D	EtherN	Slot F	Slot F

EXPANSION I/O CHASSIS C2 (1756-A13 13-SLOT) - PANEL M2

PS	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13
1756-PA	1756-IB	1756-IB	1756-IB	1756-IB	1756-IB	1756-OB	1756-OB	1756-OB	1756-IF	1756-IF	1756-IF	1756-IF	1756-IF
75	32	32	32	32	32	32	32	32	16	16	16	16	16
120/24	32DI 2	32DI 2	32DI 2	32DI 2	32DI 2	32DO 2	32DO 2	32DO 2	16AI D	16AI D	16AI D	16AI D	16AI D

EXPANSION RACKS C3, C4, C5 (PANELS M3 & M4)

CHASSIS C3 (Panel M3):
Slot 1..2 (DI), Slot 3 (DO), Slot 4..5 (AI), Slot 13 (8AO: 1756-OF8)

CHASSIS C4 (Panel M4):
Slot 1..3 (DI), Slot 4 (DO), Slot 5..6 (AI)

CHASSIS C5 (Panel M4):
Slot 1..2 (DI), Slot 3 (DO), Slot 4 (AI), Slot 5 (DO), Slot 6 (DI)

IS BARRIERS SUITE:
Pepperl+Fuchs / Rockwell 937T Series (937THDISTDC1, 937THAITXPDC1)

INTERPOSING RELAYS:
Allen-Bradley 700-HL Series 24VDC Relays with LED & Diode

FIELD JUNCTION BOX INTERCONNECTIONS & HAZARDOUS ZONES

Field Junction Box	Interconnections & Hazardous Zones
JB-401 (Cooker Section 1):	118 I/O Points 16 AI, 6 BUS, 42 DI, 54 DO
JB-402 (Cooker Section 2):	64 I/O Points 12 AI, 2 AO, 40 DI, 10 DO
JB-601 (Flash Tank 1):	96 I/O Points 14 AI, 38 DI, 44 DO
JB-602 (Flash Tank 2):	102 I/O Points 16 AI, 4 AO, 42 DI, 40 DO
IS-JB-603 (Explosionproof Zone):	32 Intrinsically Safe Points Ex ia IIC T4
IS-JB-608 / 612 / 618:	Hazardous Area Instrumentation & LEL Gas / Dust Monitors
RIO-200 (Remote Node):	Point I/O 1734 on Ethernet/IP Adapter

KALASIN STARCH PLANT - JET COOKER PROJECT		SYSTEMS INTEGRATION & AUTOMATION	
SYSTEM CONFIGURATION DIAGRAM (SCD) - CONTROLLOGIX 1756 MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-SCD-10001 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: 1:15	PROJECT NO: R5THS00138-JC	SHEET NO: 03 / 40

High-Availability Redundant Fiber Optic Device Level Ring (DLR) & Stratix Switch Network

SUPERVISORY LEVEL (SERVER ROOM / MCC)

PRIMARY SCADA SERVER (DELL PowerEdge)

FactoryTalk View SE Server / Redundant Historian

ENGINEERING WORKSTATION & HMI CLIENT

Studio 5000 Logix Designer V35 / RSLinx Classic

LOCAL HMI: PanelView Plus 7 15-inch (Door M1)

CORE INDUSTRIAL SWITCH (STRATIX 5700 / 1783-BMS10CGN)

Port 1-2: Fiber Optic Uplink to Plant Backbone Network

Port 3-4: Supervisory Control Network (Servers & EWS)

Port 5-6: Device Level Ring (DLR) Ring Supervisor

Port 7-8: 1783-ETAP Tap Modules & PanelView HMI

Subnet: 192.168.1.0/24 | Default Gateway: 192.168.1.1

MAIN CONTROLLER CHASSIS C1 (1756-A13 13-SLOT) - PANEL M1

PS	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13
1756-PA 75	1756-L9 50TPSXT	RESERVE	1756-IB 32	1756-IB 32	1756-IB 32	1756-IB 32	1756-OB 32	1756-OB 32	1756-IF 16	1756-IF 16	1756-EN 2TR	1756-N2	1756-N2
120/24	Contro	Spare	32DI 2	32DI 2	32DI 2	32DI 2	32DO 2	32DO 2	16AI D	16AI D	EtherN	Slot F	Slot F

EXPANSION I/O CHASSIS C2 (1756-A13 13-SLOT) - PANEL M2

PS	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13
1756-PA	1756-IB	1756-IB	1756-IB	1756-IB	1756-IB	1756-OB	1756-OB	1756-OB	1756-IF	1756-IF	1756-IF	1756-IF	1756-IF
75	32	32	32	32	32	32	32	32	16	16	16	16	16
120/24	32DI 2	32DI 2	32DI 2	32DI 2	32DI 2	32DO 2	32DO 2	32DO 2	16AI D	16AI D	16AI D	16AI D	16AI D

EXPANSION RACKS C3, C4, C5 (PANELS M3 & M4)

CHASSIS C3 (Panel M3):

Slot 1..2 (DI), Slot 3 (DO), Slot 4..5 (AI), Slot 13 (8AO: 1756-OF8)

CHASSIS C4 (Panel M4):

Slot 1..3 (DI), Slot 4 (DO), Slot 5..6 (AI)

CHASSIS C5 (Panel M4):

Slot 1..2 (DI), Slot 3 (DO), Slot 4 (AI), Slot 5 (DO), Slot 6 (DI)

IS BARRIERS SUITE:

Pepperl+Fuchs / Rockwell 937T Series (937THDISTDC1, 937THAITXPDC1)

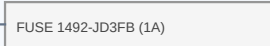
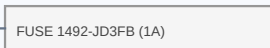
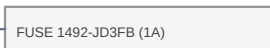
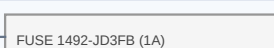
INTERPOSING RELAYS:

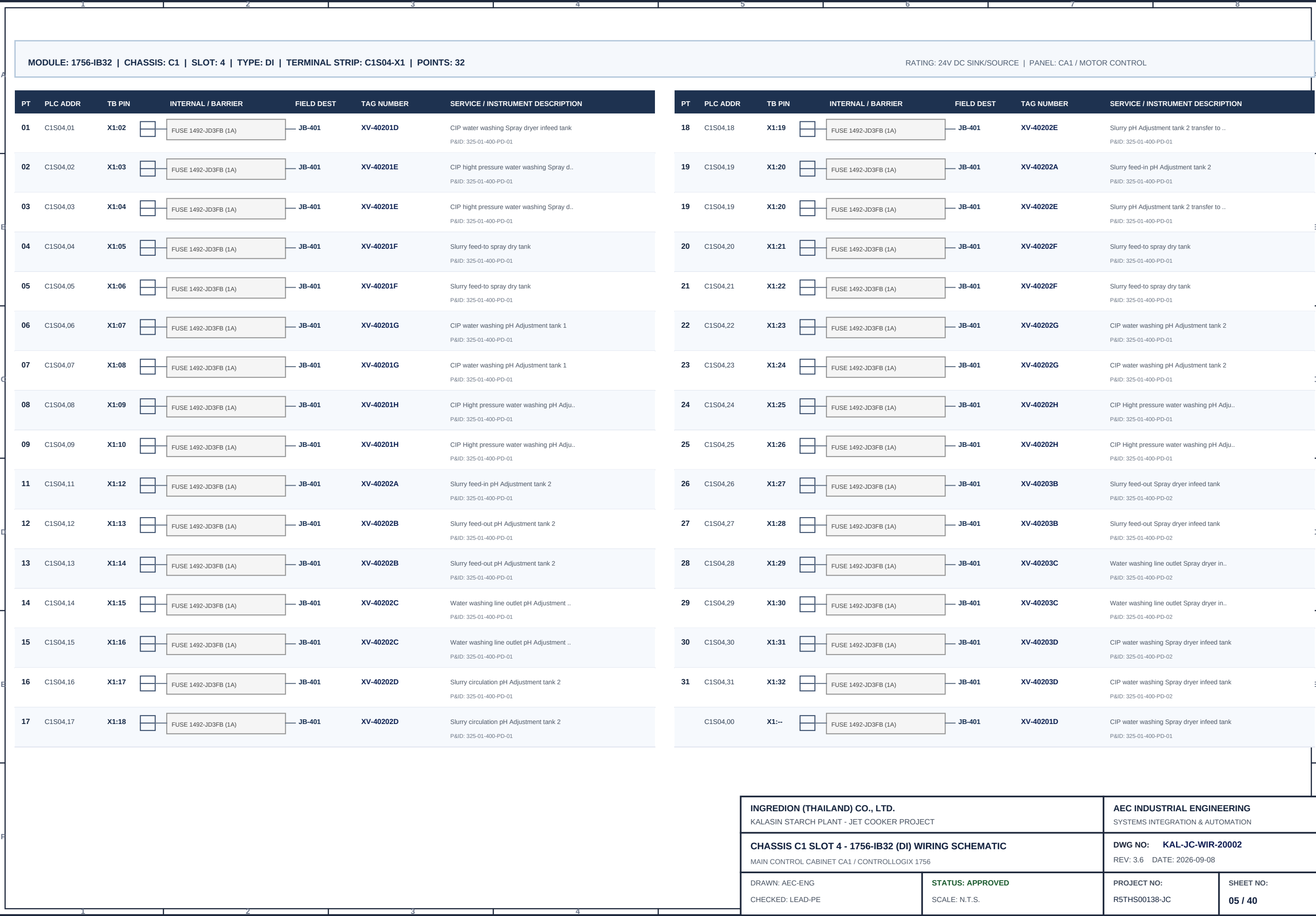
Allen-Bradley 700-HL Series 24VDC Relays with LED & Diode

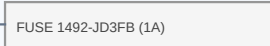
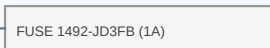
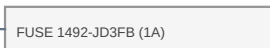
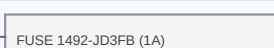
FIELD JUNCTION BOX INTERCONNECTIONS & HAZARDOUS ZONES

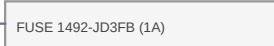
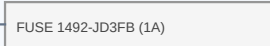
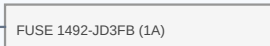
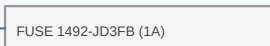
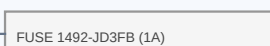
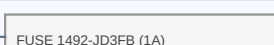
JB-401 (Cooker Section 1):	118 I/O Points 16 AI, 6 BUS, 42 DI, 54 DO
JB-402 (Cooker Section 2):	64 I/O Points 12 AI, 2 AO, 40 DI, 10 DO
JB-601 (Flash Tank 1):	96 I/O Points 14 AI, 38 DI, 44 DO
JB-602 (Flash Tank 2):	102 I/O Points 16 AI, 4 AO, 42 DI, 40 DO
IS-JB-603 (Explosionproof Zone):	32 Intrinsically Safe Points Ex ia IIC T4
IS-JB-608 / 612 / 618:	Hazardous Area Instrumentation & LEL Gas / Dust Monitors
RIO-200 (Remote Node):	Point I/O 1734 on Ethernet/IP Adapter

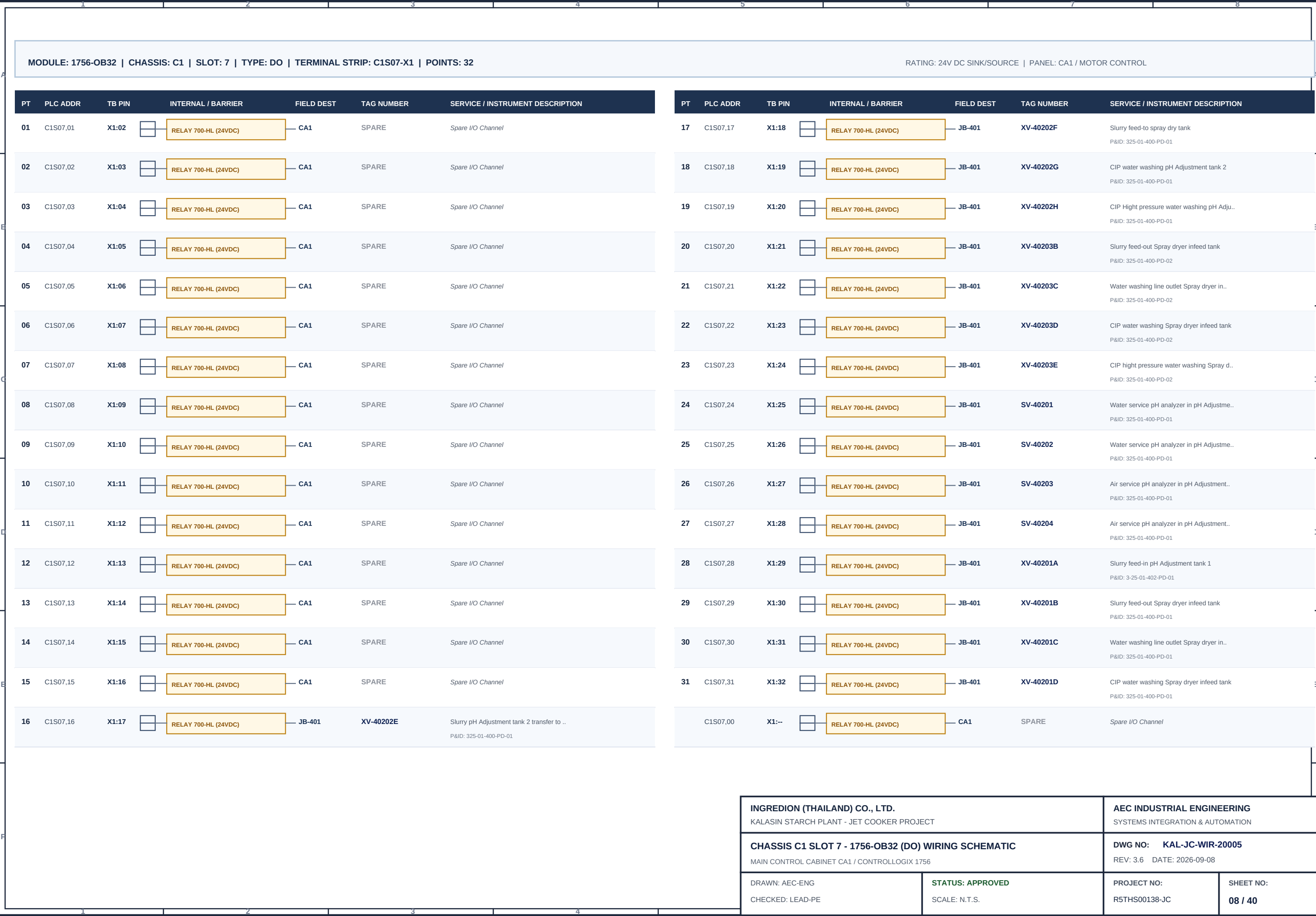
KALASIN STARCH PLANT - JET COOKER PROJECT		SYSTEMS INTEGRATION & AUTOMATION	
SYSTEM CONFIGURATION DIAGRAM (SCD) - CONTROLLOGIX 1756 MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-SCD-10001 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: 1:15	PROJECT NO: R5THS00138-JC	SHEET NO: 03 / 40

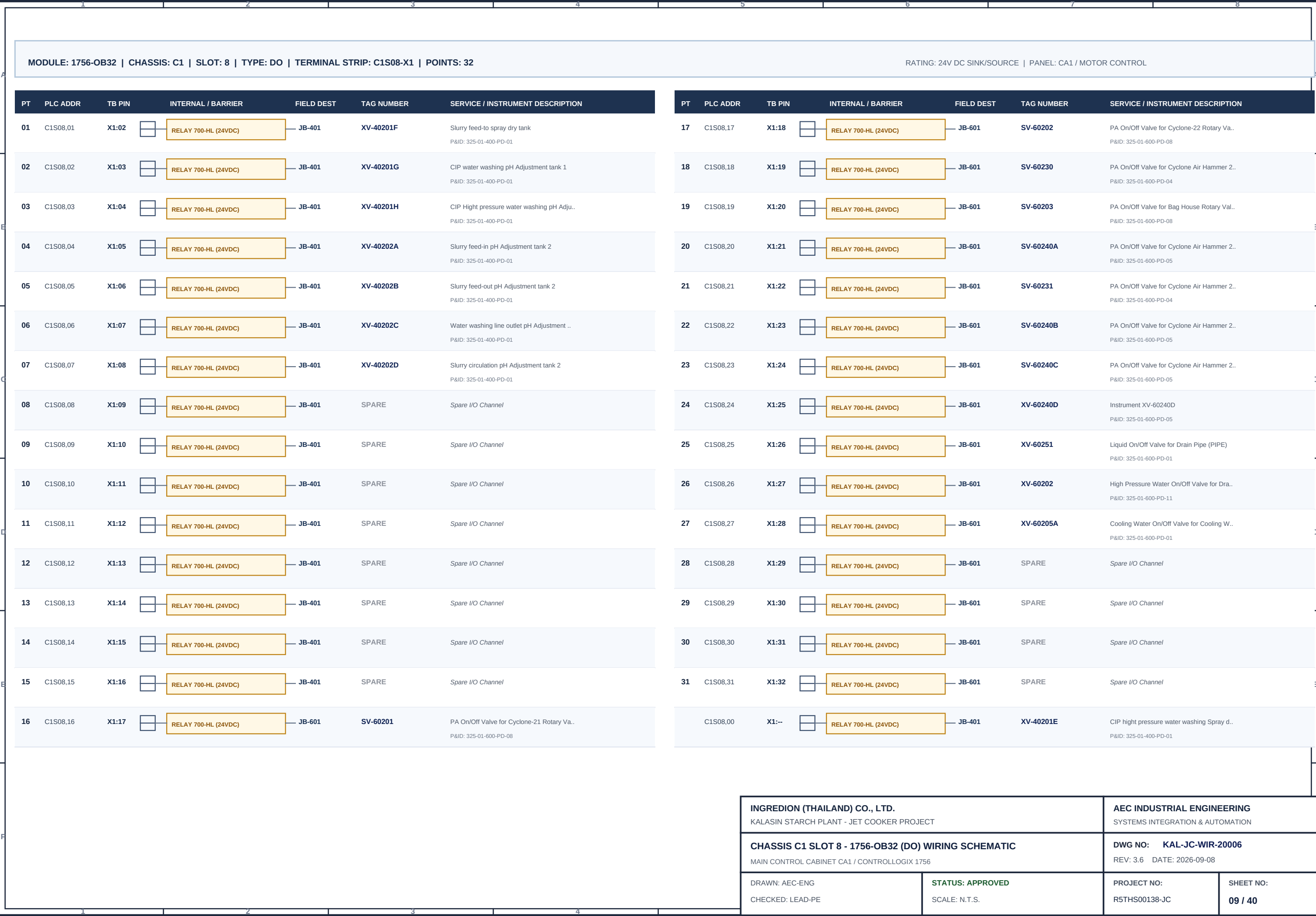
MODULE: 1756-IB32 CHASSIS: C1 SLOT: 3 TYPE: DI TERMINAL STRIP: C1S03-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL																					
PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION				PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION					
01								C1S03,01	X1:02		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				17								C1S03,17	X1:18		 FUSE 1492-JD3FB (1A)		JB-401	FSL-40202	Slurry low flow P&ID: 3-25-01-402-PD-01			
02								C1S03,02	X1:03		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				18								C1S03,18	X1:19		 FUSE 1492-JD3FB (1A)		JB-401	FT-40201	Slurry flowrate & density check transfer.. P&ID: 325-01-400-PD-02			
03								C1S03,03	X1:04		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				19								C1S03,19	X1:20		 FUSE 1492-JD3FB (1A)		JB-401	FT-40202	Feed Slurry Flow Transmitter for Pipe (P.. P&ID: 325-01-400-PD-02			
04								C1S03,04	X1:05		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				20								C1S03,20	X1:21		 FUSE 1492-JD3FB (1A)		JB-401	HV-40201	Slurry Drain line P&ID: 325-01-400-PD-01			
05								C1S03,05	X1:06		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				21								C1S03,21	X1:22		 FUSE 1492-JD3FB (1A)		JB-401	HV-40202	Slurry Drain line P&ID: 325-01-400-PD-01			
06								C1S03,06	X1:07		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				22								C1S03,22	X1:23		 FUSE 1492-JD3FB (1A)		JB-401	HV-40203	Slurry Drain tank TS-40201 P&ID: 325-01-400-PD-01			
07								C1S03,07	X1:08		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				23								C1S03,23	X1:24		 FUSE 1492-JD3FB (1A)		JB-401	HV-40204	Slurry Drain tank TS-40202 P&ID: 325-01-400-PD-01			
08								C1S03,08	X1:09		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				24								C1S03,24	X1:25		 FUSE 1492-JD3FB (1A)		JB-401	HV-40205	Slurry Drain line P&ID: 325-01-400-PD-01			
09								C1S03,09	X1:10		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				25								C1S03,25	X1:26		 FUSE 1492-JD3FB (1A)		JB-401	HV-40206	Slurry Drain tank TS-40203 P&ID: 325-01-400-PD-01			
10								C1S03,10	X1:11		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				26								C1S03,26	X1:27		 FUSE 1492-JD3FB (1A)		JB-401	XV-40201A	Slurry feed-in pH Adjustment tank 1 P&ID: 3-25-01-402-PD-01			
11								C1S03,11	X1:12		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				27								C1S03,27	X1:28		 FUSE 1492-JD3FB (1A)		JB-401	XV-40201A	Slurry feed-in pH Adjustment tank 1 P&ID: 3-25-01-402-PD-01			
12								C1S03,12	X1:13		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				28								C1S03,28	X1:29		 FUSE 1492-JD3FB (1A)		JB-401	XV-40201B	Slurry feed-out Spray dryer infeed tank P&ID: 325-01-400-PD-01			
13								C1S03,13	X1:14		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				29								C1S03,29	X1:30		 FUSE 1492-JD3FB (1A)		JB-401	XV-40201B	Slurry feed-out Spray dryer infeed tank P&ID: 325-01-400-PD-01			
14								C1S03,14	X1:15		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				30								C1S03,30	X1:31		 FUSE 1492-JD3FB (1A)		JB-401	XV-40201C	Water washing line outlet Spray dryer in.. P&ID: 325-01-400-PD-01			
15								C1S03,15	X1:16		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				31								C1S03,31	X1:32		 FUSE 1492-JD3FB (1A)		JB-401	XV-40201C	Water washing line outlet Spray dryer in.. P&ID: 325-01-400-PD-01			
16								C1S03,16	X1:17		 FUSE 1492-JD3FB (1A)		JB-401	FSL-40201	Slurry low flow P&ID: 3-25-01-402-PD-01				C1S03,00								X1:--		 FUSE 1492-JD3FB (1A)		CA1	SPARE	Spare I/O Channel				
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT																AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION																					
CHASSIS C1 SLOT 3 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756																DWG NO: KAL-JC-WIR-20001 REV: 3.6 DATE: 2026-09-08																					
DRAWN: AEC-ENG CHECKED: LEAD-PE								STATUS: APPROVED SCALE: N.T.S.								PROJECT NO: R5THS00138-JC				SHEET NO: 04 / 40																	



MODULE: 1756-IB32 CHASSIS: C1 SLOT: 5 TYPE: DI TERMINAL STRIP: C1S05-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL															
PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01							C1S05,01	X1:02			FUSE 1492-JD3FB (1A)	JB-401	XV-40203E	CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-02		17							C1S05,17	X1:18			FUSE 1492-JD3FB (1A)	JB-601	FS-60205	Cooling Water Return Flow Switch for Pos.. P&ID: 325-01-600-PD-01	
02							C1S05,02	X1:03			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		18							C1S05,18	X1:19			FUSE 1492-JD3FB (1A)	JB-601	LSL-60201	Level low switch P&ID: 325-01-600-PD-11	
03							C1S05,03	X1:04			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		19							C1S05,19	X1:20			FUSE 1492-JD3FB (1A)	JB-601	SV-60201	PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08	
04							C1S05,04	X1:05			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		20							C1S05,20	X1:21			FUSE 1492-JD3FB (1A)	JB-601	SV-60201	PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08	
05							C1S05,05	X1:06			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		21							C1S05,21	X1:22			FUSE 1492-JD3FB (1A)	JB-601	SV-60202	PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08	
06							C1S05,06	X1:07			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		22							C1S05,22	X1:23			FUSE 1492-JD3FB (1A)	JB-601	SV-60202	PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08	
07							C1S05,07	X1:08			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		23							C1S05,23	X1:24			FUSE 1492-JD3FB (1A)	JB-601	SV-60203	PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08	
08							C1S05,08	X1:09			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		24							C1S05,24	X1:25			FUSE 1492-JD3FB (1A)	JB-601	SV-60203	PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08	
09							C1S05,09	X1:10			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		25							C1S05,25	X1:26			FUSE 1492-JD3FB (1A)	JB-601	VS-60205	Vibration Gear Box Vibration Switch for .. P&ID: 325-01-600-PD-01	
10							C1S05,10	X1:11			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		26							C1S05,26	X1:27			FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel	
11							C1S05,11	X1:12			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		27							C1S05,27	X1:28			FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel	
12							C1S05,12	X1:13			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		28							C1S05,28	X1:29			FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel	
13							C1S05,13	X1:14			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		29							C1S05,29	X1:30			FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel	
14							C1S05,14	X1:15			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		30							C1S05,30	X1:31			FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel	
15							C1S05,15	X1:16			FUSE 1492-JD3FB (1A)	JB-401	SPARE	Spare I/O Channel		31							C1S05,31	X1:32			FUSE 1492-JD3FB (1A)	JB-601	SPARE	Spare I/O Channel	
16							C1S05,16	X1:17			FUSE 1492-JD3FB (1A)	JB-601	ES-60202	E-Stop for PUPD-11 (Positive Displaceme.. P&ID: 325-01-600-PD-01		C1S05,00							X1:--			FUSE 1492-JD3FB (1A)	JB-401	XV-40203E	CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-02		

MODULE: 1756-IB32 CHASSIS: C1 SLOT: 6 TYPE: DI TERMINAL STRIP: C1S06-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL															
PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT							PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01							C1S06,01	X1:02			JB-601	XA-60202B	General Fault Controlled Stop for XSUP-5.. P&ID: 325-01-600-PD-08		17							C1S06,17	X1:18			JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-05			
02							C1S06,02	X1:03			JB-601	XA-60202C	General Armed Status Indicator for XSUP-.. P&ID: 325-01-600-PD-08		18							C1S06,18	X1:19			JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-08			
03							C1S06,03	X1:04			JB-601	XA-60202D	Power Fault Status Indicator for XSUP-56.. P&ID: 325-01-600-PD-08		19							C1S06,19	X1:20			JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-05			
04							C1S06,04	X1:05			JB-601	XA-60202E	Dryer CHAM-21 Emergency Stop Actuation f.. P&ID: 325-01-600-PD-08		20							C1S06,20	X1:21			JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-08			
05							C1S06,05	X1:06			JB-601	XA-60202F	CHAM-21 Controlled Stop Trouble (No Prot.. P&ID: 325-01-600-PD-08		21							C1S06,21	X1:22			JB-601	YA-60205	Fault Command P&ID: 325-01-600-PD-01			
06							C1S06,06	X1:07			JB-601	XA-60202G	CYCL-21/22 Emergency Stop Actuation for .. P&ID: 325-01-600-PD-08		22							C1S06,22	X1:23			JB-601	YI-60205	Run Command P&ID: 325-01-600-PD-01			
07							C1S06,07	X1:08			JB-601	XA-60202H	CYCL-21/22 Controlled Stop Trouble (No P.. P&ID: 325-01-600-PD-08		23							C1S06,23	X1:24			JB-601	SPARE	Spare I/O Channel			
08							C1S06,08	X1:09			JB-601	XA-60202I	Instrument XA-60202I P&ID: 325-01-600-PD-08		24							C1S06,24	X1:25			JB-601	SPARE	Spare I/O Channel			
09							C1S06,09	X1:10			JB-601	XA-60202J	BGHS-21 Emergency Stop Actuation for XSU.. P&ID: 325-01-600-PD-08		25							C1S06,25	X1:26			JB-601	SPARE	Spare I/O Channel			
10							C1S06,10	X1:11			JB-601	XA-60202K	BGHS-21 Controlled Stop Trouble (No Prot.. P&ID: 325-01-600-PD-08		26							C1S06,26	X1:27			JB-601	SPARE	Spare I/O Channel			
11							C1S06,11	X1:12			JB-601	XA-60202L	CYCL-31 Emergency Stop Actuation for XSU.. P&ID: 325-01-600-PD-08		27							C1S06,27	X1:28			JB-601	SPARE	Spare I/O Channel			
12							C1S06,12	X1:13			JB-601	XA-60202M	CYCL-31 Controlled Stop Trouble (No Prot.. P&ID: 325-01-600-PD-08		28							C1S06,28	X1:29			JB-601	SPARE	Spare I/O Channel			
13							C1S06,13	X1:14			JB-601	XV-60202	High Pressure Water On/Off Valve for Dra.. P&ID: 325-01-600-PD-11		29							C1S06,29	X1:30			JB-601	SPARE	Spare I/O Channel			
14							C1S06,14	X1:15			JB-601	XV-60202	High Pressure Water On/Off Valve for Dra.. P&ID: 325-01-600-PD-11		30							C1S06,30	X1:31			JB-601	SPARE	Spare I/O Channel			
15							C1S06,15	X1:16			JB-601	XV-60205A	Cooling Water On/Off Valve for Cooling W.. P&ID: 325-01-600-PD-01		31							C1S06,31	X1:32			JB-601	SPARE	Spare I/O Channel			
16							C1S06,16	X1:17			JB-601	XV-60205A	Cooling Water On/Off Valve for Cooling W.. P&ID: 325-01-600-PD-01		C1S06,00							X1:--			JB-601	XA-60202A	General Alarm Emergency Stop Actuation f.. P&ID: 325-01-600-PD-08				

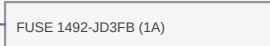
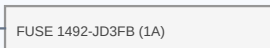
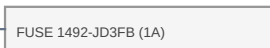
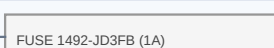


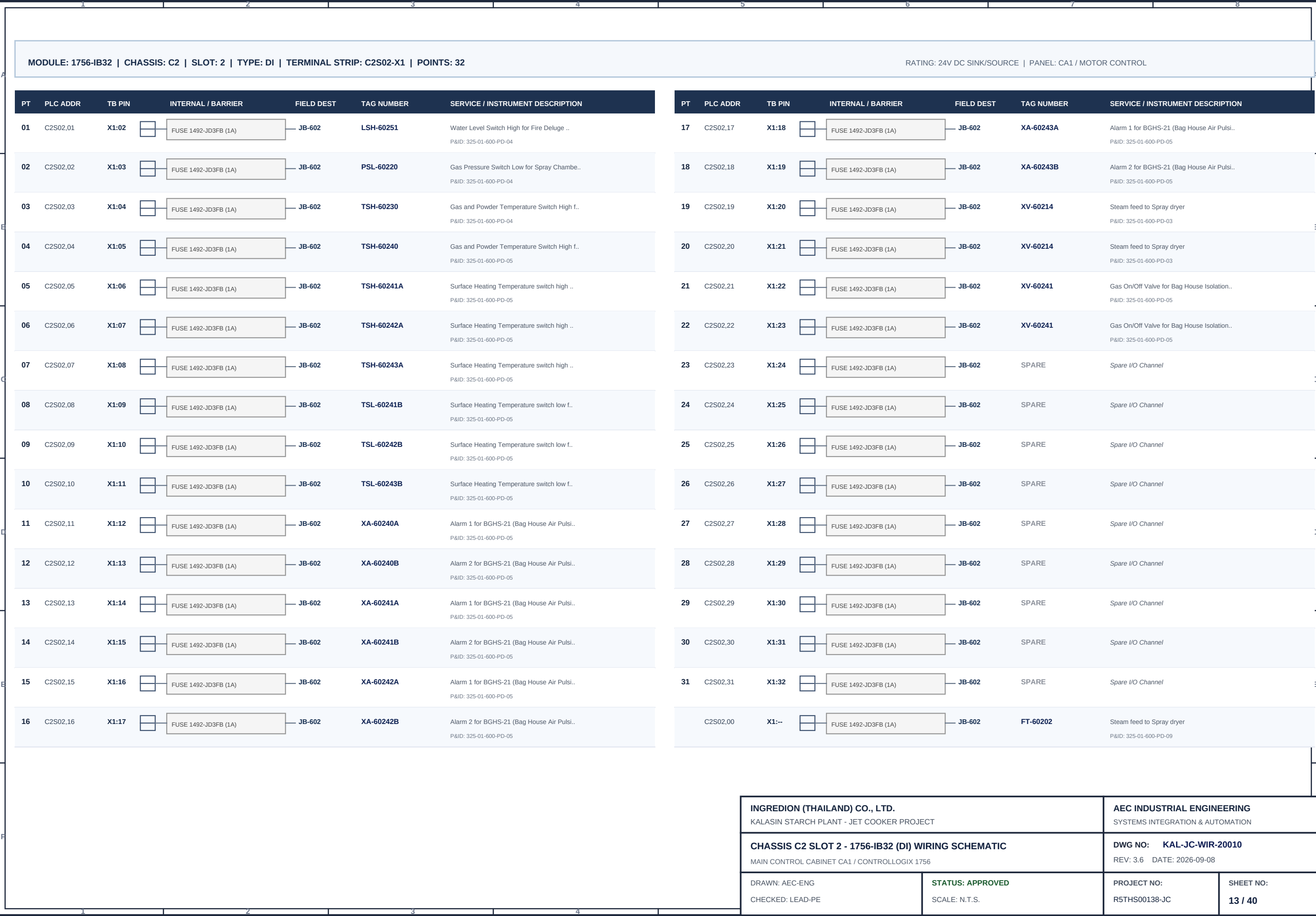


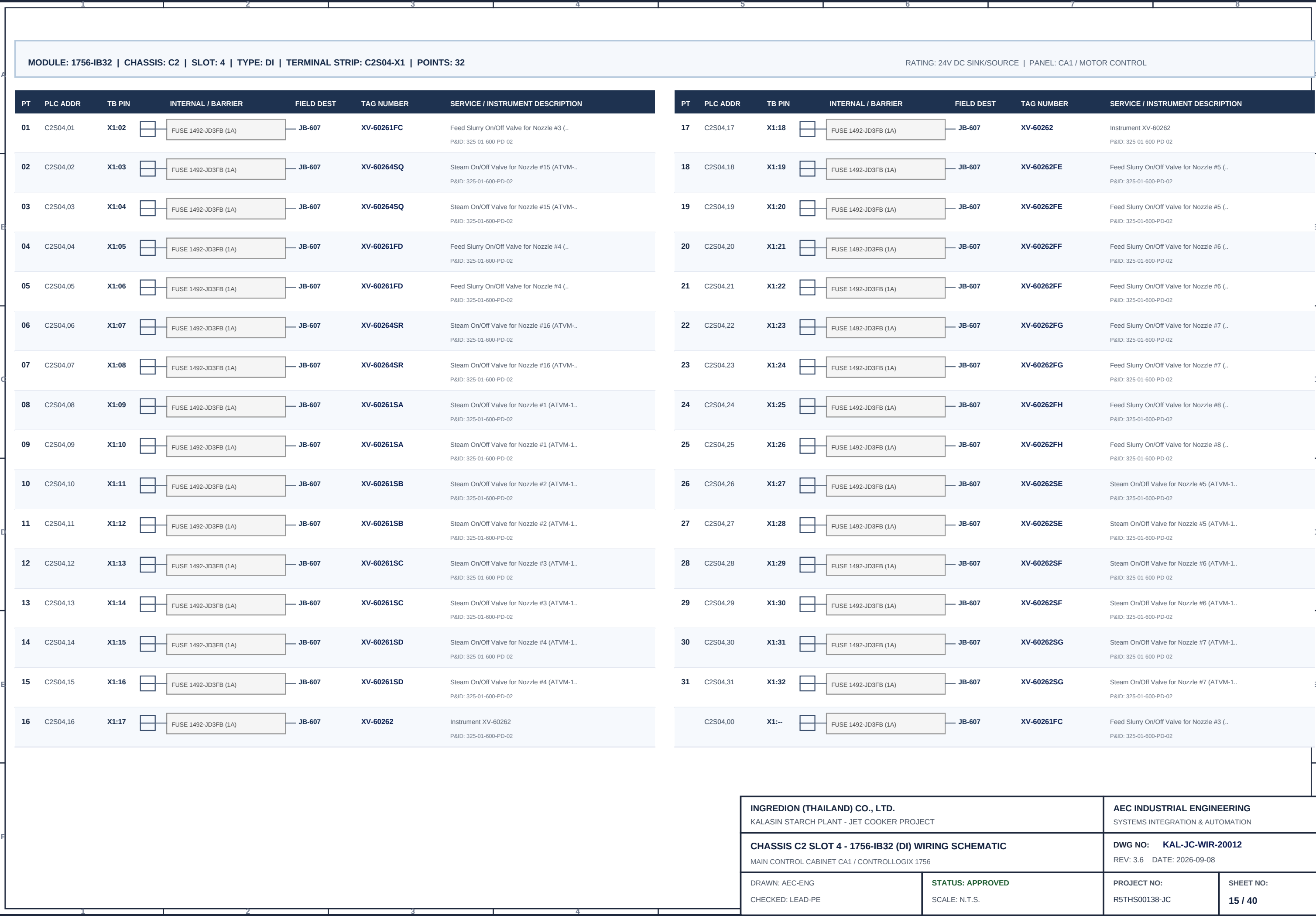
MODULE: 1756-IF16 CHASSIS: C1 SLOT: 9 TYPE: AI TERMINAL STRIP: C1S09-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C1S09,01	X1:02		LOOP POWERED (2-W)	JB-401	FT-40201	Slurry flowrate & density check transfer.. P&ID: 325-01-400-PD-02	
02	C1S09,02	X1:03		LOOP POWERED (2-W)	JB-401	FT-40202	Feed Slurry Flow Transmitter for Pipe (P.. P&ID: 325-01-400-PD-02	
03	C1S09,03	X1:04		LOOP POWERED (2-W)	JB-401	LT-40203	Level transmitter in Spray dryer infeed .. P&ID: 325-01-400-PD-02	
04	C1S09,04	X1:05		LOOP POWERED (2-W)	JB-401	LT-40201	Level transmitter in Spray dryer infeed .. P&ID: 325-01-400-PD-01	
05	C1S09,05	X1:06		LOOP POWERED (2-W)	JB-401	LT-40202	Level transmitter in pH Adjustment tank 2 P&ID: 325-01-400-PD-01	
06	C1S09,06	X1:07		LOOP POWERED (2-W)	JB-401	pHT-40201	pH analyzer in pH Adjustment tank 1 P&ID: 325-01-400-PD-01	
07	C1S09,07	X1:08		LOOP POWERED (2-W)	JB-401	pHT-40202	pH analyzer in pH Adjustment tank 2 P&ID: 325-01-400-PD-01	
08	C1S09,08	X1:09		LOOP POWERED (2-W)	JB-401	PT-40201	Pressure Transmitter for Inlet Filter S.. P&ID: 325-01-400-PD-02	
09	C1S09,09	X1:10		LOOP POWERED (2-W)	JB-401	PT-40202	Feed Slurry Pressure Transmitter for Pip.. P&ID: 325-01-400-PD-02	
10	C1S09,10	X1:11		LOOP POWERED (2-W)	JB-401	PT-60202	Feed Slurry Pressure Transmitter for Hig.. P&ID: 325-01-600-PD-01	
11	C1S09,11	X1:12		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
12	C1S09,12	X1:13		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
13	C1S09,13	X1:14		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
14	C1S09,14	X1:15		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
15	C1S09,15	X1:16		LOOP POWERED (2-W)	JB-401	SPARE	Spare I/O Channel	
	C1S09,00	X1:--		LOOP POWERED (2-W)	JB-401	FT-40201	Slurry flowrate & density check transfer.. P&ID: 325-01-400-PD-02	

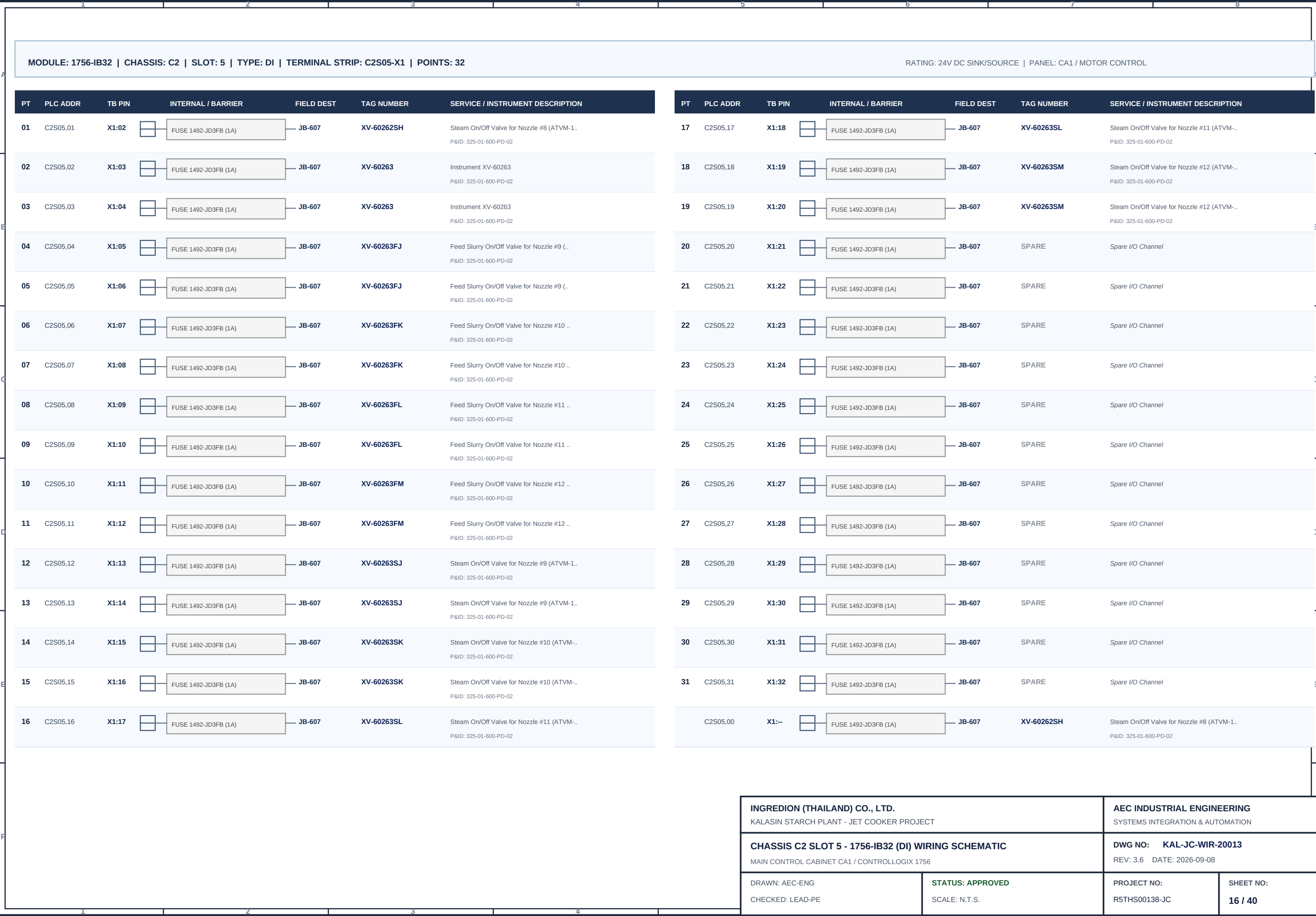
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C1 SLOT 9 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20007 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 10 / 40

MODULE: 1756-IF16 CHASSIS: C1 SLOT: 10 TYPE: AI TERMINAL STRIP: C1S10-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C1S10,01	X1:02		LOOP POWERED (2-W)	JB-601	DPT-60202	Gas Pressure Differential Transmitter f.. P&ID: 325-01-600-PD-08	
02	C1S10,02	X1:03		LOOP POWERED (2-W)	JB-601	FT-60201	Gas Flow Transmitter for Cooling Conveyl.. P&ID: 325-01-600-PD-08	
03	C1S10,03	X1:04		LOOP POWERED (2-W)	JB-601	FT-60204	Compressed air Flow P&ID: 325-01-600-PD-10	
04	C1S10,04	X1:05		LOOP POWERED (2-W)	JB-601	FT-61307	Compressed air Flow P&ID: 3-25-01-613-PD-04	
05	C1S10,05	X1:06		LOOP POWERED (2-W)	JB-601	PT-60203	Gas Pressure Transmitter for Cool Air Du.. P&ID: 325-01-600-PD-08	
06	C1S10,06	X1:07		LOOP POWERED (2-W)	JB-601	PT-60204	Pressure transmitter P&ID: 325-01-600-PD-10	
07	C1S10,07	X1:08		LOOP POWERED (2-W)	JB-601	PT-60205	Feed Slurry Pressure Transmitter for Hig.. P&ID: 325-01-600-PD-01	
08	C1S10,08	X1:09		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
09	C1S10,09	X1:10		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
10	C1S10,10	X1:11		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
11	C1S10,11	X1:12		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
12	C1S10,12	X1:13		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
13	C1S10,13	X1:14		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
14	C1S10,14	X1:15		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
15	C1S10,15	X1:16		LOOP POWERED (2-W)	JB-601	SPARE	Spare I/O Channel	
	C1S10,00	X1:--		LOOP POWERED (2-W)	JB-601	DPI-60240	Gas Pressure Differntial Pressure (Bag H.. P&ID: 325-01-600-PD-05	
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT							AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C1 SLOT 10 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756							DWG NO: KAL-JC-WIR-20008 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.			PROJECT NO: R5THS00138-JC	SHEET NO: 11 / 40

MODULE: 1756-IB32 CHASSIS: C2 SLOT: 1 TYPE: DI TERMINAL STRIP: C2S01-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL																			
PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION				PT								PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION			
01								C2S01,01	X1:02			JB-402	FS-40205	Jet Cooker P&ID: 325-01-402-PD-02				17								C2S01,17	X1:18			JB-402	XV-40205C	Jet Cooker P&ID: 325-01-402-PD-02			
02								C2S01,02	X1:03			JB-402	FT-40221	Jet Cooker P&ID: 325-01-402-PD-02				18								C2S01,18	X1:19			JB-402	XV-40205D	Jet Cooker P&ID: 325-01-402-PD-02			
03								C2S01,03	X1:04			JB-402	FT-40222	Jet Cooker P&ID: 325-01-402-PD-02				19								C2S01,19	X1:20			JB-402	XV-40205D	Jet Cooker P&ID: 325-01-402-PD-02			
04								C2S01,04	X1:05			JB-402	HV-40207	Jet Cooker P&ID: 325-01-402-PD-01				20								C2S01,20	X1:21			JB-402	XV-40205F	Jet Cooker P&ID: 325-01-402-PD-02			
05								C2S01,05	X1:06			JB-402	HV-40208	Jet Cooker P&ID: 325-01-402-PD-01				21								C2S01,21	X1:22			JB-402	XV-40205F	Jet Cooker P&ID: 325-01-402-PD-02			
06								C2S01,06	X1:07			JB-402	XV-40203F	Jet Cooker P&ID: 325-01-402-PD-02				22								C2S01,22	X1:23			JB-402	XV-40222	Jet Cooker P&ID: 325-01-402-PD-02			
07								C2S01,07	X1:08			JB-402	XV-40203F	Jet Cooker P&ID: 325-01-402-PD-02				23								C2S01,23	X1:24			JB-402	XV-40222	Jet Cooker			
08								C2S01,08	X1:09			JB-402	XV-40204B	Jet Cooker P&ID: 325-01-402-PD-01				24								C2S01,24	X1:25			JB-402	SPARE	Spare I/O Channel			
09								C2S01,09	X1:10			JB-402	XV-40204B	Jet Cooker P&ID: 325-01-402-PD-01				25								C2S01,25	X1:26			JB-402	SPARE	Spare I/O Channel			
10								C2S01,10	X1:11			JB-402	XV-40204C	Jet Cooker P&ID: 325-01-402-PD-01				26								C2S01,26	X1:27			JB-402	SPARE	Spare I/O Channel			
11								C2S01,11	X1:12			JB-402	XV-40204C	Jet Cooker P&ID: 325-01-402-PD-01				27								C2S01,27	X1:28			JB-402	SPARE	Spare I/O Channel			
12								C2S01,12	X1:13			JB-402	XV-40205A	Jet Cooker P&ID: 325-01-402-PD-02				28								C2S01,28	X1:29			JB-402	SPARE	Spare I/O Channel			
13								C2S01,13	X1:14			JB-402	XV-40205A	Jet Cooker P&ID: 325-01-402-PD-02				29								C2S01,29	X1:30			JB-402	SPARE	Spare I/O Channel			
14								C2S01,14	X1:15			JB-402	XV-40205B	Jet Cooker P&ID: 325-01-402-PD-02				30								C2S01,30	X1:31			JB-402	SPARE	Spare I/O Channel			
15								C2S01,15	X1:16			JB-402	XV-40205B	Jet Cooker P&ID: 325-01-402-PD-02				31								C2S01,31	X1:32			JB-402	SPARE	Spare I/O Channel			
16								C2S01,16	X1:17			JB-402	XV-40205C	Jet Cooker P&ID: 325-01-402-PD-02				C2S01,00								X1:--			JB-402	FS-40204	Jet Cooker P&ID: 325-01-402-PD-01				
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT																AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION																			
CHASSIS C2 SLOT 1 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756																DWG NO: KAL-JC-WIR-20009 REV: 3.6 DATE: 2026-09-08																			
DRAWN: AEC-ENG CHECKED: LEAD-PE												STATUS: APPROVED SCALE: N.T.S.												PROJECT NO: R5THS00138-JC				SHEET NO: 12 / 40							

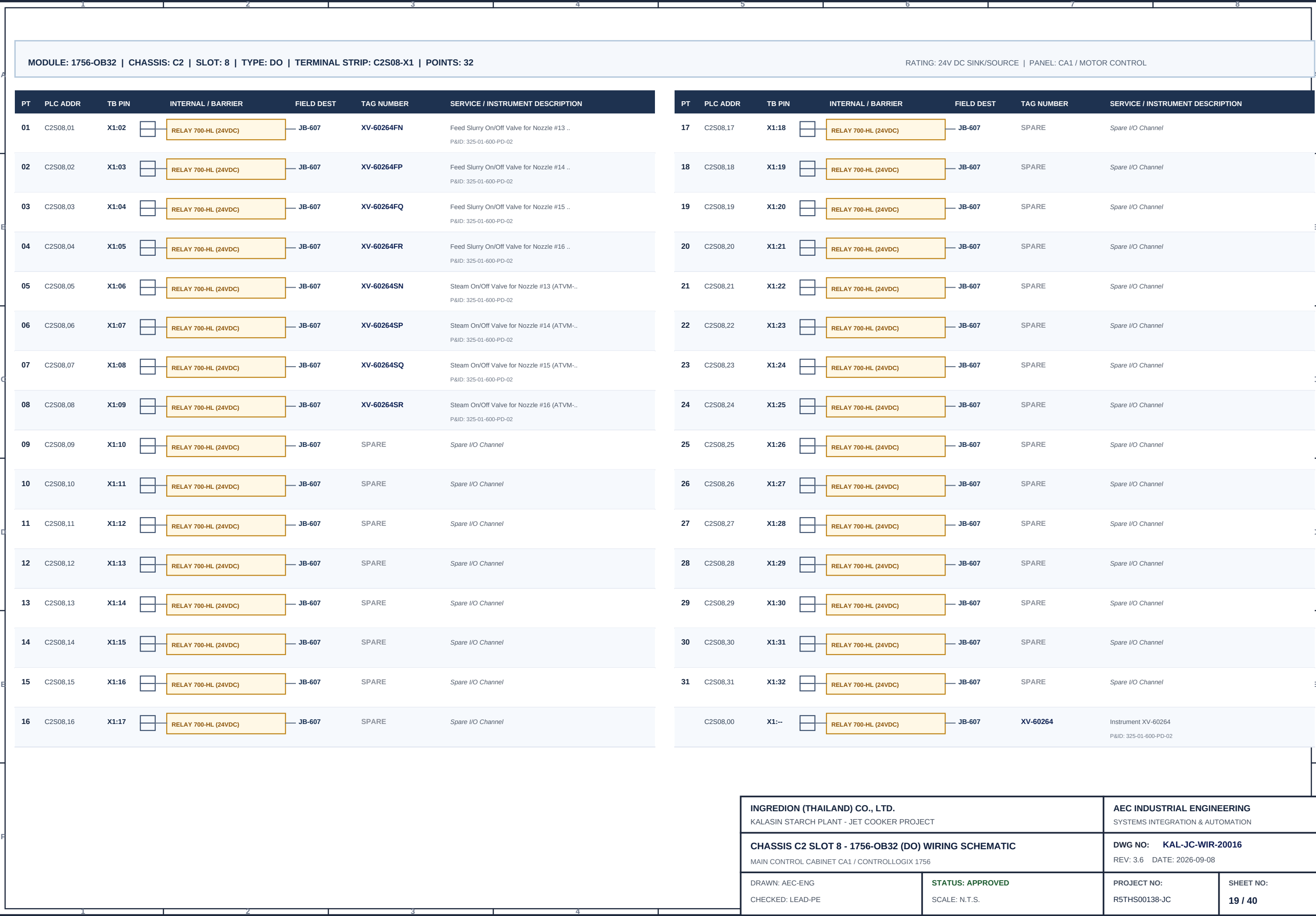






MODULE: 1756-OB32 CHASSIS: C2 SLOT: 6 TYPE: DO TERMINAL STRIP: C2S06-X1 POINTS: 32														RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL													
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION											
01	C2S06,01	X1:02		RELAY 700-HL (24VDC)	JB-402	XV-40204B	Jet Cooker P&ID: 325-01-402-PD-01		17	C2S06,17	X1:18		RELAY 700-HL (24VDC)	JB-602	XV-60214	Steam feed to Spray dryer P&ID: 325-01-600-PD-03											
02	C2S06,02	X1:03		RELAY 700-HL (24VDC)	JB-402	XV-40204C	Jet Cooker P&ID: 325-01-402-PD-01		18	C2S06,18	X1:19		RELAY 700-HL (24VDC)	JB-602	XV-60241	Gas On/Off Valve for Bag House Isolation.. P&ID: 325-01-600-PD-05											
03	C2S06,03	X1:04		RELAY 700-HL (24VDC)	JB-402	XV-40205A	Jet Cooker P&ID: 325-01-402-PD-02		19	C2S06,19	X1:20		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
04	C2S06,04	X1:05		RELAY 700-HL (24VDC)	JB-402	XV-40205B	Jet Cooker P&ID: 325-01-402-PD-02		20	C2S06,20	X1:21		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
05	C2S06,05	X1:06		RELAY 700-HL (24VDC)	JB-402	XV-40205C	Jet Cooker P&ID: 325-01-402-PD-02		21	C2S06,21	X1:22		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
06	C2S06,06	X1:07		RELAY 700-HL (24VDC)	JB-402	XV-40205D	Jet Cooker P&ID: 325-01-402-PD-02		22	C2S06,22	X1:23		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
07	C2S06,07	X1:08		RELAY 700-HL (24VDC)	JB-402	XV-40205F	Jet Cooker P&ID: 325-01-402-PD-02		23	C2S06,23	X1:24		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
08	C2S06,08	X1:09		RELAY 700-HL (24VDC)	JB-402	XV-40222	Jet Cooker P&ID: 325-01-402-PD-02		24	C2S06,24	X1:25		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
09	C2S06,09	X1:10		RELAY 700-HL (24VDC)	JB-402	SPARE	Spare I/O Channel		25	C2S06,25	X1:26		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
10	C2S06,10	X1:11		RELAY 700-HL (24VDC)	JB-402	SPARE	Spare I/O Channel		26	C2S06,26	X1:27		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
11	C2S06,11	X1:12		RELAY 700-HL (24VDC)	JB-402	SPARE	Spare I/O Channel		27	C2S06,27	X1:28		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
12	C2S06,12	X1:13		RELAY 700-HL (24VDC)	JB-402	SPARE	Spare I/O Channel		28	C2S06,28	X1:29		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
13	C2S06,13	X1:14		RELAY 700-HL (24VDC)	JB-402	SPARE	Spare I/O Channel		29	C2S06,29	X1:30		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
14	C2S06,14	X1:15		RELAY 700-HL (24VDC)	JB-402	SPARE	Spare I/O Channel		30	C2S06,30	X1:31		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
15	C2S06,15	X1:16		RELAY 700-HL (24VDC)	JB-402	SPARE	Spare I/O Channel		31	C2S06,31	X1:32		RELAY 700-HL (24VDC)	JB-602	SPARE	Spare I/O Channel											
16	C2S06,16	X1:17		RELAY 700-HL (24VDC)	JB-602	XS-60240B	Fan Command for BGHS-21 (Bag House Air P.. P&ID: 325-01-600-PD-05			C2S06,00	X1:--		RELAY 700-HL (24VDC)	JB-402	XV-40203F	Jet Cooker P&ID: 325-01-402-PD-02											
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT										AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION																	
CHASSIS C2 SLOT 6 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756										DWG NO: KAL-JC-WIR-20014 REV: 3.6 DATE: 2026-09-08																	
DRAWN: AEC-ENG CHECKED: LEAD-PE					STATUS: APPROVED SCALE: N.T.S.					PROJECT NO: R5THS00138-JC				SHEET NO: 17 / 40													

MODULE: 1756-OB32 CHASSIS: C2 SLOT: 7 TYPE: DO TERMINAL STRIP: C2S07-X1 POINTS: 32								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C2S07,01	X1:02		RELAY 700-HL (24VDC)	JB-607	XS-FH-60210C	Command Remote Reset for AHTR-21 P&ID: 325-01-600-PD-06	17	C2S07,17	X1:18		RELAY 700-HL (24VDC)	JB-607	XV-60262FH	Feed Slurry On/Off Valve for Nozzle #8 (.. P&ID: 325-01-600-PD-02
02	C2S07,02	X1:03		RELAY 700-HL (24VDC)	JB-607	XS-FH-60210D	Spare P&ID: 325-01-600-PD-06	18	C2S07,18	X1:19		RELAY 700-HL (24VDC)	JB-607	XV-60262SE	Steam On/Off Valve for Nozzle #5 (ATVM-1.. P&ID: 325-01-600-PD-02
03	C2S07,03	X1:04		RELAY 700-HL (24VDC)	JB-607	XV-60201	Steam feed to Spray dryer P&ID: 325-01-600-PD-02	19	C2S07,19	X1:20		RELAY 700-HL (24VDC)	JB-607	XV-60262SF	Steam On/Off Valve for Nozzle #6 (ATVM-1.. P&ID: 325-01-600-PD-02
04	C2S07,04	X1:05		RELAY 700-HL (24VDC)	JB-607	XV-60261	Instrument XV-60261 P&ID: 325-01-600-PD-02	20	C2S07,20	X1:21		RELAY 700-HL (24VDC)	JB-607	XV-60262SG	Steam On/Off Valve for Nozzle #7 (ATVM-1.. P&ID: 325-01-600-PD-02
05	C2S07,05	X1:06		RELAY 700-HL (24VDC)	JB-607	XV-60261FA	Feed Slurry On/Off Valve for Nozzle #1 (.. P&ID: 325-01-600-PD-02	21	C2S07,21	X1:22		RELAY 700-HL (24VDC)	JB-607	XV-60262SH	Steam On/Off Valve for Nozzle #8 (ATVM-1.. P&ID: 325-01-600-PD-02
06	C2S07,06	X1:07		RELAY 700-HL (24VDC)	JB-607	XV-60261FB	Feed Slurry On/Off Valve for Nozzle #2 (.. P&ID: 325-01-600-PD-02	22	C2S07,22	X1:23		RELAY 700-HL (24VDC)	JB-607	XV-60263	Instrument XV-60263 P&ID: 325-01-600-PD-02
07	C2S07,07	X1:08		RELAY 700-HL (24VDC)	JB-607	XV-60261FC	Feed Slurry On/Off Valve for Nozzle #3 (.. P&ID: 325-01-600-PD-02	23	C2S07,23	X1:24		RELAY 700-HL (24VDC)	JB-607	XV-60263FJ	Feed Slurry On/Off Valve for Nozzle #9 (.. P&ID: 325-01-600-PD-02
08	C2S07,08	X1:09		RELAY 700-HL (24VDC)	JB-607	XV-60261FD	Feed Slurry On/Off Valve for Nozzle #4 (.. P&ID: 325-01-600-PD-02	24	C2S07,24	X1:25		RELAY 700-HL (24VDC)	JB-607	XV-60263FK	Feed Slurry On/Off Valve for Nozzle #10 .. P&ID: 325-01-600-PD-02
09	C2S07,09	X1:10		RELAY 700-HL (24VDC)	JB-607	XV-60261SA	Steam On/Off Valve for Nozzle #1 (ATVM-1.. P&ID: 325-01-600-PD-02	25	C2S07,25	X1:26		RELAY 700-HL (24VDC)	JB-607	XV-60263FL	Feed Slurry On/Off Valve for Nozzle #11 .. P&ID: 325-01-600-PD-02
10	C2S07,10	X1:11		RELAY 700-HL (24VDC)	JB-607	XV-60261SB	Steam On/Off Valve for Nozzle #2 (ATVM-1.. P&ID: 325-01-600-PD-02	26	C2S07,26	X1:27		RELAY 700-HL (24VDC)	JB-607	XV-60263FM	Feed Slurry On/Off Valve for Nozzle #12 .. P&ID: 325-01-600-PD-02
11	C2S07,11	X1:12		RELAY 700-HL (24VDC)	JB-607	XV-60261SC	Steam On/Off Valve for Nozzle #3 (ATVM-1.. P&ID: 325-01-600-PD-02	27	C2S07,27	X1:28		RELAY 700-HL (24VDC)	JB-607	XV-60263SJ	Steam On/Off Valve for Nozzle #9 (ATVM-1.. P&ID: 325-01-600-PD-02
12	C2S07,12	X1:13		RELAY 700-HL (24VDC)	JB-607	XV-60261SD	Steam On/Off Valve for Nozzle #4 (ATVM-1.. P&ID: 325-01-600-PD-02	28	C2S07,28	X1:29		RELAY 700-HL (24VDC)	JB-607	XV-60263SK	Steam On/Off Valve for Nozzle #10 (ATVM-.. P&ID: 325-01-600-PD-02
13	C2S07,13	X1:14		RELAY 700-HL (24VDC)	JB-607	XV-60262	Instrument XV-60262 P&ID: 325-01-600-PD-02	29	C2S07,29	X1:30		RELAY 700-HL (24VDC)	JB-607	XV-60263SL	Steam On/Off Valve for Nozzle #11 (ATVM-.. P&ID: 325-01-600-PD-02
14	C2S07,14	X1:15		RELAY 700-HL (24VDC)	JB-607	XV-60262FE	Feed Slurry On/Off Valve for Nozzle #5 (.. P&ID: 325-01-600-PD-02	30	C2S07,30	X1:31		RELAY 700-HL (24VDC)	JB-607	XV-60263SM	Steam On/Off Valve for Nozzle #12 (ATVM-.. P&ID: 325-01-600-PD-02
15	C2S07,15	X1:16		RELAY 700-HL (24VDC)	JB-607	XV-60262FF	Feed Slurry On/Off Valve for Nozzle #6 (.. P&ID: 325-01-600-PD-02	31	C2S07,31	X1:32		RELAY 700-HL (24VDC)	JB-607	SPARE	Spare I/O Channel
16	C2S07,16	X1:17		RELAY 700-HL (24VDC)	JB-607	XV-60262FG	Feed Slurry On/Off Valve for Nozzle #7 (.. P&ID: 325-01-600-PD-02		C2S07,00	X1:--		RELAY 700-HL (24VDC)	JB-607	SV-60207	Instr. Air On/Off Valve for Camera #4 (A.. P&ID: 325-01-600-PD-07
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT												AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION			
CHASSIS C2 SLOT 7 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756												DWG NO: KAL-JC-WIR-20015 REV: 3.6 DATE: 2026-09-08			
DRAWN: AEC-ENG CHECKED: LEAD-PE						STATUS: APPROVED SCALE: N.T.S.						PROJECT NO: R5THS00138-JC		SHEET NO: 18 / 40	



MODULE: 1756-IF16 CHASSIS: C2 SLOT: 10 TYPE: AI TERMINAL STRIP: C2S10-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C2S10,01	X1:02		LOOP POWERED (2-W)	JB-602	DPT-60203	Gas Pressure Differential Transmitter f.. P&ID: 325-01-600-PD-03	
02	C2S10,02	X1:03		LOOP POWERED (2-W)	JB-602	FT-60202	Steam feed to Spray dryer P&ID: 325-01-600-PD-09	
03	C2S10,03	X1:04		LOOP POWERED (2-W)	JB-602	PT-60210	Gas Pressure Transmitter for Hot Air Duc.. P&ID: 325-01-600-PD-03	
04	C2S10,04	X1:05		LOOP POWERED (2-W)	JB-602	PT-60215	15% Polypropylene Glycol Pressure Indica.. P&ID: 325-01-600-PD-03	
05	C2S10,05	X1:06		LOOP POWERED (2-W)	JB-602	PT-60216	Gas Pressure Transmitter for Combustion .. P&ID: 325-01-600-PD-03	
06	C2S10,06	X1:07		LOOP POWERED (2-W)	JB-602	PT-60240	Gas Pressure Transmitter for Bag House E.. P&ID: 325-01-600-PD-05	
07	C2S10,07	X1:08		LOOP POWERED (2-W)	JB-602	PT-60241	Gas Pressure Transmitter for Exhaust Fan.. P&ID: 325-01-600-PD-05	
08	C2S10,08	X1:09		LOOP POWERED (2-W)	JB-602	TT-60210	Gas Temperature Element for Hot Air Duct.. P&ID: 325-01-600-PD-03	
09	C2S10,09	X1:10		LOOP POWERED (2-W)	JB-602	TT-60210A	Instrument TT-60210A P&ID: 325-01-600-PD-03	
10	C2S10,10	X1:11		LOOP POWERED (2-W)	JB-602	TT-60210B	Instrument TT-60210B P&ID: 325-01-600-PD-03	
11	C2S10,11	X1:12		LOOP POWERED (2-W)	JB-602	TT-60211	Gas Temperature Element for Hot Air Duct.. P&ID: 325-01-600-PD-03	
12	C2S10,12	X1:13		LOOP POWERED (2-W)	JB-602	TT-60212A	Bearing Temperature Transmitter for Main.. P&ID: 325-01-600-PD-03	
13	C2S10,13	X1:14		LOOP POWERED (2-W)	JB-602	TT-60212B	Bearing Temperature Transmitter for Main.. P&ID: 325-01-600-PD-03	
14	C2S10,14	X1:15		LOOP POWERED (2-W)	JB-602	TT-60214	Steam Temperature Transmitter for Cold A.. P&ID: 325-01-600-PD-03	
15	C2S10,15	X1:16		LOOP POWERED (2-W)	JB-602	TT-60215A	Liquid Temperature Transmitter for Heat .. P&ID: 325-01-600-PD-03	
	C2S10,00	X1:--		LOOP POWERED (2-W)	JB-602	AT-60241	Dust Analyzer Transmitter for Exhaust Du.. P&ID: 325-01-600-PD-05	

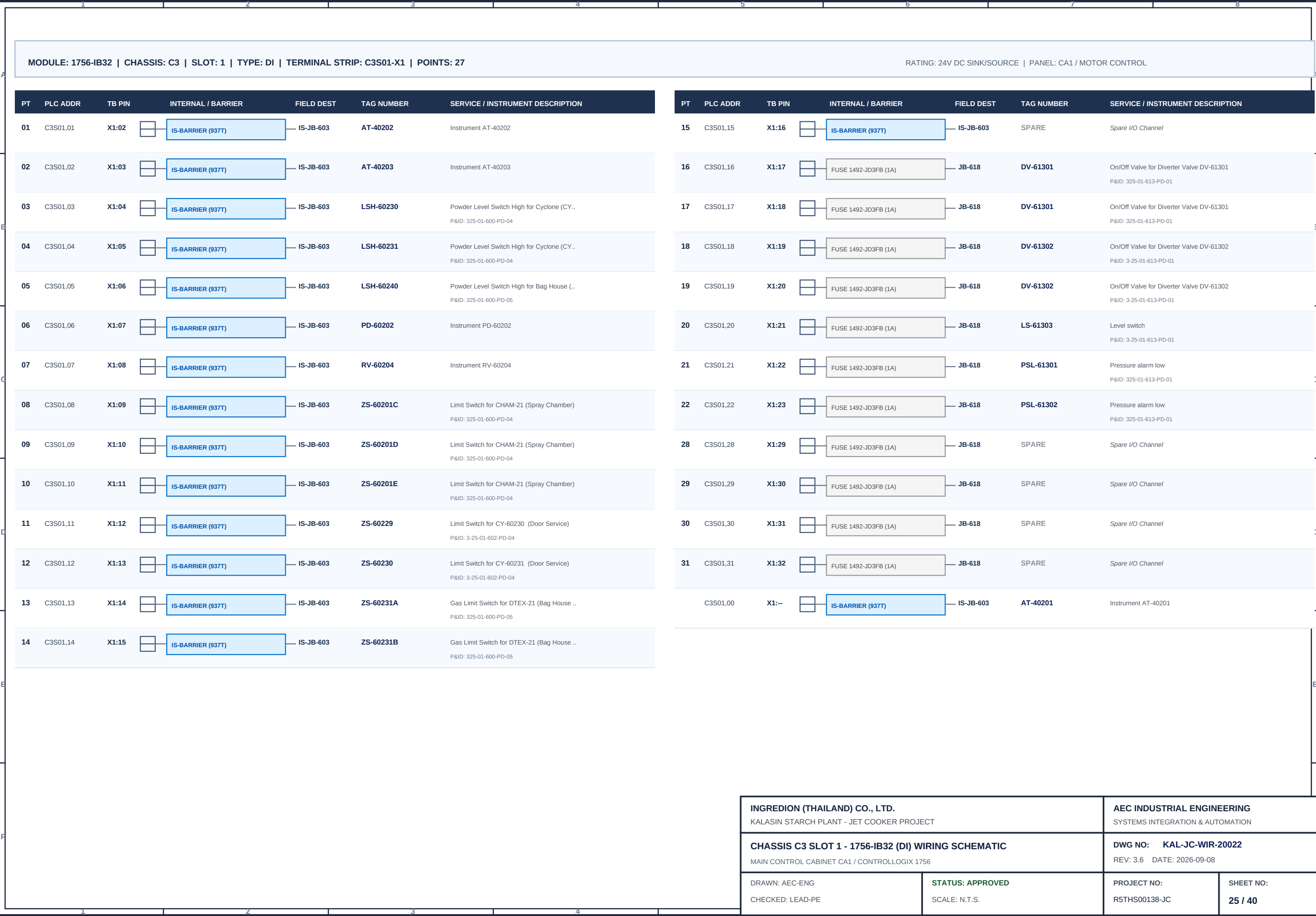
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C2 SLOT 10 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20018 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 21 / 40

MODULE: 1756-IF16 CHASSIS: C2 SLOT: 12 TYPE: AI TERMINAL STRIP: C2S12-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C2S12,01	X1:02		LOOP POWERED (2-W)	JB-607	FT-60221	Cool air Flow P&ID: 325-01-600-PD-04	
02	C2S12,02	X1:03		LOOP POWERED (2-W)	JB-607	FT-60261FA	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
03	C2S12,03	X1:04		LOOP POWERED (2-W)	JB-607	FT-60261FB	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
04	C2S12,04	X1:05		LOOP POWERED (2-W)	JB-607	FT-60261FC	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
05	C2S12,05	X1:06		LOOP POWERED (2-W)	JB-607	FT-60261FD	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
06	C2S12,06	X1:07		LOOP POWERED (2-W)	JB-607	FT-60262FE	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
07	C2S12,07	X1:08		LOOP POWERED (2-W)	JB-607	FT-60262FF	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
08	C2S12,08	X1:09		LOOP POWERED (2-W)	JB-607	FT-60262FG	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
09	C2S12,09	X1:10		LOOP POWERED (2-W)	JB-607	FT-60262FH	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
10	C2S12,10	X1:11		LOOP POWERED (2-W)	JB-607	FT-60263FJ	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
11	C2S12,11	X1:12		LOOP POWERED (2-W)	JB-607	FT-60263FK	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
12	C2S12,12	X1:13		LOOP POWERED (2-W)	JB-607	FT-60263FL	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
13	C2S12,13	X1:14		LOOP POWERED (2-W)	JB-607	FT-60263FM	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
14	C2S12,14	X1:15		LOOP POWERED (2-W)	JB-607	FT-60264FN	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
15	C2S12,15	X1:16		LOOP POWERED (2-W)	JB-607	FT-60264FP	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
	C2S12,00	X1:--		LOOP POWERED (2-W)	JB-607	FT-60207	Instr. Air Flow Transmitter for Camera #.. P&ID: 325-01-600-PD-07	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C2 SLOT 12 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20020 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 23 / 40

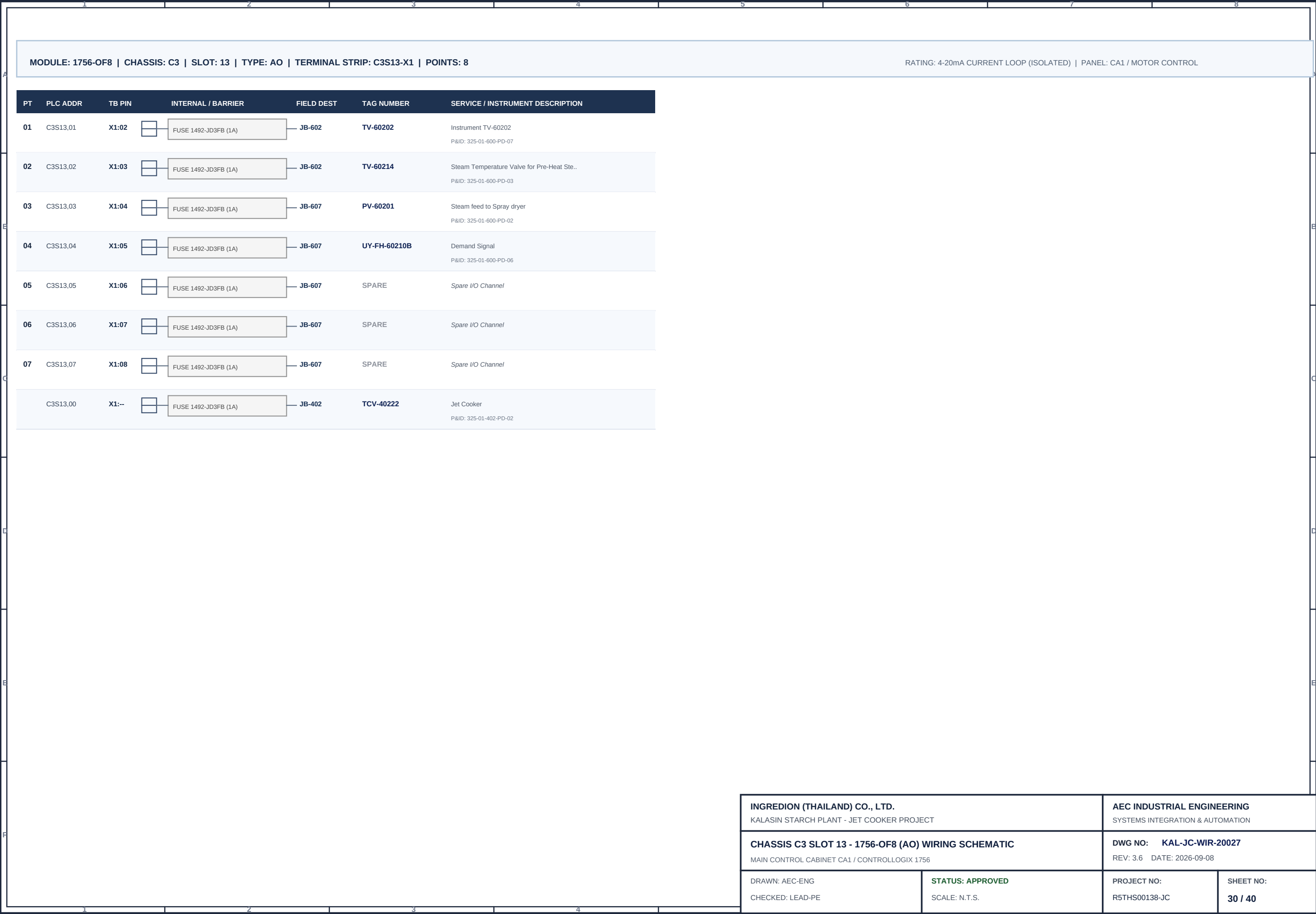
MODULE: 1756-IF16 CHASSIS: C2 SLOT: 13 TYPE: AI TERMINAL STRIP: C2S13-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C2S13,01	X1:02		LOOP POWERED (2-W)	JB-607	FT-60264FR	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	
02	C2S13,02	X1:03		LOOP POWERED (2-W)	JB-607	PT-60201	Steam feed to Spray dryer P&ID: 3-25-01-602-PD-02	
03	C2S13,03	X1:04		LOOP POWERED (2-W)	JB-607	PT-60207	Instr. Air Pressure Transmitter for Came.. P&ID: 325-01-600-PD-07	
04	C2S13,04	X1:05		LOOP POWERED (2-W)	JB-607	PT-60221	Gas Pressure Transmitter for Cooling Air.. P&ID: 325-01-600-PD-02	
05	C2S13,05	X1:06		LOOP POWERED (2-W)	JB-607	PT-60221	Gas Pressure Transmitter for Cooling Air.. P&ID: 325-01-600-PD-04	
06	C2S13,06	X1:07		LOOP POWERED (2-W)	JB-607	TI-FH-60210	Gas Burner Firing Rate Temp Signal to AH.. P&ID: 325-01-600-PD-06	
07	C2S13,07	X1:08		LOOP POWERED (2-W)	JB-607	TT-60201	Steam feed to Spray dryer P&ID: 325-01-600-PD-02	
08	C2S13,08	X1:09		LOOP POWERED (2-W)	JB-607	TT-60202	Compress air temp to camera P&ID: 325-01-600-PD-07	
09	C2S13,09	X1:10		LOOP POWERED (2-W)	JB-607	TT-60207	Instr. Air Temperature Transmitter for A.. P&ID: 325-01-600-PD-07	
10	C2S13,10	X1:11		LOOP POWERED (2-W)	JB-607	UY-FH-60210A	Comb fan VFD Speed Control P&ID: 325-01-600-PD-06	
11	C2S13,11	X1:12		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
12	C2S13,12	X1:13		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
13	C2S13,13	X1:14		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
14	C2S13,14	X1:15		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
15	C2S13,15	X1:16		LOOP POWERED (2-W)	JB-607	SPARE	Spare I/O Channel	
	C2S13,00	X1:--		LOOP POWERED (2-W)	JB-607	FT-60264FQ	Feed Slurry Flow Transmitter for Nozzle .. P&ID: 325-01-600-PD-02	

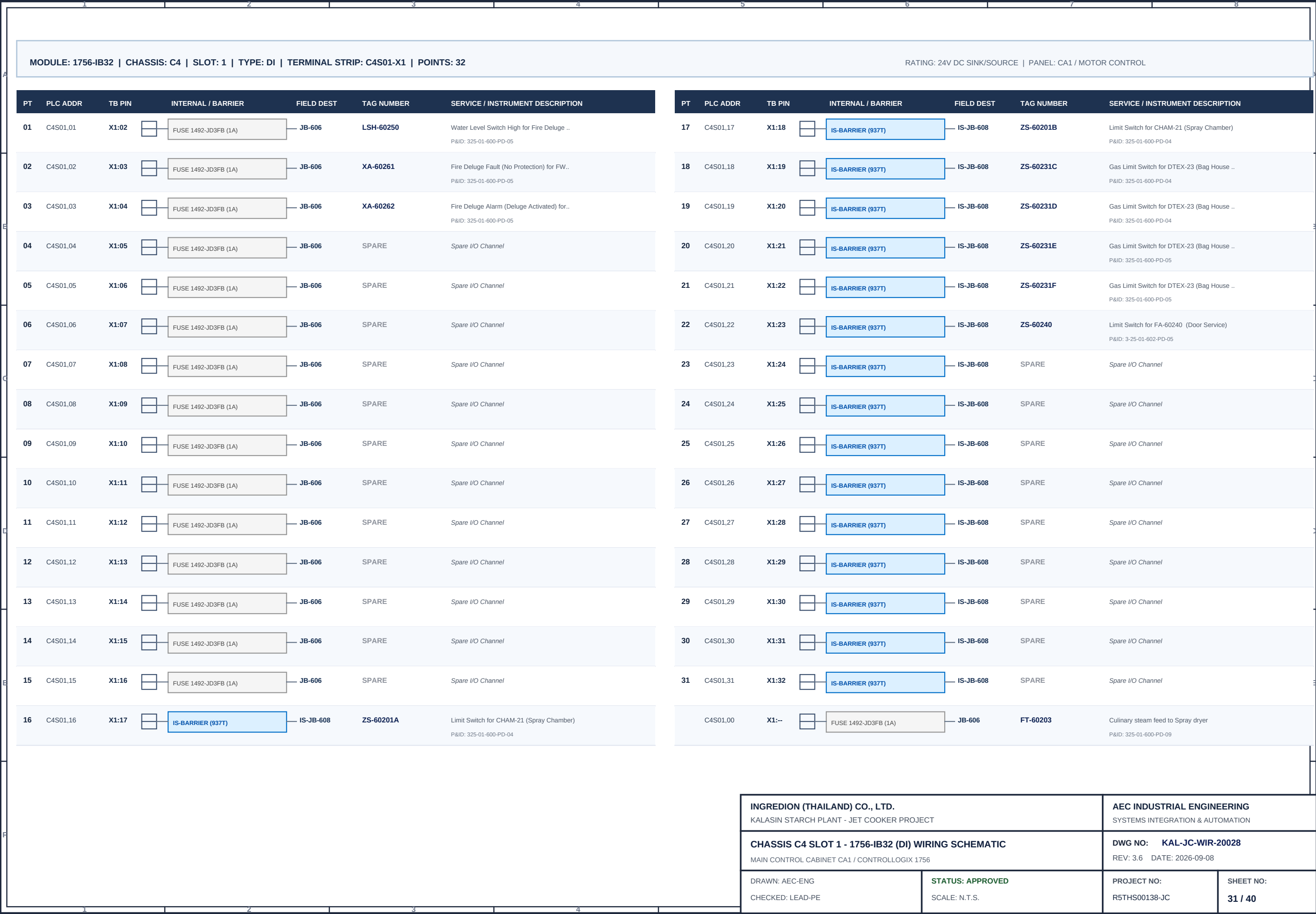
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C2 SLOT 13 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20021 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 24 / 40



MODULE: 1756-OB32 CHASSIS: C3 SLOT: 3 TYPE: DO TERMINAL STRIP: C3S03-X1 POINTS: 32														RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL													
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION		PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION											
01	C3S03,01	X1:02		RELAY 700-HL (24VDC)	JB-618	DV-61302	On/Off Valve for Diverter Valve DV-61302 P&ID: 3-25-01-613-PD-01		17	C3S03,17	X1:18		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
02	C3S03,02	X1:03		RELAY 700-HL (24VDC)	JB-618	SV-61301	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-01		18	C3S03,18	X1:19		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
03	C3S03,03	X1:04		RELAY 700-HL (24VDC)	JB-618	SV-61302	Air service P&ID: 325-01-613-PD-01		19	C3S03,19	X1:20		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
04	C3S03,04	X1:05		RELAY 700-HL (24VDC)	JB-618	SV-61303	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-01		20	C3S03,20	X1:21		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
05	C3S03,05	X1:06		RELAY 700-HL (24VDC)	JB-618	SV-61304	PA On/Off Valve for Conveying Cooler Cyc.. P&ID: 325-01-613-PD-01		21	C3S03,21	X1:22		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
06	C3S03,06	X1:07		RELAY 700-HL (24VDC)	JB-618	SV-61305	Air service P&ID: 325-01-613-PD-01		22	C3S03,22	X1:23		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
07	C3S03,07	X1:08		RELAY 700-HL (24VDC)	JB-618	SV-61306	Instrument SV-61306 P&ID: 3-25-01-613-PD-01		23	C3S03,23	X1:24		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
08	C3S03,08	X1:09		RELAY 700-HL (24VDC)	JB-618	XV-61301	Product feed-out Packing silo P&ID: 3-25-01-613-PD-01		24	C3S03,24	X1:25		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
09	C3S03,09	X1:10		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel		25	C3S03,25	X1:26		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
10	C3S03,10	X1:11		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel		26	C3S03,26	X1:27		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
11	C3S03,11	X1:12		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel		27	C3S03,27	X1:28		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
12	C3S03,12	X1:13		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel		28	C3S03,28	X1:29		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
13	C3S03,13	X1:14		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel		29	C3S03,29	X1:30		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
14	C3S03,14	X1:15		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel		30	C3S03,30	X1:31		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
15	C3S03,15	X1:16		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel		31	C3S03,31	X1:32		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel											
16	C3S03,16	X1:17		RELAY 700-HL (24VDC)	JB-618	SPARE	Spare I/O Channel			C3S03,00	X1:--		RELAY 700-HL (24VDC)	JB-618	DV-61301	On/Off Valve for Diverter Valve DV-61301 P&ID: 325-01-613-PD-01											

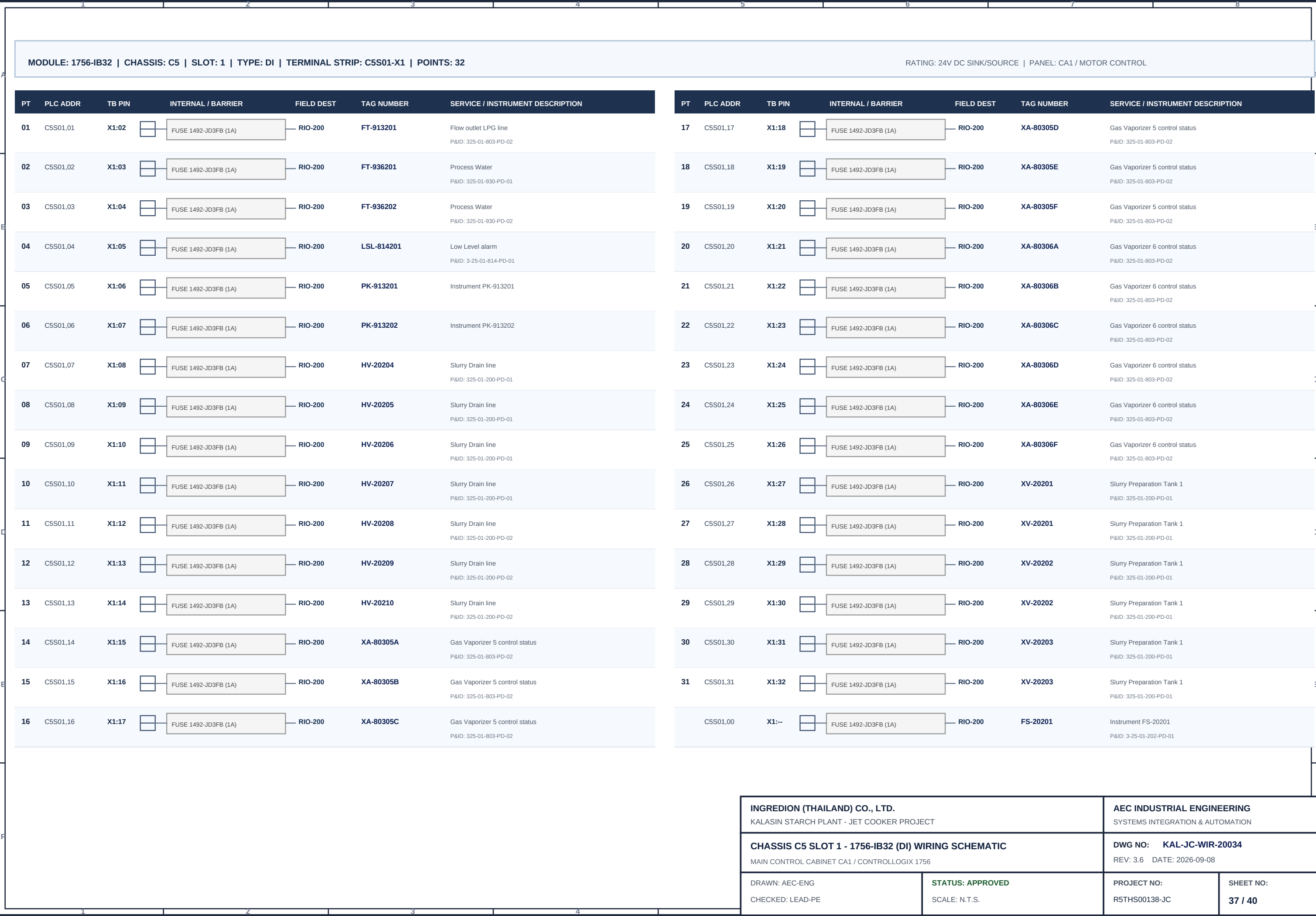
MODULE: 1756-IF16 CHASSIS: C3 SLOT: 5 TYPE: AI TERMINAL STRIP: C3S05-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C3S05,01	X1:02		LOOP POWERED (2-W)	JB-618	PT-61306	Instrument PT-61306 P&ID: 3-25-01-613-PD-01	
02	C3S05,02	X1:03		LOOP POWERED (2-W)	JB-618	TT-61303	Instrument TT-61303 P&ID: 3-25-01-613-PD-01	
03	C3S05,03	X1:04		LOOP POWERED (2-W)	JB-618	TT-61304	Instrument TT-61304 P&ID: 3-25-01-613-PD-01	
04	C3S05,04	X1:05		IS-TX-BARRIER (937T)	IS-JB-618	DPT-61301	Vent filter pressure P&ID: 325-01-613-PD-01	
05	C3S05,05	X1:06		IS-TX-BARRIER (937T)	IS-JB-618	DPT-61302	Vent filter pressure P&ID: 325-01-613-PD-01	
06	C3S05,06	X1:07		IS-TX-BARRIER (937T)	IS-JB-618	LT-61301	Level powder in silo P&ID: 325-01-613-PD-01	
07	C3S05,07	X1:08		IS-TX-BARRIER (937T)	IS-JB-618	PT-61301	Pressure in silo P&ID: 325-01-613-PD-01	
08	C3S05,08	X1:09		IS-TX-BARRIER (937T)	IS-JB-618	PT-61303	Powder Pressure Transmitter for Cooling .. P&ID: 325-01-613-PD-01	
09	C3S05,09	X1:10		IS-TX-BARRIER (937T)	IS-JB-618	PT-61304	Gas Pressure Transmitter for Storage Bin.. P&ID: 325-01-613-PD-01	
10	C3S05,10	X1:11		IS-TX-BARRIER (937T)	IS-JB-618	PT-61305	Gas Pressure Transmitter for Exhaust Duc.. P&ID: 325-01-613-PD-01	
11	C3S05,11	X1:12		IS-TX-BARRIER (937T)	IS-JB-618	TT-61301	Gas Temperature Transmitter for Cold Air.. P&ID: 325-01-613-PD-01	
12	C3S05,12	X1:13		IS-TX-BARRIER (937T)	IS-JB-618	TT-61302	Gas and Powder Temperature Transmitter f.. P&ID: 325-01-613-PD-01	
13	C3S05,13	X1:14		IS-TX-BARRIER (937T)	IS-JB-618	SPARE	Spare I/O Channel	
14	C3S05,14	X1:15		IS-TX-BARRIER (937T)	IS-JB-618	SPARE	Spare I/O Channel	
15	C3S05,15	X1:16		IS-TX-BARRIER (937T)	IS-JB-618	SPARE	Spare I/O Channel	
	C3S05,00	X1:--		LOOP POWERED (2-W)	JB-618	PT-61303	Powder Pressure Transmitter for Cooling .. P&ID: 3-25-01-613-PD-01	
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT							AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C3 SLOT 5 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756							DWG NO: KAL-JC-WIR-20026 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.			PROJECT NO: R5THS00138-JC	SHEET NO: 29 / 40

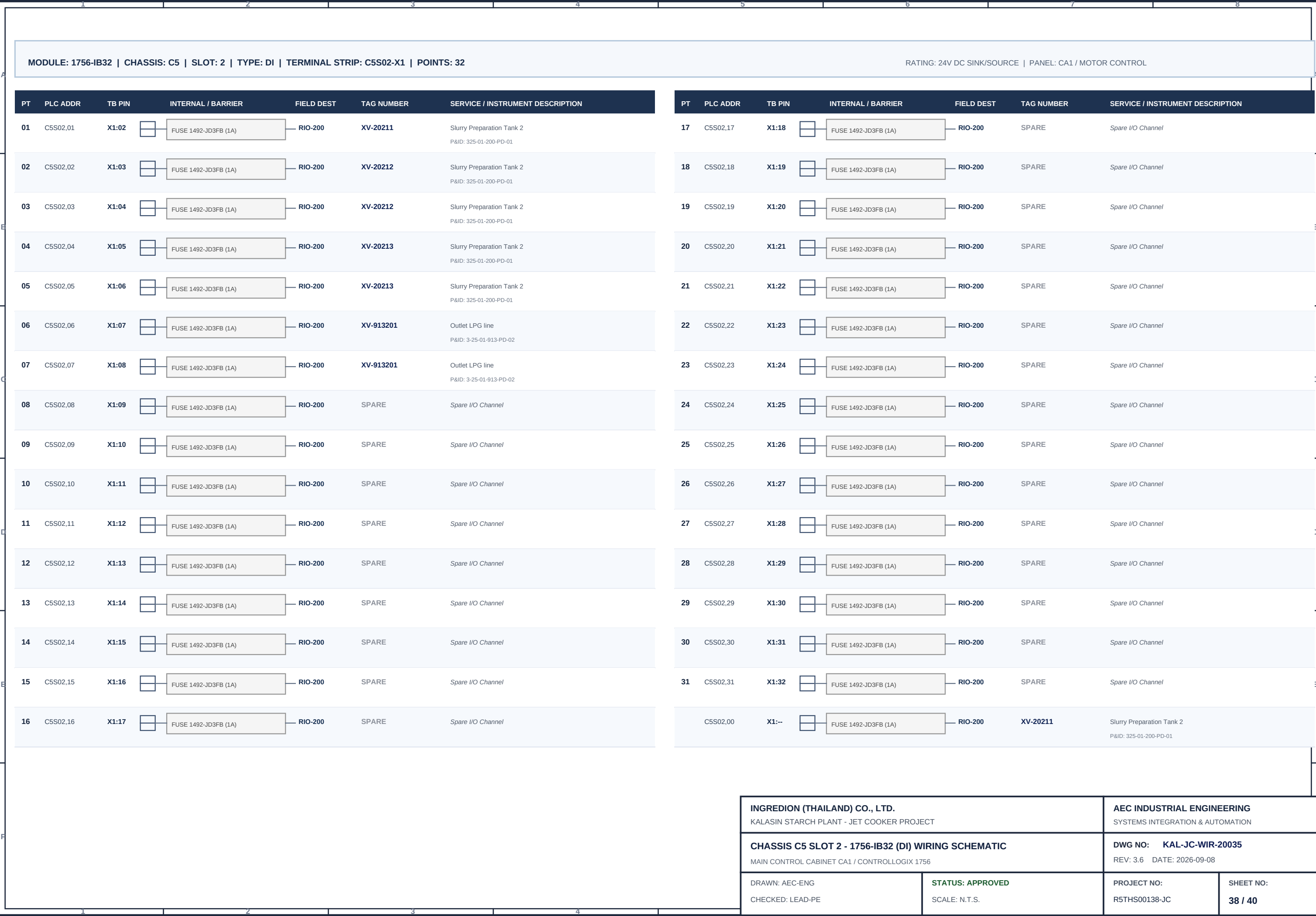


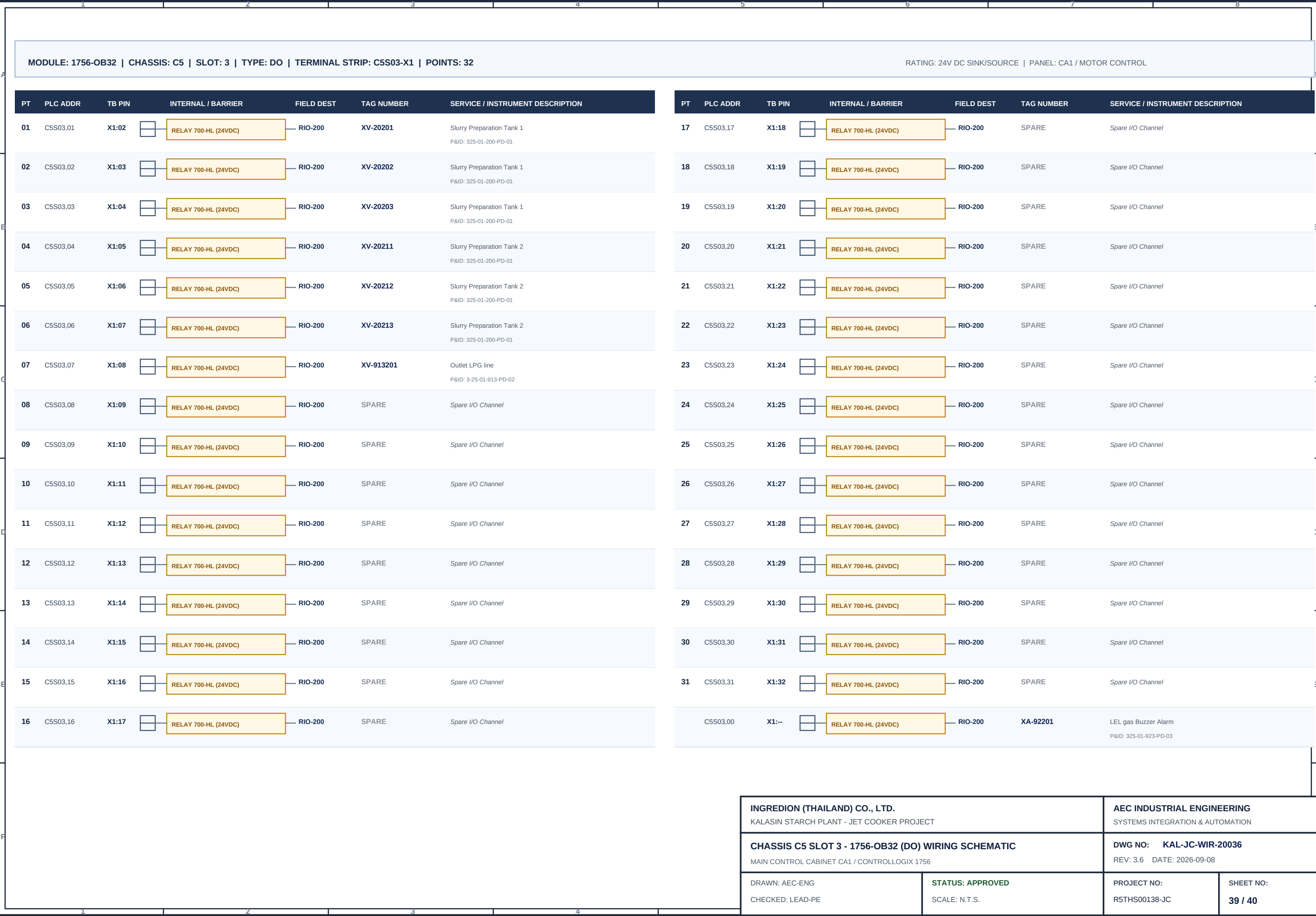


MODULE: 1756-IB32 CHASSIS: C4 SLOT: 3 TYPE: DI TERMINAL STRIP: C4S03-X1 POINTS: 31								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C4S03,01	X1:02		IS-BARRIER (937T)	IS-JB-612	ISV-61304	Explosion isolation flap valve ISV-61304 P&ID: 325-01-613-PD-05	18	C4S03,18	X1:19		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
02	C4S03,02	X1:03		IS-BARRIER (937T)	IS-JB-612	LSH-61304	Big bag level alarm high P&ID: 325-01-613-PD-03	19	C4S03,19	X1:20		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
03	C4S03,03	X1:04		IS-BARRIER (937T)	IS-JB-612	LSH-61305	Big bag level alarm high P&ID: 325-01-613-PD-05	20	C4S03,20	X1:21		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
04	C4S03,04	X1:05		IS-BARRIER (937T)	IS-JB-612	LSH-61307	High Level alarm reject screen P&ID: 3-25-01-613-PD-01	21	C4S03,21	X1:22		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
05	C4S03,05	X1:06		IS-BARRIER (937T)	IS-JB-612	LSH-61312	Level alarm high P&ID: 325-01-613-PD-05	22	C4S03,22	X1:23		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
06	C4S03,06	X1:07		IS-BARRIER (937T)	IS-JB-612	PSE-61302	Explosion Vent PSE-61302 P&ID: 325-01-613-PD-03	23	C4S03,23	X1:24		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
07	C4S03,07	X1:08		IS-BARRIER (937T)	IS-JB-612	PSE-61303	Explosion Vent PSE-61303 P&ID: 325-01-613-PD-05	24	C4S03,24	X1:25		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
09	C4S03,09	X1:10		IS-BARRIER (937T)	IS-JB-612	RV-61301	Instrument RV-61301	25	C4S03,25	X1:26		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
10	C4S03,10	X1:11		IS-BARRIER (937T)	IS-JB-612	RV-61304	Instrument RV-61304	26	C4S03,26	X1:27		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
11	C4S03,11	X1:12		IS-BARRIER (937T)	IS-JB-612	RV-61305	Instrument RV-61305	27	C4S03,27	X1:28		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
12	C4S03,12	X1:13		IS-BARRIER (937T)	IS-JB-612	SC-61301	Instrument SC-61301	28	C4S03,28	X1:29		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
13	C4S03,13	X1:14		IS-BARRIER (937T)	IS-JB-612	SC-61302	Instrument SC-61302	29	C4S03,29	X1:30		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
14	C4S03,14	X1:15		IS-BARRIER (937T)	IS-JB-612	ZS-61312	Position Magnet Door P&ID: 325-01-613-PD-01	30	C4S03,30	X1:31		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
15	C4S03,15	X1:16		IS-BARRIER (937T)	IS-JB-612	ZS-61313	Position Magnet Door P&ID: 325-01-613-PD-01	31	C4S03,31	X1:32		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel
16	C4S03,16	X1:17		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel		C4S03,00	X1:--		IS-BARRIER (937T)	IS-JB-612	ISV-61303	Explosion isolation flap valve ISV-61303 P&ID: 325-01-613-PD-03
17	C4S03,17	X1:18		IS-BARRIER (937T)	IS-JB-612	SPARE	Spare I/O Channel								
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT								AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION							
CHASSIS C4 SLOT 3 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756								DWG NO: KAL-JC-WIR-20030 REV: 3.6 DATE: 2026-09-08							
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.				PROJECT NO: R5THS00138-JC		SHEET NO: 33 / 40					

MODULE: 1756-OB32 CHASSIS: C4 SLOT: 4 TYPE: DO TERMINAL STRIP: C4S04-X1 POINTS: 32								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C4S04,01	X1:02		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	17	C4S04,17	X1:18		RELAY 700-HL (24VDC)	JB-612	PVM-61301	Start / Stop Command for FVNR PV-61301 (.. P&ID: 325-01-613-PD-03
02	C4S04,02	X1:03		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	18	C4S04,18	X1:19		RELAY 700-HL (24VDC)	JB-612	SV-61307	Instrument SV-61307 P&ID: 3-25-01-613-PD-01
03	C4S04,03	X1:04		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	19	C4S04,19	X1:20		RELAY 700-HL (24VDC)	JB-612	SV-61308	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-03
04	C4S04,04	X1:05		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	20	C4S04,20	X1:21		RELAY 700-HL (24VDC)	JB-612	SV-61309	Air service P&ID: 325-01-613-PD-03
05	C4S04,05	X1:06		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	21	C4S04,21	X1:22		RELAY 700-HL (24VDC)	JB-612	SV-61310	Timer compress air pluse solenoid valve P&ID: 325-01-613-PD-05
06	C4S04,06	X1:07		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	22	C4S04,22	X1:23		RELAY 700-HL (24VDC)	JB-612	SV-61311	Air service P&ID: 325-01-613-PD-05
07	C4S04,07	X1:08		RELAY 700-HL (24VDC)	JB-606	SPARE	Spare I/O Channel	23	C4S04,23	X1:24		RELAY 700-HL (24VDC)	JB-612	XS-61301A	Filter plant in operation P&ID: 325-01-613-PD-02
08	C4S04,08	X1:09		IS-TX-BARRIER (937T)	IS-JB-608	XA-60201	LEL gas Buzzer Alarm P&ID: 325-01-600-PD-03	24	C4S04,24	X1:25		RELAY 700-HL (24VDC)	JB-612	XS-61301B	Material transport in operation P&ID: 325-01-613-PD-02
09	C4S04,09	X1:10		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	25	C4S04,25	X1:26		RELAY 700-HL (24VDC)	JB-612	XS-61301C	Belt plant in operation P&ID: 325-01-613-PD-02
10	C4S04,10	X1:11		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	26	C4S04,26	X1:27		RELAY 700-HL (24VDC)	JB-612	XS-61301D	Sequence Stop P&ID: 325-01-613-PD-02
11	C4S04,11	X1:12		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	27	C4S04,27	X1:28		RELAY 700-HL (24VDC)	JB-612	XS-61301E	Status from Customer Emergency stop P&ID: 325-01-613-PD-02
12	C4S04,12	X1:13		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	28	C4S04,28	X1:29		RELAY 700-HL (24VDC)	JB-612	XS-61301F	Operation Release P&ID: 325-01-613-PD-02
13	C4S04,13	X1:14		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	29	C4S04,29	X1:30		RELAY 700-HL (24VDC)	JB-612	XV-61302	Dust collector feed vent P&ID: 325-01-613-PD-02
14	C4S04,14	X1:15		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	30	C4S04,30	X1:31		RELAY 700-HL (24VDC)	JB-612	XV-61303	Dust collector feed from packer P&ID: 325-01-613-PD-02
15	C4S04,15	X1:16		IS-TX-BARRIER (937T)	IS-JB-608	SPARE	Spare I/O Channel	31	C4S04,31	X1:32		RELAY 700-HL (24VDC)	JB-612	SPARE	Spare I/O Channel
16	C4S04,16	X1:17		RELAY 700-HL (24VDC)	JB-612	PVM-61301	Start / Stop Command for FVNR PV-61301 (.. P&ID: 325-01-613-PD-03		C4S04,00	X1:--		RELAY 700-HL (24VDC)	JB-606	SV-60220	PA On/Off Valve for Chamber PSE purge ai.. P&ID: 325-01-600-PD-04
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT												AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION			
CHASSIS C4 SLOT 4 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756												DWG NO: KAL-JC-WIR-20031 REV: 3.6 DATE: 2026-09-08			
DRAWN: AEC-ENG CHECKED: LEAD-PE						STATUS: APPROVED SCALE: N.T.S.						PROJECT NO: R5THS00138-JC		SHEET NO: 34 / 40	







MODULE: 1756-IF16 CHASSIS: C5 SLOT: 4 TYPE: AI TERMINAL STRIP: C5S04-X1 POINTS: 16							RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL	
PT	PLC ADDR	TB PIN	INTERNAL / BARRIER		FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	
01	C5S04,01	X1:02		LOOP POWERED (2-W)	RIO-200	FT-913201	Flow outlet LPG line P&ID: 325-01-803-PD-02	
02	C5S04,02	X1:03		LOOP POWERED (2-W)	RIO-200	FT-913201	Flow outlet LPG line P&ID: 325-01-803-PD-02	
03	C5S04,03	X1:04		LOOP POWERED (2-W)	RIO-200	FT-936201	Process Water P&ID: 325-01-930-PD-01	
04	C5S04,04	X1:05		LOOP POWERED (2-W)	RIO-200	FT-936202	Process Water P&ID: 325-01-930-PD-02	
05	C5S04,05	X1:06		LOOP POWERED (2-W)	RIO-200	LT-913201	Level LPG in storage P&ID: 325-01-803-PD-01	
06	C5S04,06	X1:07		LOOP POWERED (2-W)	RIO-200	PT-913201	Pressure outlet LPG line P&ID: 325-01-803-PD-02	
07	C5S04,07	X1:08		LOOP POWERED (2-W)	RIO-200	XT-92201	LEL gas analyzer P&ID: 325-01-923-PD-03	
08	C5S04,08	X1:09		LOOP POWERED (2-W)	RIO-200	XV-913201	Outlet LPG line P&ID: 3-25-01-913-PD-02	
09	C5S04,09	X1:10		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
10	C5S04,10	X1:11		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
11	C5S04,11	X1:12		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
12	C5S04,12	X1:13		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
13	C5S04,13	X1:14		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel P&ID: 3-25-01-913-PD-02	
14	C5S04,14	X1:15		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel	
15	C5S04,15	X1:16		LOOP POWERED (2-W)	RIO-200	SPARE	Spare I/O Channel	
	C5S04,00	X1:--		LOOP POWERED (2-W)	RIO-200	EXT-BIILER	Instrument EXT-BIILER P&ID: External Remote Setpoint	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
CHASSIS C5 SLOT 4 - 1756-IF16 (AI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756		DWG NO: KAL-JC-WIR-20037 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 40 / 40

KALASIN STARCH PLANT - JET COOKER PROJECT

STANDARD INSTRUMENT HOOK-UP & TYPICAL ELECTRICAL LOOP DRAWINGS

Based on Equipment Packages in 11-instrument Manual/ & Master I/O Mapping Rev 3.6

DRAWING PACKAGE OBJECTIVES & ENGINEERING STANDARDS

Project Scope:	Jet Cooker Slurry & Flash Tank Instrumentation
Client / Owner:	INGREDION (THAILAND) CO., LTD.
Plant Location:	Kalasin Starch Manufacturing Plant, Thailand
System Integrator:	AEC INDUSTRIAL ENGINEERING CO., LTD.
Covered Packages:	23 Complete Equipment Types (CS-DS-01 to CS-DS-22)
Drawing Contents:	Piping Hook-Up, Loop Wiring Schematic & BOM
Electrical Standard:	Rockwell ControlLogix 1756 Architecture (24V DC / 4-20mA)
Hazardous Areas:	Ex ia / Ex d Classifications (937T Barriers & Glands)
CAD Standard:	1:1 Metric Millimeters (AutoCAD DXF R2010 Layered)
Piping Standard:	ASME B16.5 Cl.150/300 RF Flanges & NPT Threads
Document Status:	APPROVED ENGINEERING TEMPLATES (Rev 3.6)

DRAWING SYMBOLOGY & CAD LAYER CONVENTIONS

Process Piping:
Heavy solid line (0.7mm), showing process flow direction
Impulse Tubing:
Medium dashed line, 1/2" OD seamless SS316 tubing
Instrument Air:
Pneumatic air supply line, 6.0 bar dry filtered air
Analog Signal (AI/AO):
4-20mA HART loop, shielded twisted pair (Belden 8761)
Discrete Signal (DI/DO):
24V DC wetted contact or solenoid drive (Belden 9533)
IS Barrier (Ex ia):
Sky Blue block, Rockwell 937T Series Intrinsically Safe
Interposing Relay:
Orange block, Allen-Bradley 700-HL Series 24VDC Relay
Terminal Strip:
Green circle/box, Marshalling Panel CA1 (C#S##-X1)
Earth Grounding:
Protective Earth (PE) & Instrument Isolated Clean Earth (IE)

PREPARED BY

AEC INDUSTRIAL ENG.

Lead Instrumentation Eng.

CHECKED BY

AEC AUTOMATION LEAD

System Architect & PE

VERIFIED BY

PROJECT ELECTRICAL PE

Licensed Professional Eng.

APPROVED BY

INGREDION PROJECT MGR.

Client Automation Lead

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECT

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

INSTRUMENT HOOK-UP & LOOP DRAWING DIRECTORY
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM

DWG NO: KAL-JC-TYP-000
REV: 3.6 DATE: 2026-09-08

DRAWN: AEC-ENG
CHECKED: LEAD-PE

STATUS: APPROVED
SCALE: N.T.S.

PROJECT NO:
R5THS00138-JC

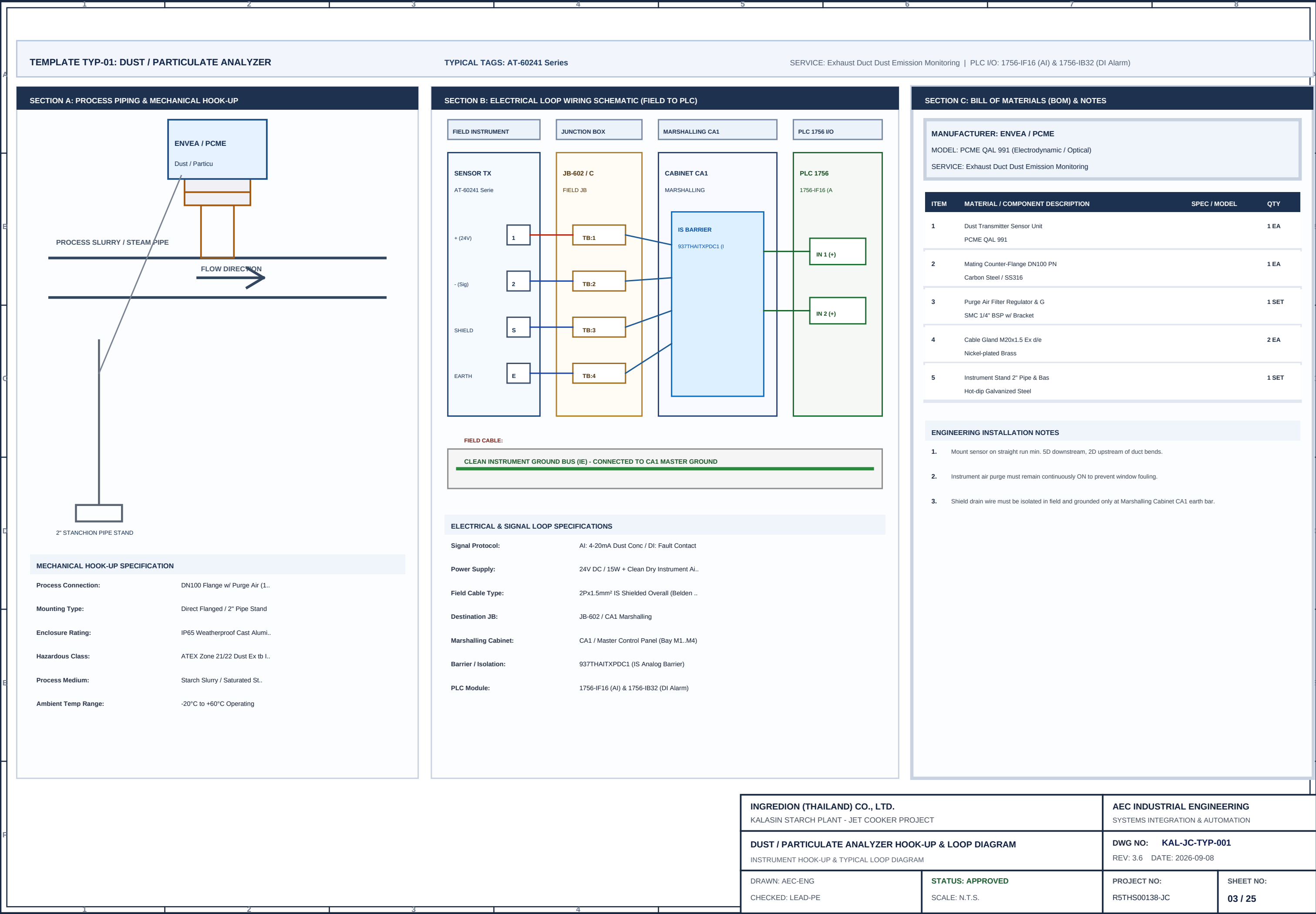
SHEET NO:
01 / 25

MASTER DRAWING INDEX - 23 INSTRUMENT HOOK-UP & LOOP DIAGRAM TEMPLATES

SHT	DRAWING NO	EQUIPMENT CATEGORY / TEMPLATE TITLE	REV	STATUS
03	KAL-JC-TYP-001	DUST / PARTICULATE ANALYZER HOOK-UP & LO.. Mfr: ENVEA / PCME Model: PCME QAL 991 (Electrodynamic	3.6	APPROVED
04	KAL-JC-TYP-002	COMBUSTIBLE LEL GAS DETECTOR HOOK-UP & L.. Mfr: Sensidyne / SmartGas Model: 925FGD Explosionproof Hydroc	3.6	APPROVED
05	KAL-JC-TYP-003	pH ANALYTICAL MEASUREMENT LOOP HOOK-UP &.. Mfr: Mettler Toledo Model: M300G2 4-Wire Transmitter +	3.6	APPROVED
06	KAL-JC-TYP-004	DIFFERENTIAL PRESSURE GAUGE HOOK-UP & ME.. Mfr: Ashcroft Model: Model 1132 (Piston-Type Diff	3.6	APPROVED
07	KAL-JC-TYP-005	DIFFERENTIAL PRESSURE TRANSMITTER HOOK-U.. Mfr: Endress+Hauser Model: Deltabar PMD55B (Piezoresist	3.6	APPROVED
08	KAL-JC-TYP-006	CORIOLIS MASS FLOWMETER HOOK-UP & LOOP D.. Mfr: Endress+Hauser Model: Promass E 300 (8E3B50 / Dual	3.6	APPROVED
09	KAL-JC-TYP-007	ELECTROMAGNETIC FLOWMETER HOOK-UP & LOOP.. Mfr: Endress+Hauser Model: Promag P 300 (5P3B50 / 80 /	3.6	APPROVED
10	KAL-JC-TYP-008	VORTEX STEAM FLOWMETER HOOK-UP & LOOP DI.. Mfr: Endress+Hauser Model: Prowirl F 200 (7F2C80 / 1H w	3.6	APPROVED
11	KAL-JC-TYP-009	THERMAL MASS GAS FLOWMETER HOOK-UP & LOO.. Mfr: Endress+Hauser Model: t-mass F 300 / I 300 (6F3B /	3.6	APPROVED
12	KAL-JC-TYP-010	ULTRASONIC CLAMP-ON FLOWMETER HOOK-UP & .. Mfr: Keyence Model: FD-H Series (FD-H20 / FD-H32	3.6	APPROVED
13	KAL-JC-TYP-011	MANUAL HAND VALVE POSITION SWITCH HOOK-U.. Mfr: TeroFox / APL-HKC Model: TF-20DFS Ball Valve + APL-21	3.6	APPROVED
14	KAL-JC-TYP-012	LIQUID LEVEL SWITCH (VIBRATING FORK) HOO.. Mfr: Endress+Hauser Model: Liquiphant FTL51B (Tuning Fo	3.6	APPROVED
15	KAL-JC-TYP-013	SOLID LEVEL SWITCH (POWDER FORK) HOOK-UP.. Mfr: Endress+Hauser Model: Soliphant FTM51 (Vibrating F	3.6	APPROVED

SHT	DRAWING NO	EQUIPMENT CATEGORY / TEMPLATE TITLE	REV	STATUS
16	KAL-JC-TYP-014	RADAR LEVEL TRANSMITTER (80 GHz) HOOK-UP.. Mfr: Endress+Hauser Model: Micropilot FMR67B (80 GHz Hi	3.6	APPROVED
17	KAL-JC-TYP-015	PRESSURE GAUGE & SIPHON HOOK-UP & MECHAN.. Mfr: Ashcroft Model: Model T5500 (100mm Dial All-	3.6	APPROVED
18	KAL-JC-TYP-016	ELECTRONIC PRESSURE SWITCH HOOK-UP & LOO.. Mfr: Endress+Hauser Model: Ceraphant PTC31B (Ceramic Se	3.6	APPROVED
19	KAL-JC-TYP-017	PRESSURE TRANSMITTER HOOK-UP & LOOP DIAG.. Mfr: Endress+Hauser Model: Cerabar PMP51B / PMC51B (Met	3.6	APPROVED
20	KAL-JC-TYP-018	BIMETAL TEMPERATURE GAUGE & THERMOWELL H.. Mfr: Ashcroft Model: Model FI (100mm Dial Everyan	3.6	APPROVED
21	KAL-JC-TYP-019	TEMPERATURE TRANSMITTER (RTD Pt100) HOOK.. Mfr: Endress+Hauser Model: iTEMP TMT71 + Omnigrad M TM4	3.6	APPROVED
22	KAL-JC-TYP-020	PNEUMATIC ON-OFF CONTROL VALVE HOOK-UP &.. Mfr: TeroFox / El-O-Matic Model: TF-20DFS Ball Valve + F-Seri	3.6	APPROVED
23	KAL-JC-TYP-021	MODULATING GLOBE CONTROL VALVE HOOK-UP &.. Mfr: Samson Model: Type 3241 Globe Valve + Trov	3.6	APPROVED
24	KAL-JC-TYP-022	ELECTRONIC TEMPERATURE SWITCH HOOK-UP & .. Mfr: Endress+Hauser Model: Thermophant TTR35 (Compact P	3.6	APPROVED
25	KAL-JC-TYP-023	PNEUMATIC TURBINE VIBRATOR HOOK-UP & LOO.. Mfr: Netter Vibration Model: NCT 5 Pneumatic Turbine Vibr	3.6	APPROVED

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
INSTRUMENT TEMPLATE DRAWING DIRECTORY INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM		DWG NO: KAL-JC-TYP-000-IDX REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 02 / 25

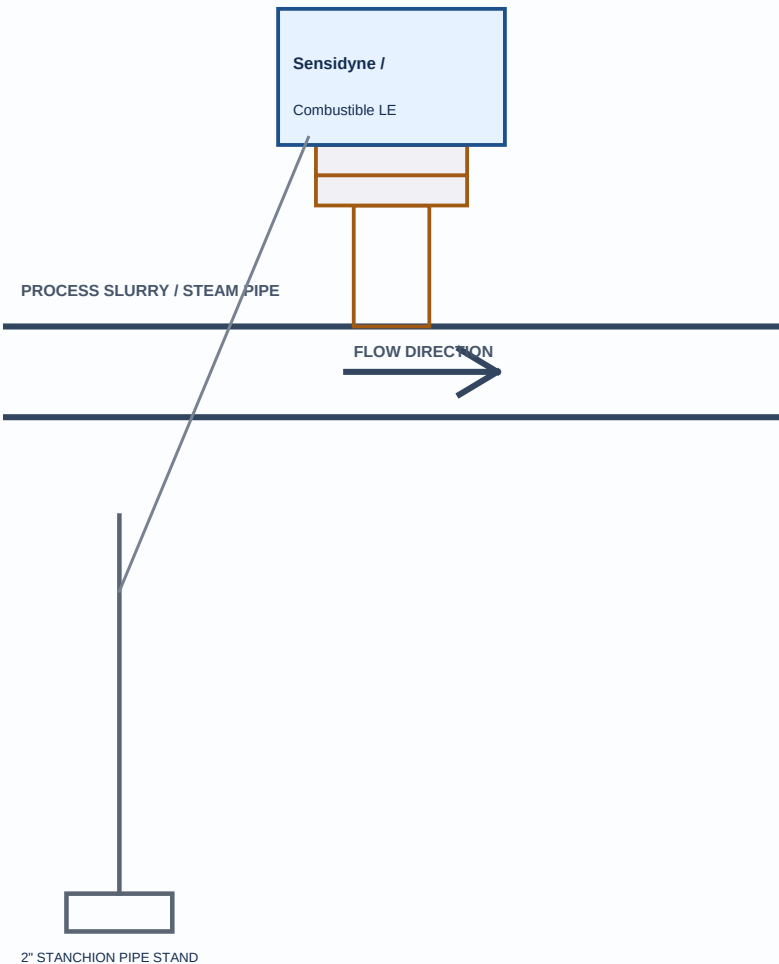


TEMPLATE TYP-02: COMBUSTIBLE LEL GAS ANALYZER

TYPICAL TAGS: AIT-60101 / AIT-60201

SERVICE: Explosion Hazard Lower Explosive Limit (LEL) Monitor | PLC I/O: 1756-IF16 (AI LEL) & 1756-IB32 (DI Alarm)

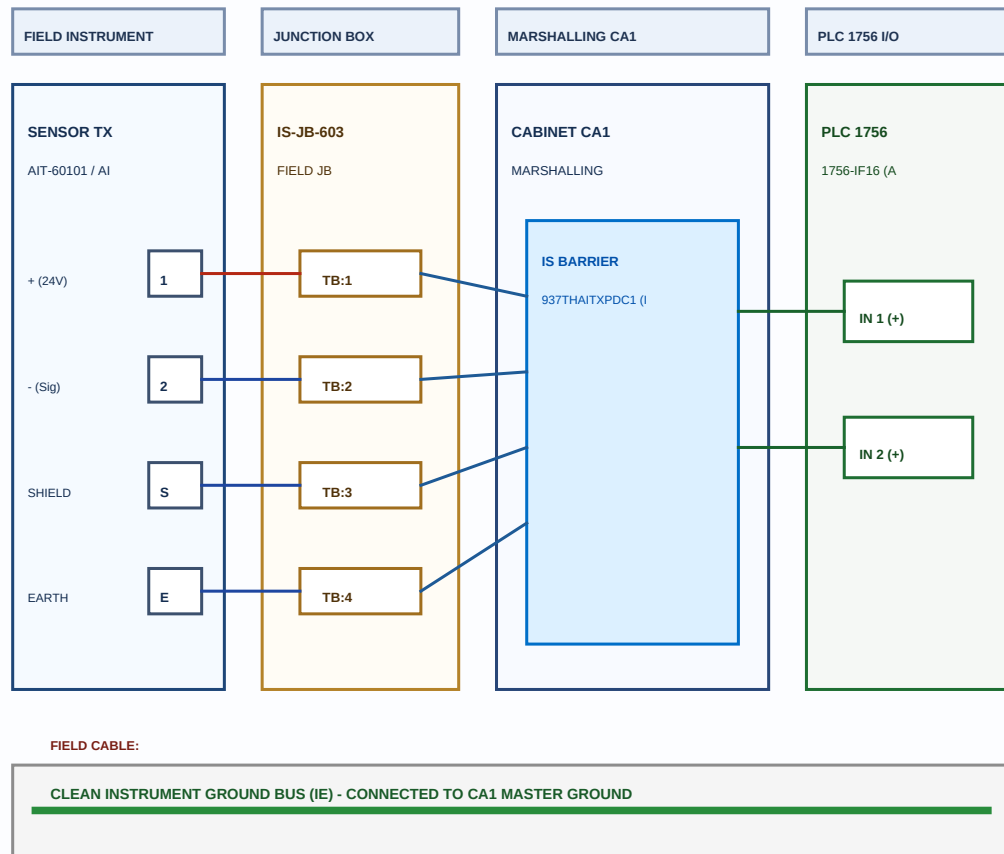
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	Ambient Air Diffusion / Spla..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66 Heavy-duty Ex..
Hazardous Class:	Class I, Div 1, Grp BCD / Ex..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 0-100% LEL (4-20mA) / DI: Gas Alarm
Power Supply:	24V DC (18-32V DC, 200mA max)
Field Cable Type:	3Cx1.5mm² Shielded Fire-Resistant (Bel..
Destination JB:	IS-JB-603 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (AI LEL) & 1756-IB32 (DI Ala..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: SENSIDYNE / SMARTGAS

MODEL: 925FGD Explosionproof Hydrocarbon Gas Detector

SERVICE: Explosion Hazard Lower Explosive Limit (LEL) Monitor

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Gas Detector Transmitter 925FG Sensidyne Infrared / Catalytic		1 EA
2	Weather Guard / Dust Protectio Hydrophobic SS316		1 EA
3	Calibration Adapter & Cup Molded PVC / Viton O-ring		1 EA
4	Explosionproof Cable Gland 3/4 SS316 Flameproof Ex d		1 EA
5	Wall / Stanchion Mounting Brac SS304 Universal Bracket		1 SET

ENGINEERING INSTALLATION NOTES

- Install 0.3m to 0.5m above floor for gases heavier than air, or near ceiling for lighter gases.
- Field sensor must undergo zero and span gas calibration prior to plant commissioning.
- Wiring must run in rigid galvanized steel conduit or approved armored flameproof cable.

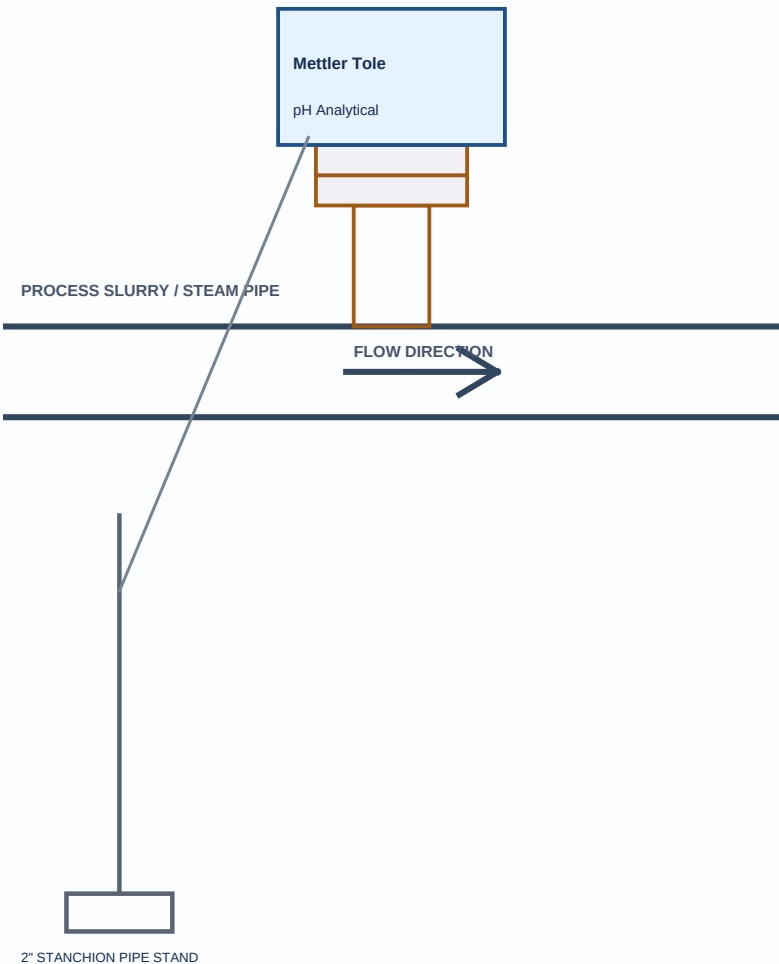
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONCOMBUSTIBLE LEL GAS DETECTOR HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-002
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
04 / 25

TEMPLATE TYP-03: PH ANALYTICAL LOOP

TYPICAL TAGS: AIT-40101 / AIT-40201

SERVICE: Cooker Slurry & Flash Tank pH Neutralization | PLC I/O: 1756-IF16 (AI pH) & 1756-IB32 (DI Alarms)

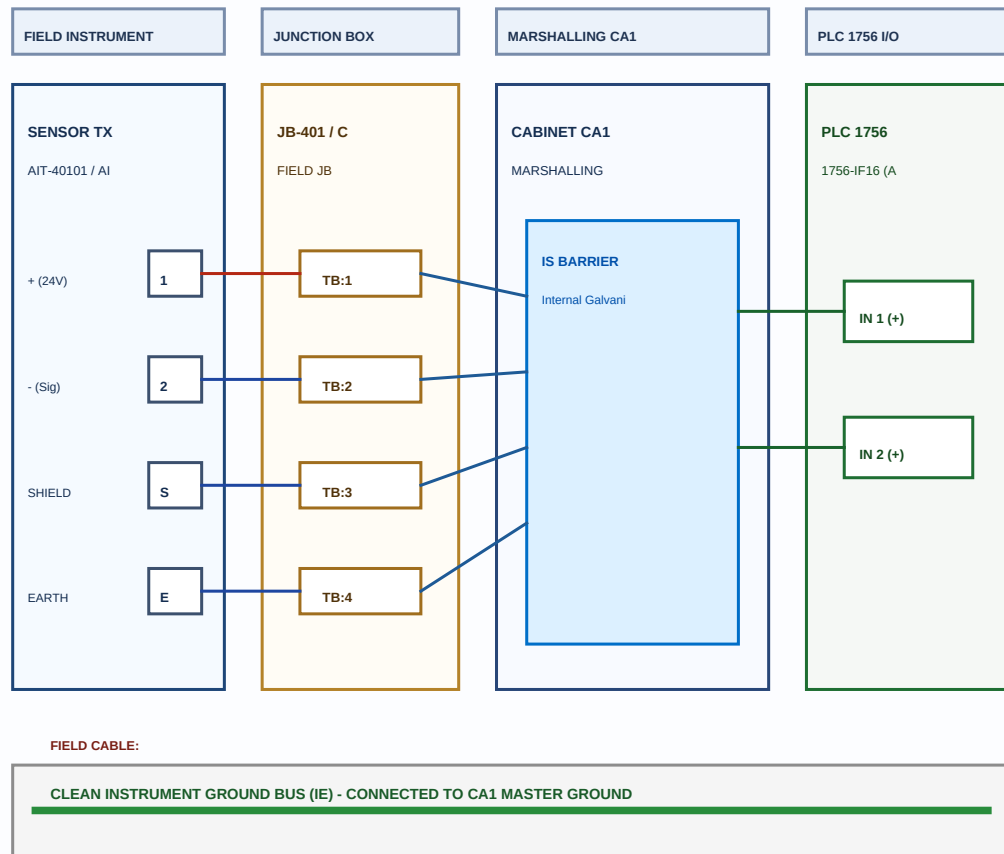
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN25 Weld-in Socket (Incline..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP65 Polycarbonate..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6/T4..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 0-14 pH (4-20mA HART) / AI: Temp
Power Supply:	24V DC / 10W (Optional 100-240V AC)
Field Cable Type:	VP6 High-Z Cable to Tx + 2Px1.5mm² Shi..
Destination JB:	JB-401 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation (4-Wire Tx)
PLC Module:	1756-IF16 (AI pH) & 1756-IB32 (DI Alar..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: METTLER TOLEDO

MODEL: M300G2 4-Wire Transmitter + InPro3250i + InTrac 777P

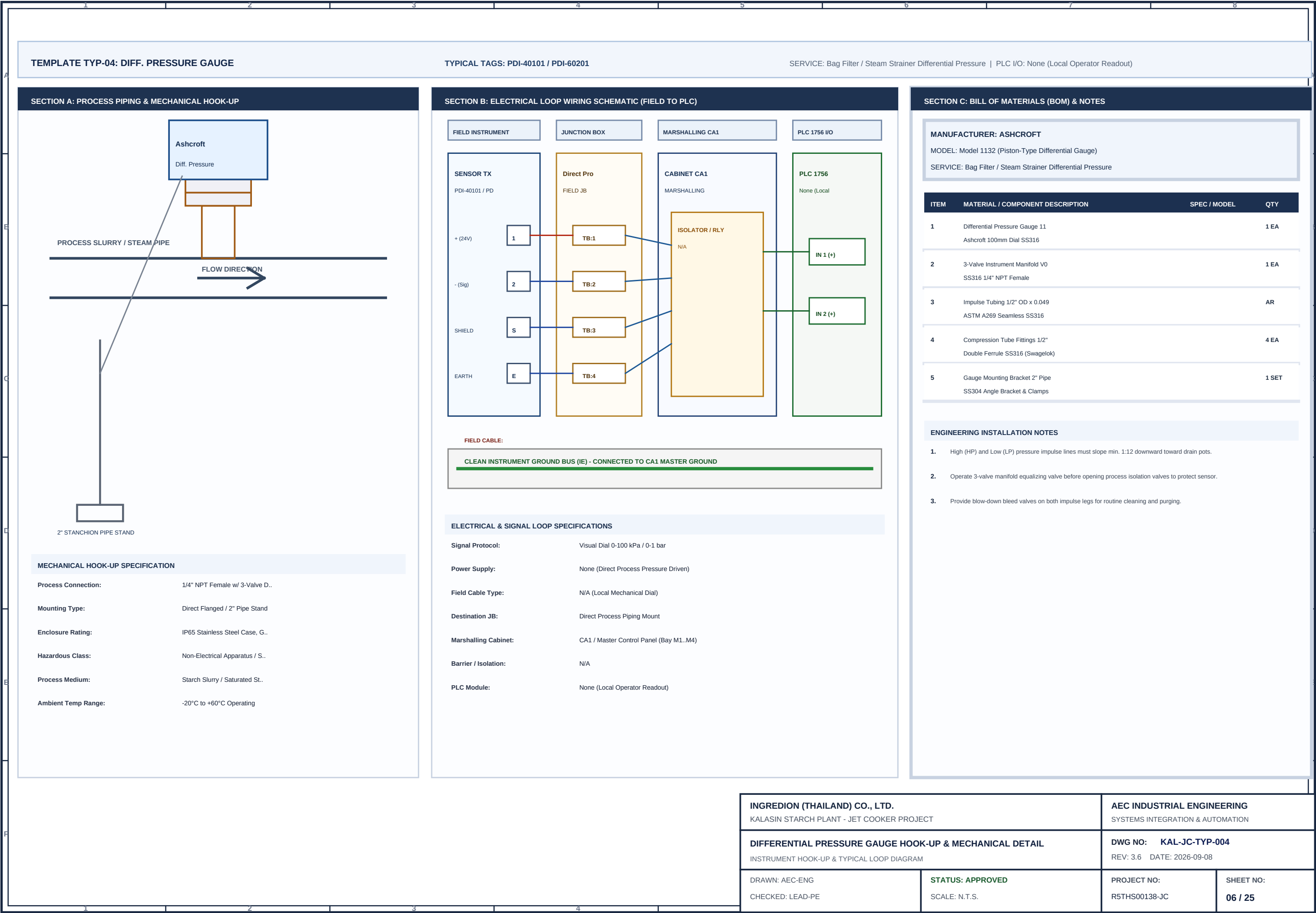
SERVICE: Cooker Slurry & Flash Tank pH Neutralization

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	pH Multi-Parameter Transmitter Mettler Toledo M300 4-Wire		1 EA
2	Intelligent pH Sensor InPro325 Pre-pressurized Liquid Gel		1 EA
3	Retractable Housing InTrac 777 SS316L / Hastelloy C w/ Flange		1 EA
4	Weld-in Socket 40mm / DN25 (15 SS316L Sloped Socket		1 EA
5	2" Pipe Mounting Kit for M300G SS304 Bracket & U-Bolts		1 SET

ENGINEERING INSTALLATION NOTES

- Weld socket must be installed inclined at min. 15° above horizontal to prevent air bubble entrapment.
- Use retractable housing safety interlock valve before extracting probe during process operation.
- Ground transmitter chassis directly to plant PE bar with min. 4mm² copper grounding strap.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTpH ANALYTICAL MEASUREMENT LOOP HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-003
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
05 / 25

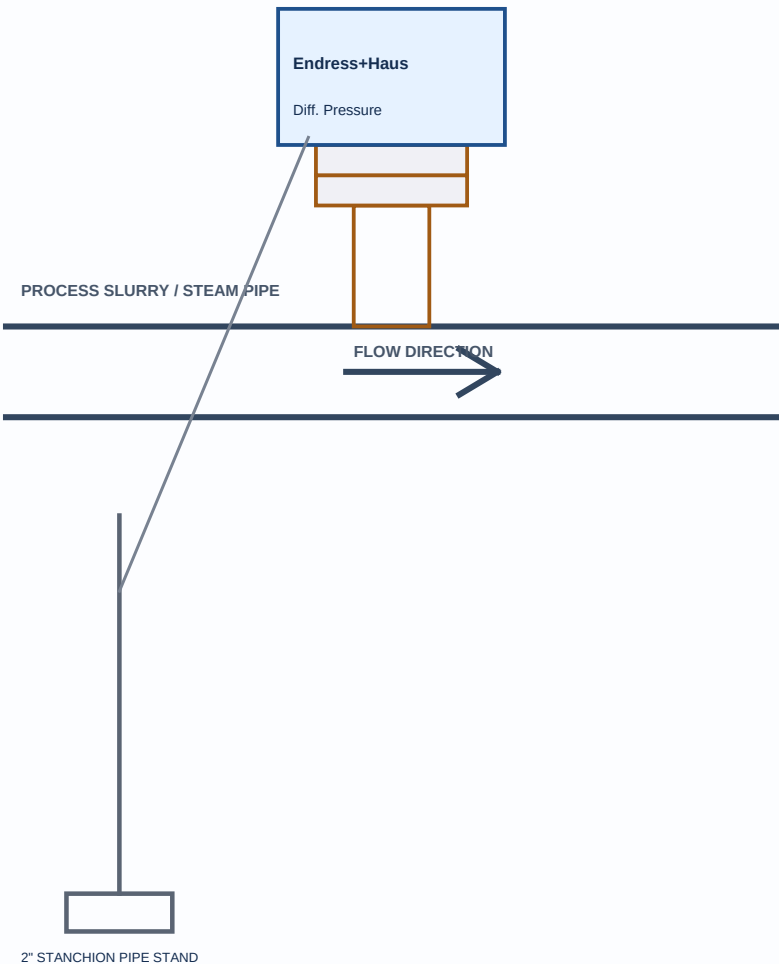


TEMPLATE TYP-05: DIFF. PRESSURE TRANSMITTER

TYPICAL TAGS: DPT-60201 / DPT-60901

SERVICE: Filter Differential Pressure / Orifice Flow Drop | PLC I/O: 1756-IF16 (16-Point Diff Analog Input)

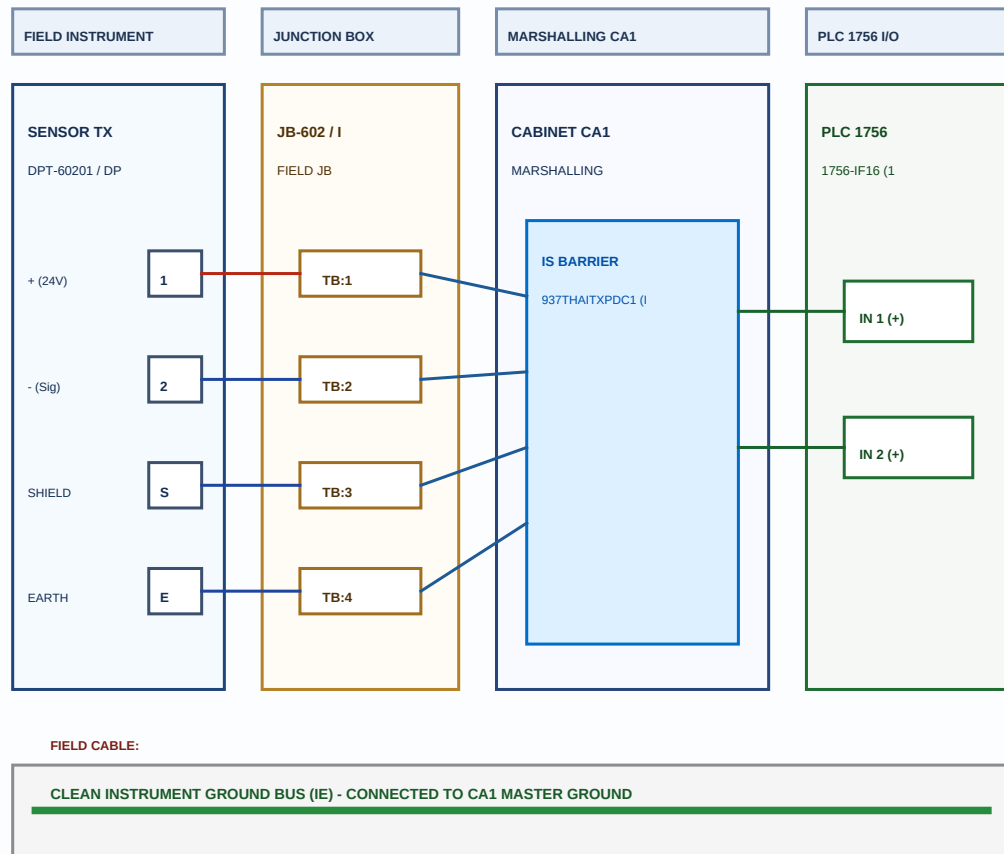
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/4" NPT Female w/ 5-Valve M..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual-Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Process Delta-P (0-500 mbar)
Power Supply:	24V DC via 1756-IF16 Analog Input Loop
Field Cable Type:	1Px1.5mm ² Twisted Shielded (Belden 876..
Destination JB:	JB-602 / IS-JB-603
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Diff Analog Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Deltabar PMD55B (Piezoresistive / Ceramic)

SERVICE: Filter Differential Pressure / Orifice Flow Drop

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Deltabar PMD55B Transmitter E+H Ceramic / SS316 Diaphragm		1 EA
2	5-Valve Manifold DA63M (Flange SS316 Integrated Bleed Valves		1 EA
3	2" Pipe Mounting Bracket Kit SS304 Stanchion Bracket & U-Bolt		1 SET
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-Plated Brass Armored Glan		1 EA
5	Impulse Tubing 1/2" OD x 0.049 ASTM A269 Seamless SS316		AR

ENGINEERING INSTALLATION NOTES

- Mount transmitter below process tapping points for liquid service to maintain filled impulse lines.
- 5-valve manifold sequence: Open equalizing, open HP & LP blocks, close equalizing, check zero.
- Connect 2-wire loop to PLC 1756-IF16 terminals: (+) to 24V loop power, (-) to Channel Input.

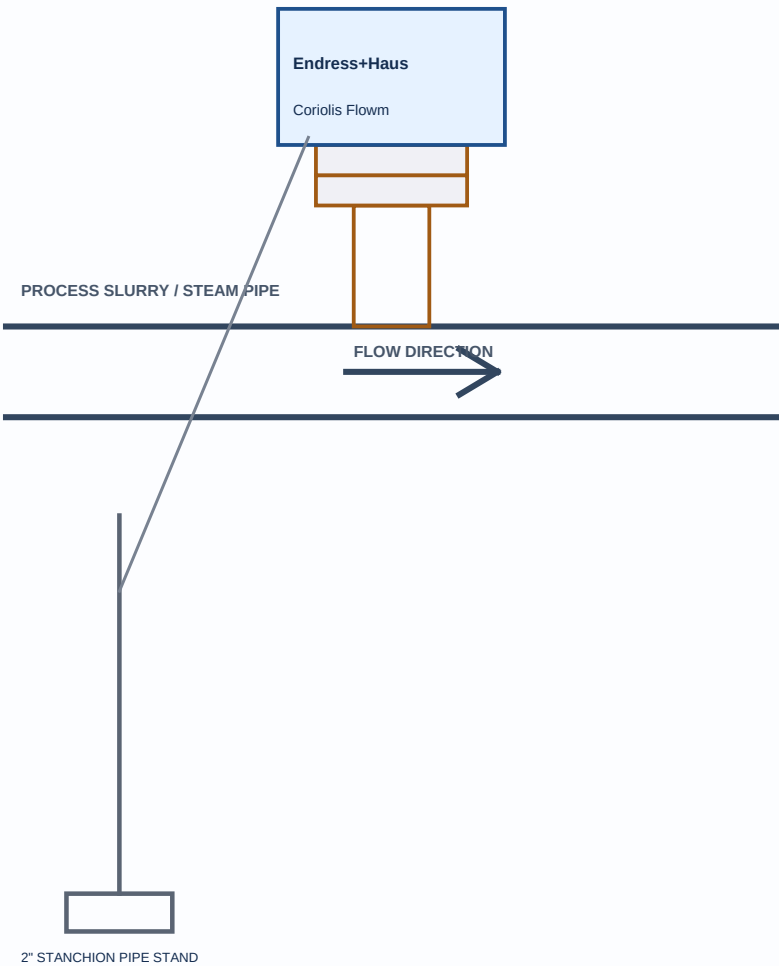
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDIFFERENTIAL PRESSURE TRANSMITTER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-005
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
07 / 25

TEMPLATE TYP-06: CORIOLIS FLOWMETER

TYPICAL TAGS: FT-40201 Series

SERVICE: Starch Slurry High-Accuracy Mass Flow & Density | PLC I/O: 1756-IF16 (Ch. 1 Mass Flow / Ch. 2 Density)

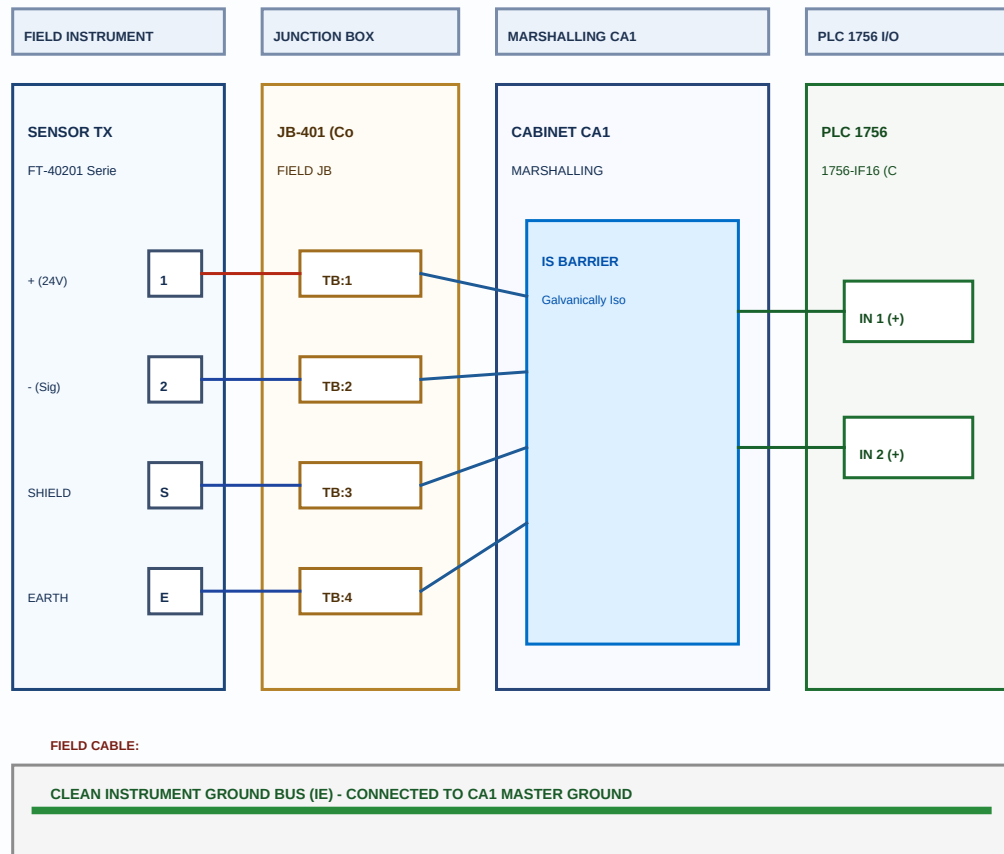
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN50 (2") ASME B16.5 Cl.150 ..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP67 Die-cast Aluminum Trans..
Hazardous Class:	ATEX II 2G Ex db eb ia IIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Mass (kg/h) & Density (g/c..
Power Supply:	24V DC / 10W (Transmitter Electronics)
Field Cable Type:	2Px1.5mm² Shielded Overall (Belden 877..
Destination JB:	JB-401 (Coker Section 1)
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Galvanically Isolated Active 4-20mA Ou..
PLC Module:	1756-IF16 (Ch. 1 Mass Flow / Ch. 2 Den..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Promass E 300 (8E3B50 / Dual Tube SS316L)

SERVICE: Starch Slurry High-Accuracy Mass Flow & Density

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Promass E 300 Coriolis Flowmet E+H DN50 Cl.150 SS316L Dual Tube		1 EA
2	Companion Counter-Flanges DN50 ASTM A182 F316 / RF Flanges		2 EA
3	PTFE / Spiral Wound Gaskets 316SS with PTFE Filler		2 EA
4	Cable Glands M20x1.5 Ex d/e Nickel-plated Brass Armored Glan		2 EA
5	Pipe Support Clamps & Structur Heavy Duty Rigidity Anchors		2 SET

ENGINEERING INSTALLATION NOTES

- Sensor tubes must be installed pointing downwards (flag-down) for liquid slurry to prevent gas collection.
- Support pipeline rigidly on both upstream and downstream sides within 2 pipe diameters of meter flanges.
- Do not clamp or attach pipe supports directly to the Coriolis sensor housing.

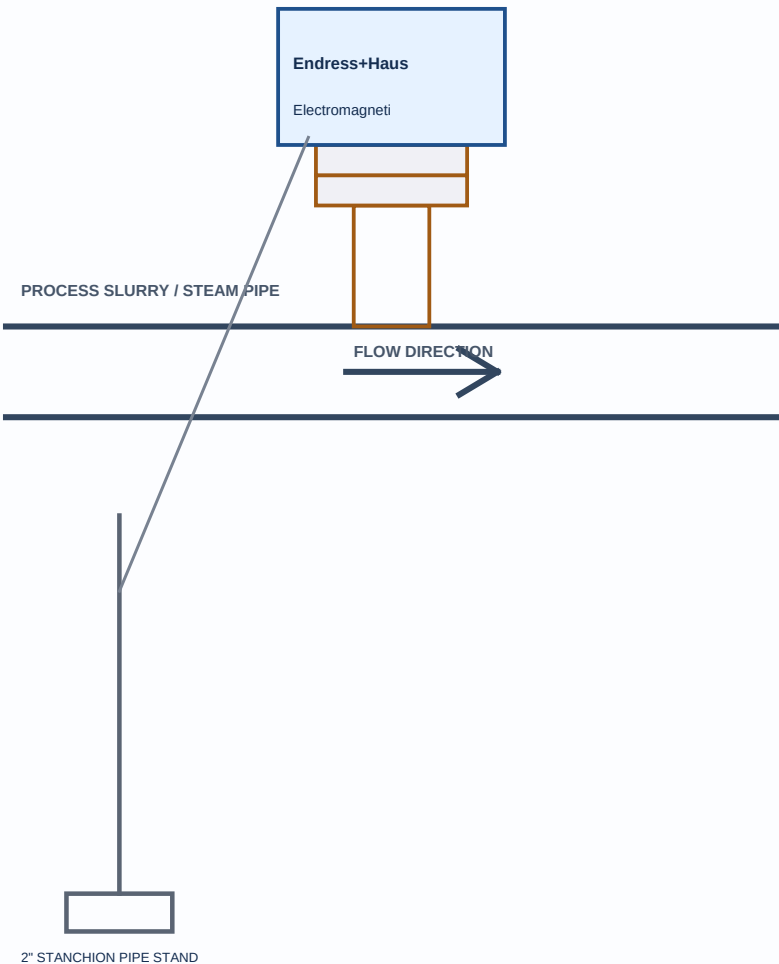
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONCORIOLIS MASS FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-006
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
08 / 25

TEMPLATE TYP-07: ELECTROMAGNETIC FLOWMETER

TYPICAL TAGS: FT-40202 / FT-60201

SERVICE: Slurry & Process Water Volumetric Flow Measurement | PLC I/O: 1756-IF16 (4-20mA Flow) & 1756-IB32 (Pulse Counter)

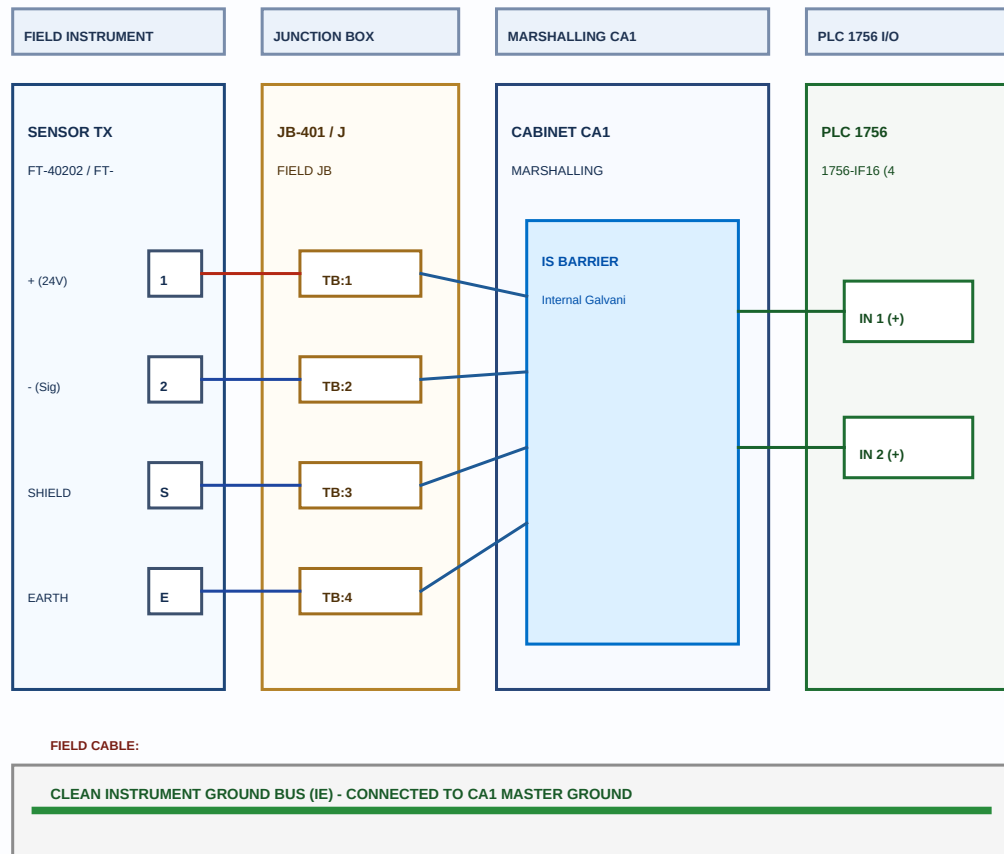
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN50/80/100 ASME CL150 RF F..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP67 Aluminum Tran..
Hazardous Class:	ATEX II 2G Ex eb Iia IIC T6....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-150 m³/h) + Scaled Pulses
Power Supply:	24V DC / 8W (4-Wire Electronics)
Field Cable Type:	1Px1.5mm² (Analog) + 1Px1.5mm² (Pulse/..
Destination JB:	JB-401 / JB-601
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation
PLC Module:	1756-IF16 (4-20mA Flow) & 1756-IB32 (P..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Promag P 300 (5P3B50 / 80 / 1H PTFE Liner)

SERVICE: Slurry & Process Water Volumetric Flow Measurement

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Promag P 300 Magmeter Sensor & E+H PTFE Lined, Hastelloy C Elec		1 EA
2	Grounding Rings / Discs SS316L E+H Grounding Electrodes Kit		2 EA
3	PTFE Envelope Flange Gaskets Chemical Resistant Gaskets		2 EA
4	Cable Glands M20x1.5 Ex d/e Nickel-plated Brass Glands		2 EA
5	Grounding Cables 6mm² Green/Ye Copper Earthing Strap with Lugs		2 EA

ENGINEERING INSTALLATION NOTES

- Straight piping requirements: min. 5D upstream and 2D downstream of flowmeter.
- Install grounding rings between meter flanges and pipe counter-flanges to ensure reference ground.
- Pipe must always remain 100% full of conductive liquid for accurate flow measurement.

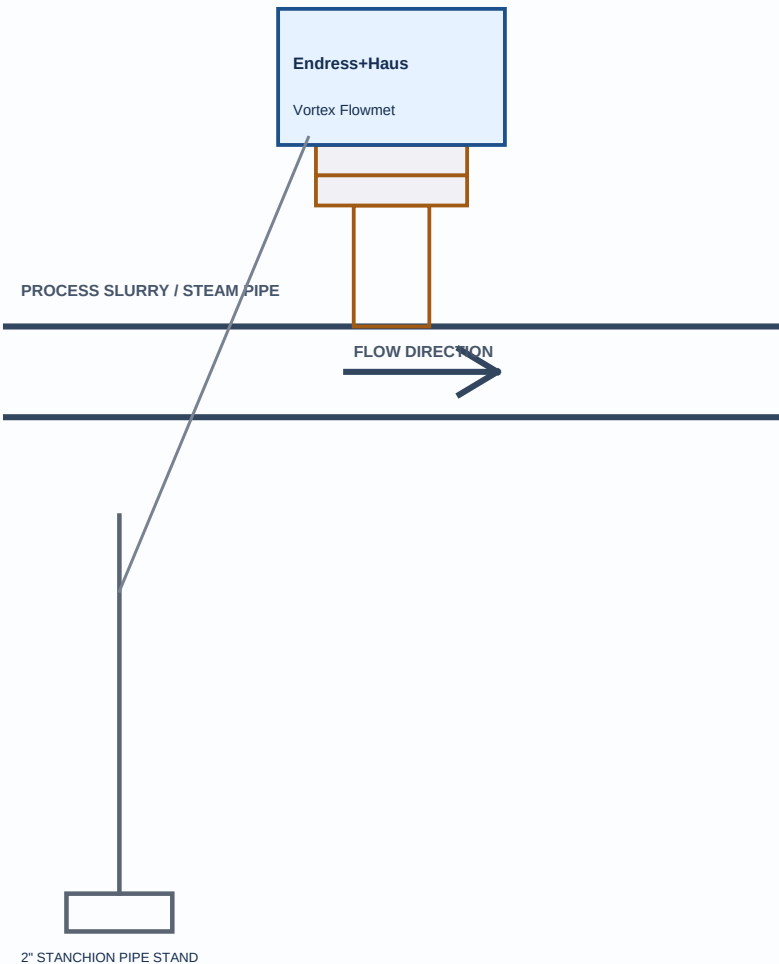
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTROMAGNETIC FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-007
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
09 / 25

TEMPLATE TYP-08: VORTEX FLOWMETER

TYPICAL TAGS: FT-60101 / FT-60202

SERVICE: High-Pressure Saturated Steam Flow Measurement | PLC I/O: 1756-IF16 (Compensated Mass Flow Steam)

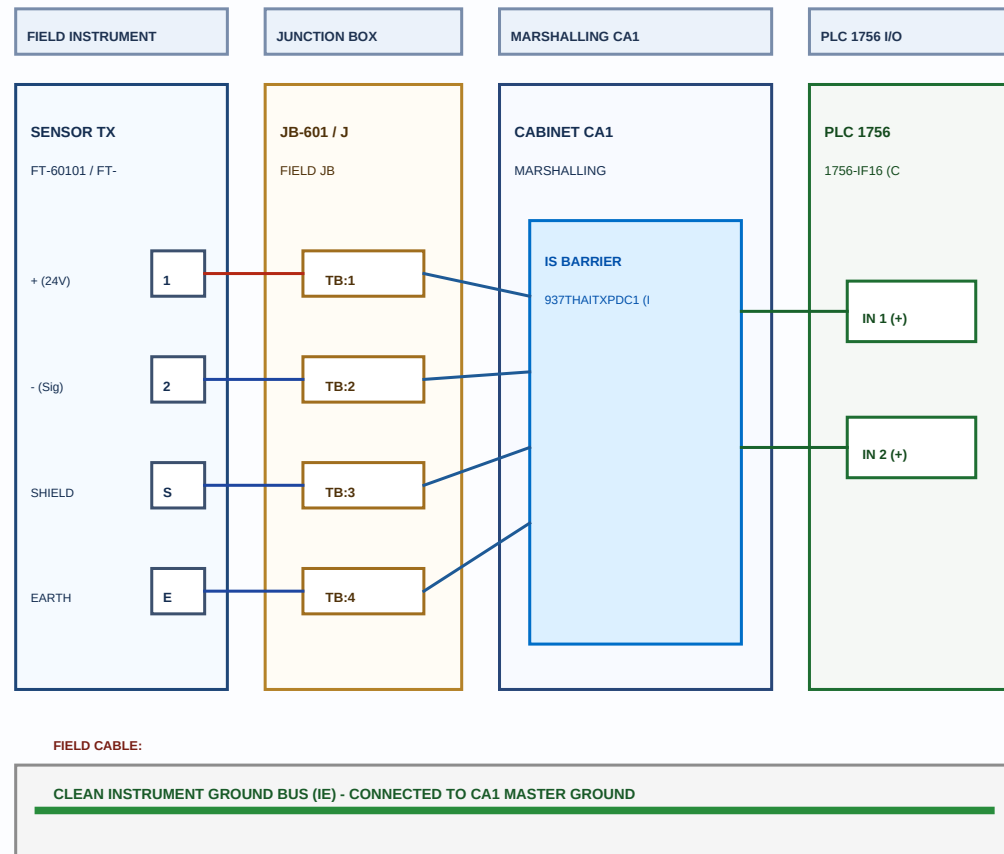
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN80/100 (3"/4") ASME Cl.300..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	Dual-Compartment IP67 Die-ca..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Steam Flow (0-10,000 kg/h)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Shielded (High Temp ..
Destination JB:	JB-601 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Loop Powered Repeater)
PLC Module:	1756-IF16 (Compensated Mass Flow Steam)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Prowirl F 200 (7F2C80 / 1H with Temp Sensor)

SERVICE: High-Pressure Saturated Steam Flow Measurement

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Prowirl F 200 Flanged Vortex M E+H SS316 Cast Body w/ DSC Senso		1 EA
2	Counter-Flanges DN80 Cl.300 RF ASTM A105 / A182 F316 Flanges		2 EA
3	Spiral Wound Gaskets Cl.300 Gr Graphite Filled SS316 Rings		2 EA
4	Cable Gland M20x1.5 Ex d/e Hig Brass Nickel-Plated Gland		1 EA
5	Pipeline Insulation Spacer & S Thermal Extension Neck Jacket		1 SET

ENGINEERING INSTALLATION NOTES

- Straight piping: min. 15D upstream and 5D downstream from flow disturbers (valves, reducers).
- Insulate pipe line up to meter neck, but leave transmitter electronics housing uninsulated.
- Verify that internal integrated Pt100 temperature compensation is activated in HART configuration.

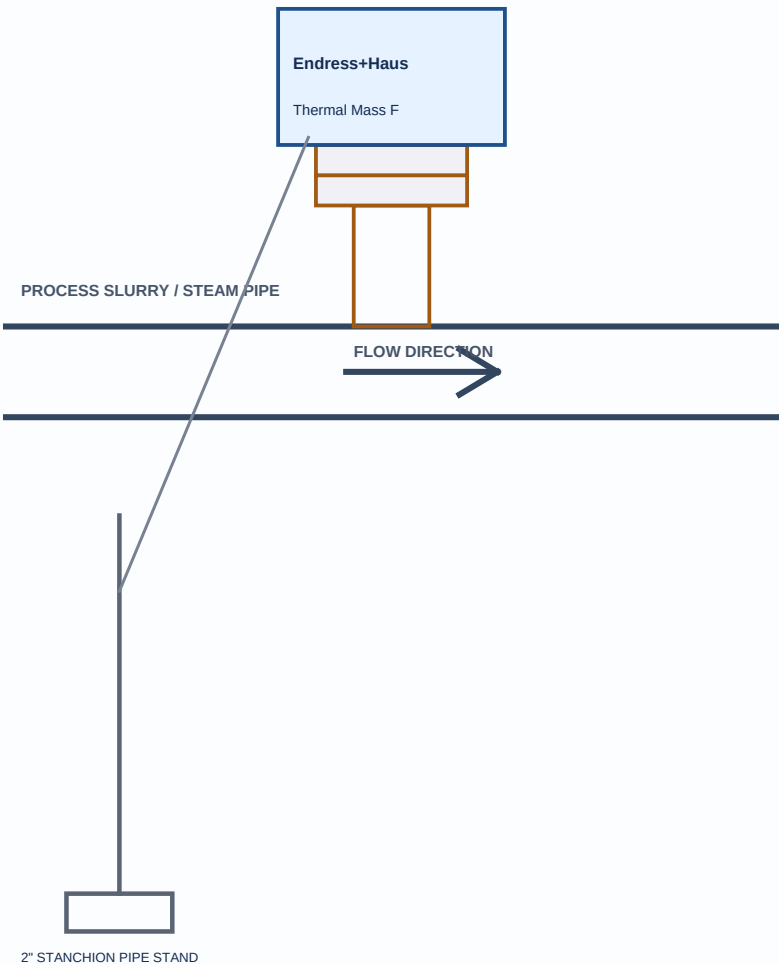
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONVORTEX STEAM FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-008
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
10 / 25

TEMPLATE TYP-09: THERMAL MASS FLOWMETER

TYPICAL TAGS: FT-80301 Series

SERVICE: LPG / Natural Gas Boiler Fuel Consumption | PLC I/O: 1756-IF16 (AI Gas Flow) & 1756-IB32 (DI Alarm)

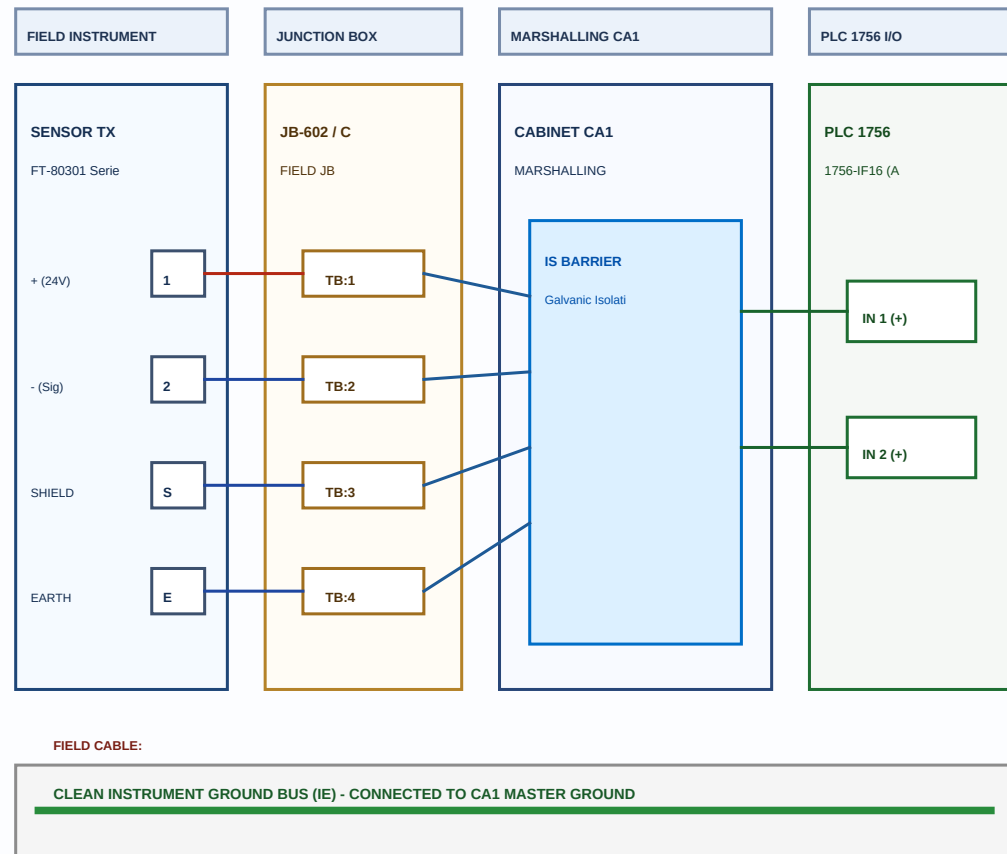
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN100 Flange / 1" NPT Retrac..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP67 Aluminum Encl..
Hazardous Class:	ATEX II 2G Ex db eb ia IIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Standard Nm³/h Flow
Power Supply:	24V DC / 12W (4-Wire Electronics)
Field Cable Type:	2Px1.5mm² Shielded Armored (Belden 8777)
Destination JB:	JB-602 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Galvanic Isolation / Ex d Installation
PLC Module:	1756-IF16 (AI Gas Flow) & 1756-IB32 (D..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: t-mass F 300 / I 300 (6F3B / 6I3B Insertion)

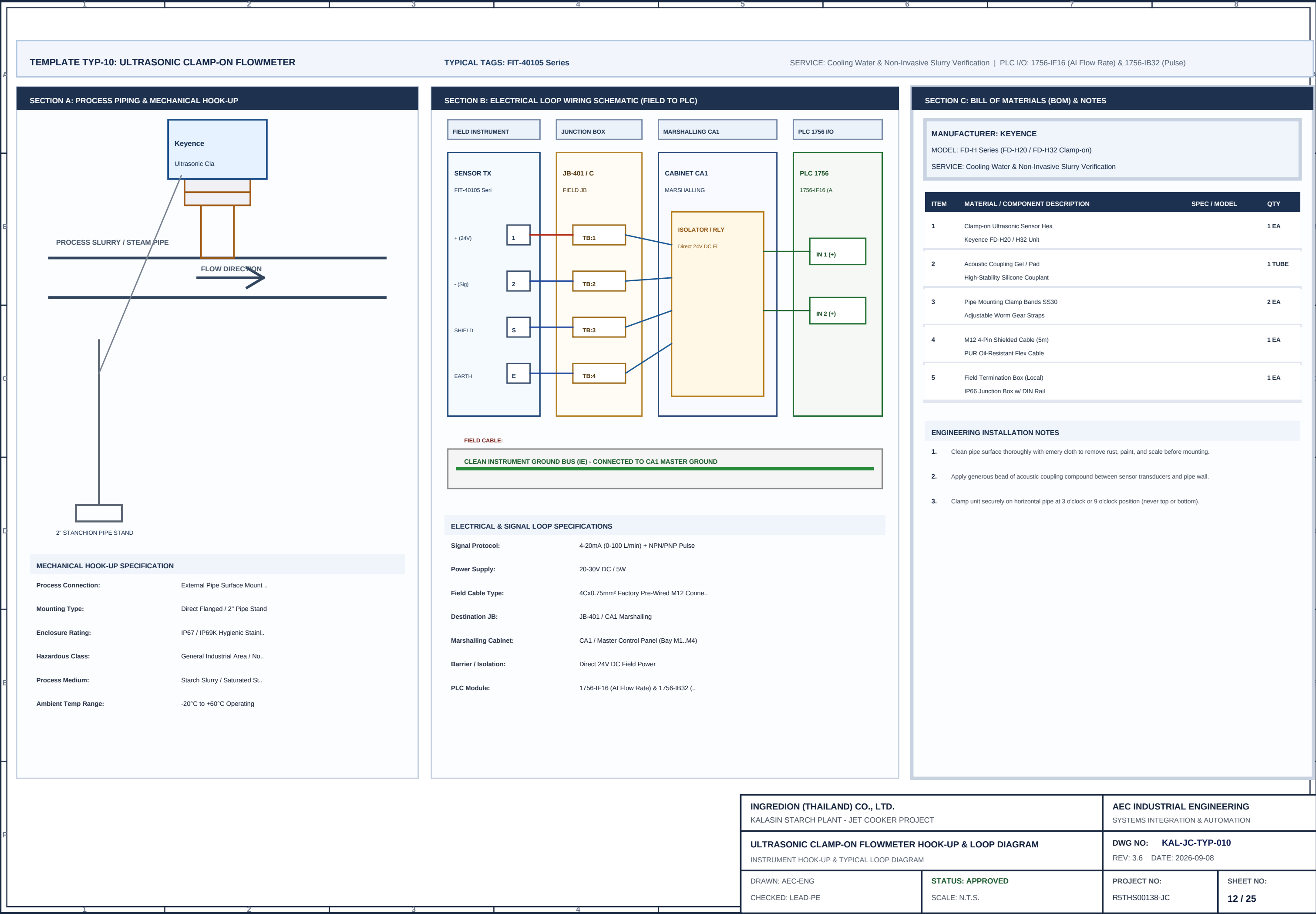
SERVICE: LPG / Natural Gas Boiler Fuel Consumption

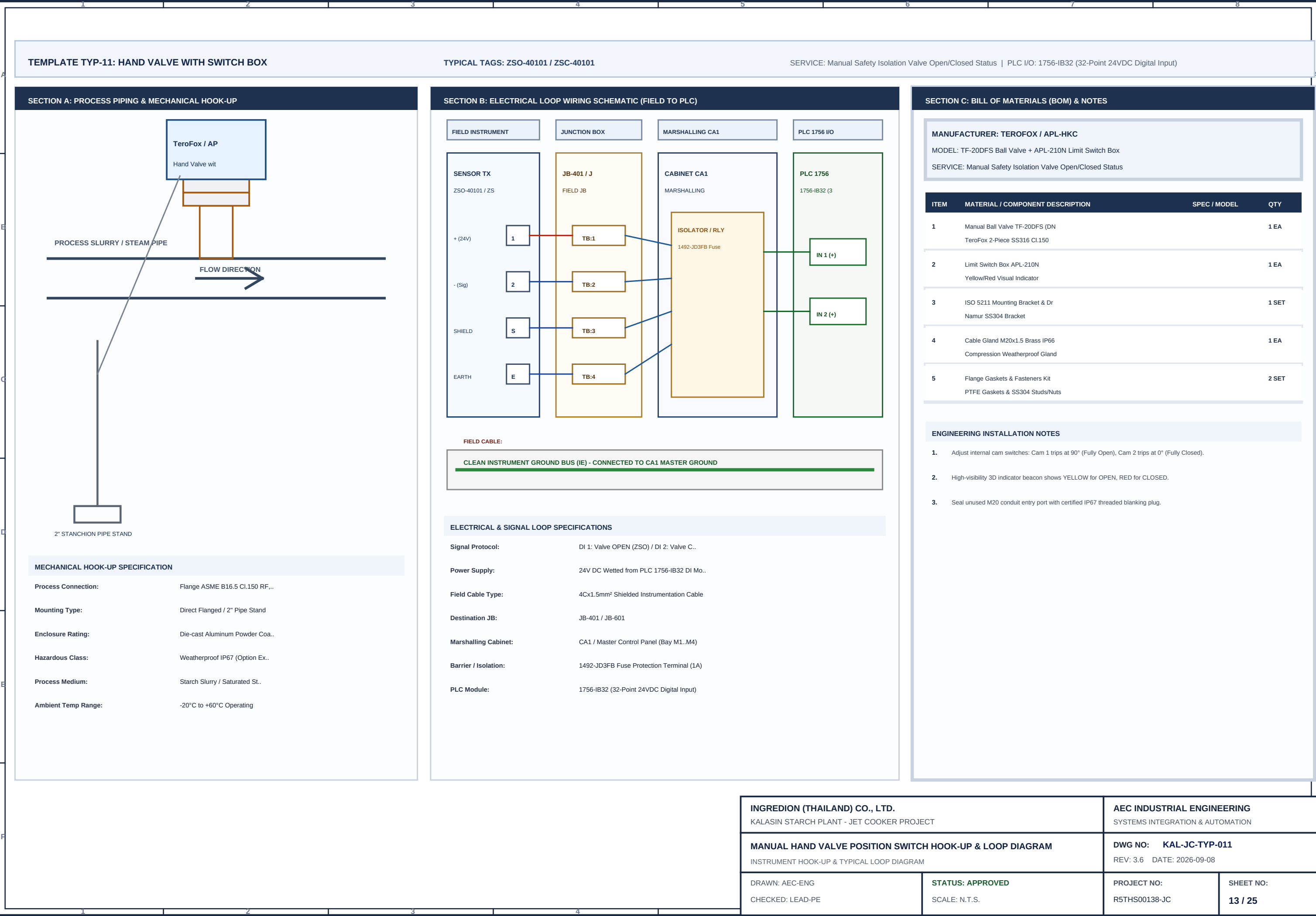
ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	t-mass 300 Thermal Flowmeter T E+H Insertion Probe w/ Hot-Tap		1 EA
2	Weld-in Boss 1" NPT Female Thr ASTM A105 / Carbon Steel Boss		1 EA
3	Isolation Ball Valve 1" Full B SS316 Cl.300 Gas Ball Valve		1 EA
4	Cable Gland M20x1.5 Ex d Flame Brass Gland for Armored Cable		2 EA
5	Grounding Cable 6mm² w/ Lug Copper Earthing Conductor		1 EA

ENGINEERING INSTALLATION NOTES

- Install probe at centerline of pipe (0.5 x Inside Diameter) matching flow direction arrow exactly.
- Verify straight pipe run: min. 20D upstream and 5D downstream of insertion tap.
- Equipotential bonding must be connected between meter body and gas piping system.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTTHERMAL MASS GAS FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-009
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
11 / 25



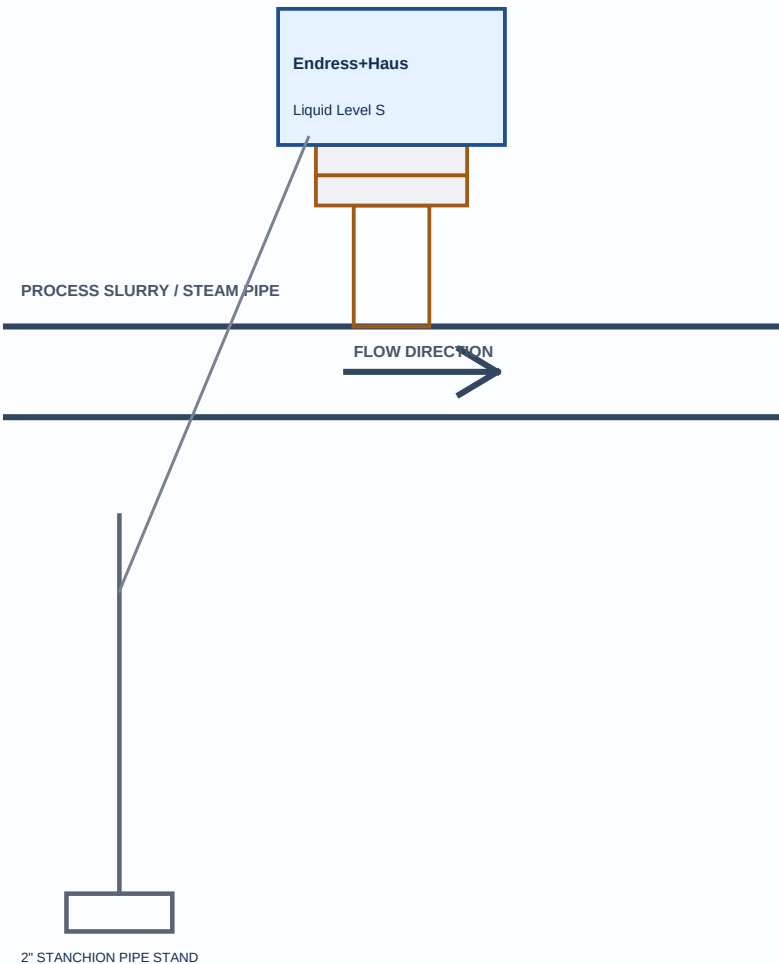


TEMPLATE TYP-12: LIQUID LEVEL SWITCH

TYPICAL TAGS: LSH-40101 / LSL-40101

SERVICE: High / Low Slurry Level Interlock & Pump Protection | PLC I/O: 1756-IB32 (24V DC Sinking/Sourcing Digital Input)

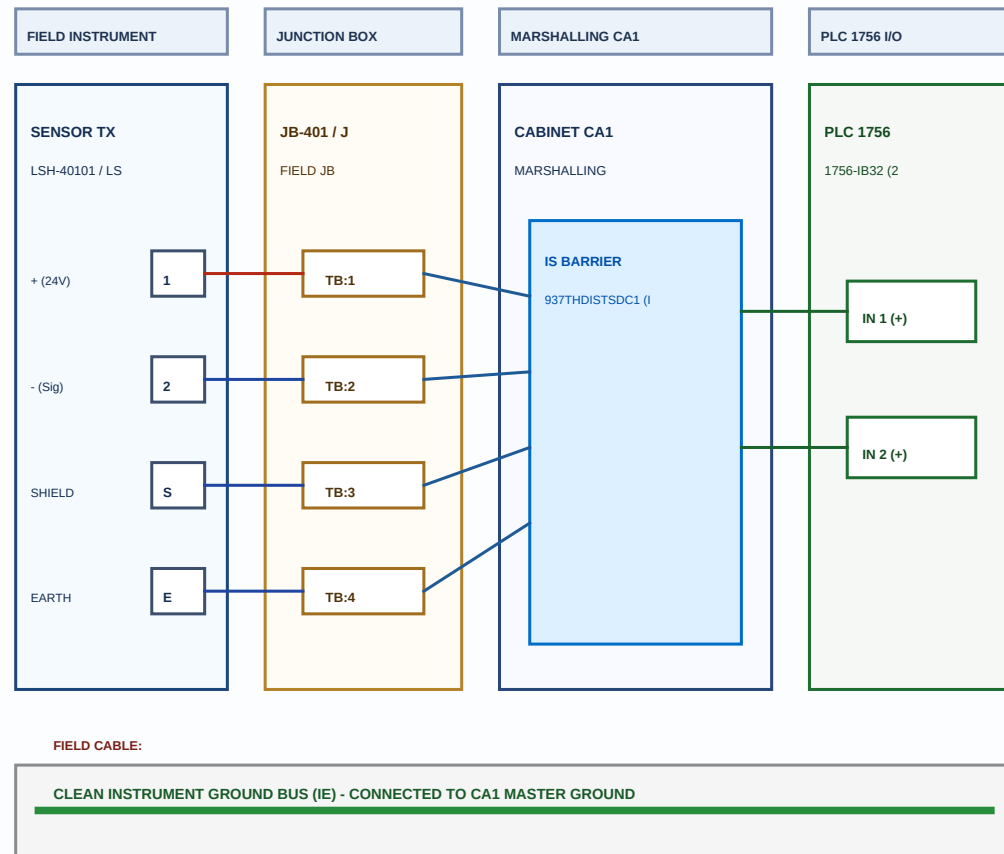
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	3/4" NPT Male Threaded / Tri..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Stainless ..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6.....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Level Alarm (Dry / Submerged ..
Power Supply:	24V DC (10-55V DC, PNP Mode)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THDISTSDC1 (IS Digital Input Barrier)
PLC Module:	1756-IB32 (24V DC Sinking/Sourcing Dig..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER
MODEL: Liquiphant FTL51B (Tuning Fork SS316L)
SERVICE: High / Low Slurry Level Interlock & Pump Protection

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Liquiphant FTL51B Level Switch E+H Tuning Fork SS316L (100mm)		1 EA
2	Welding Boss 3/4" NPT Half Cou SS316 Tank Welding Socket		1 EA
3	Cable Gland M20x1.5 Ex d/e Nickel-plated Brass Armored Glan		1 EA
4	Viton / EPDM Thread Seal Ring Process Sealing Gasket		1 EA
5	Grounding Strap 4mm² with Lugs Copper Flexible Earth Braid		1 EA

ENGINEERING INSTALLATION NOTES

- Install fork with tines aligned vertically to prevent slurry buildup on top of vibrating blades.
- Set fail-safe mode switch: MAX (Overfill Protection) or MIN (Dry Run Pump Protection).
- Keep fork at least 50mm away from tank walls, internal baffles, and agitator blades.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECT

LIQUID LEVEL SWITCH (VIBRATING FORK) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM

DRAWN: AEC-ENG
CHECKED: LEAD-PE

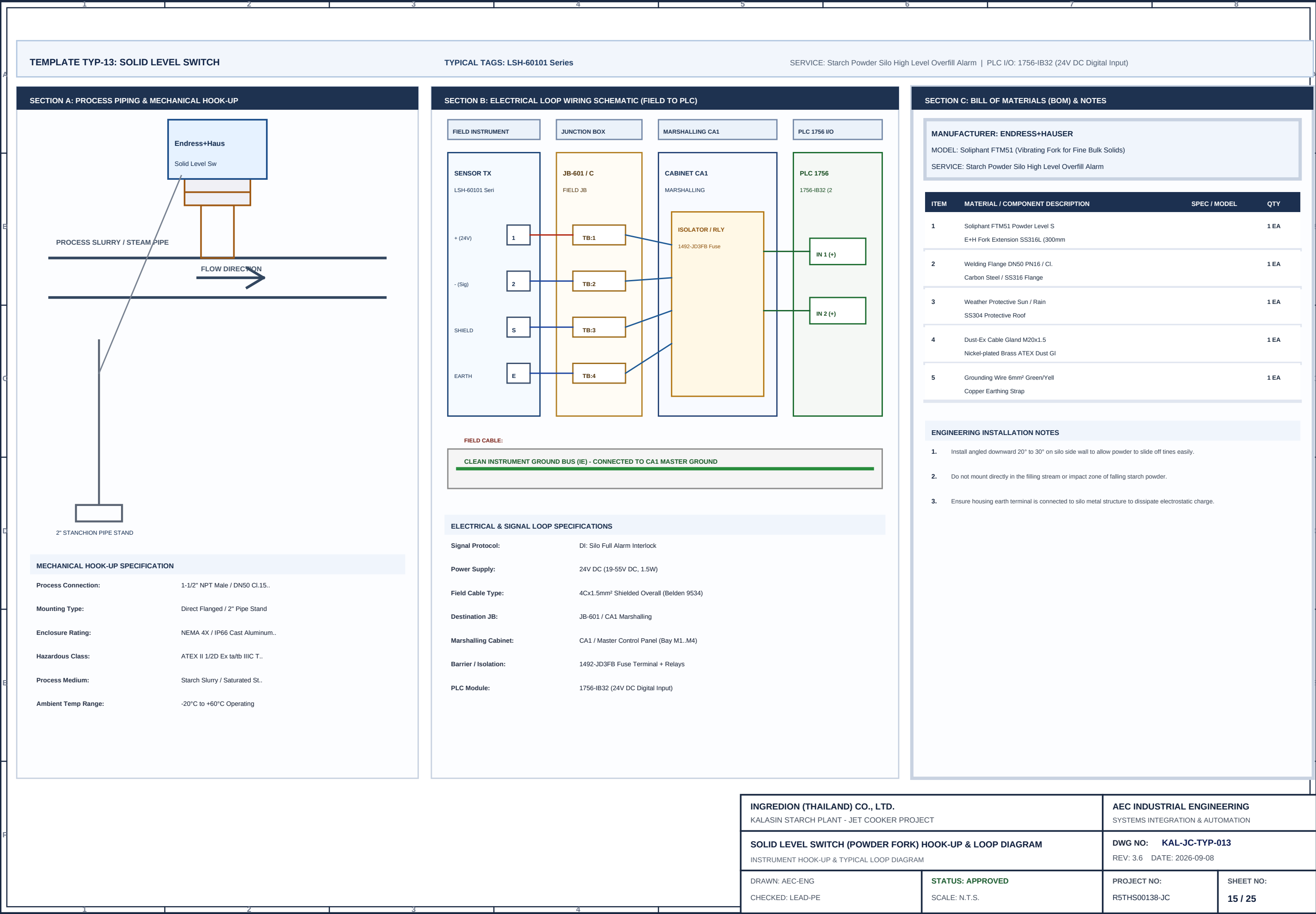
STATUS: APPROVED
SCALE: N.T.S.

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

DWG NO: KAL-JC-TYP-012
REV: 3.6 DATE: 2026-09-08

PROJECT NO:
R5THS00138-JC

SHEET NO:
14 / 25

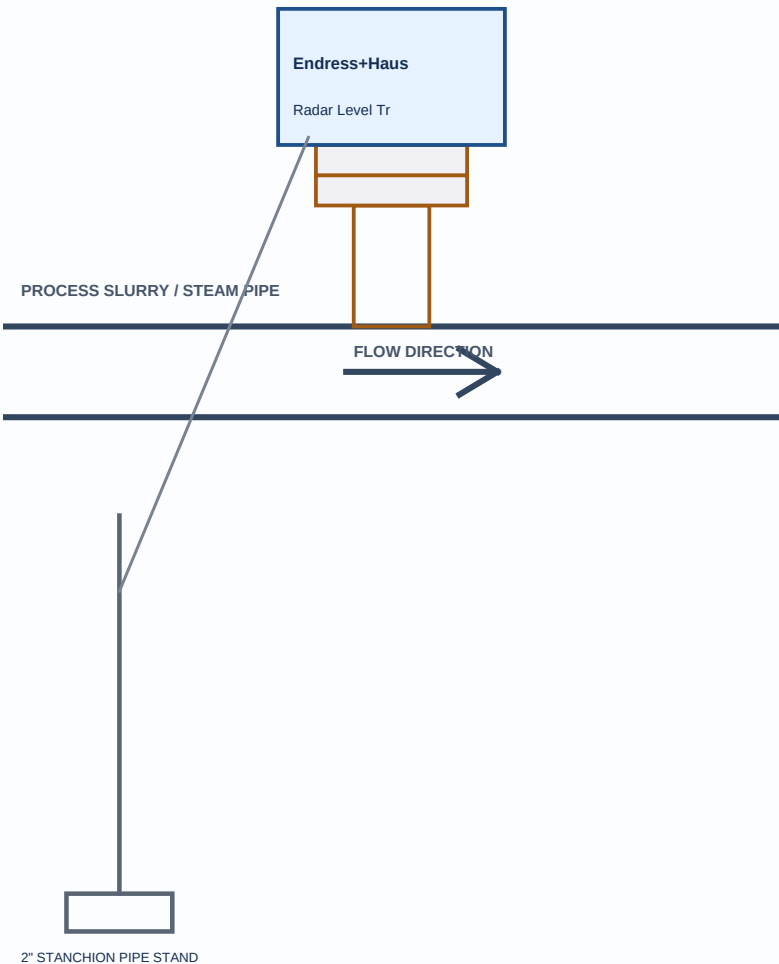


TEMPLATE TYP-14: RADAR LEVEL TRANSMITTER

TYPICAL TAGS: LT-40101 / LT-40201

SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level | PLC I/O: 1756-IF16 (16-Point Differential AI)

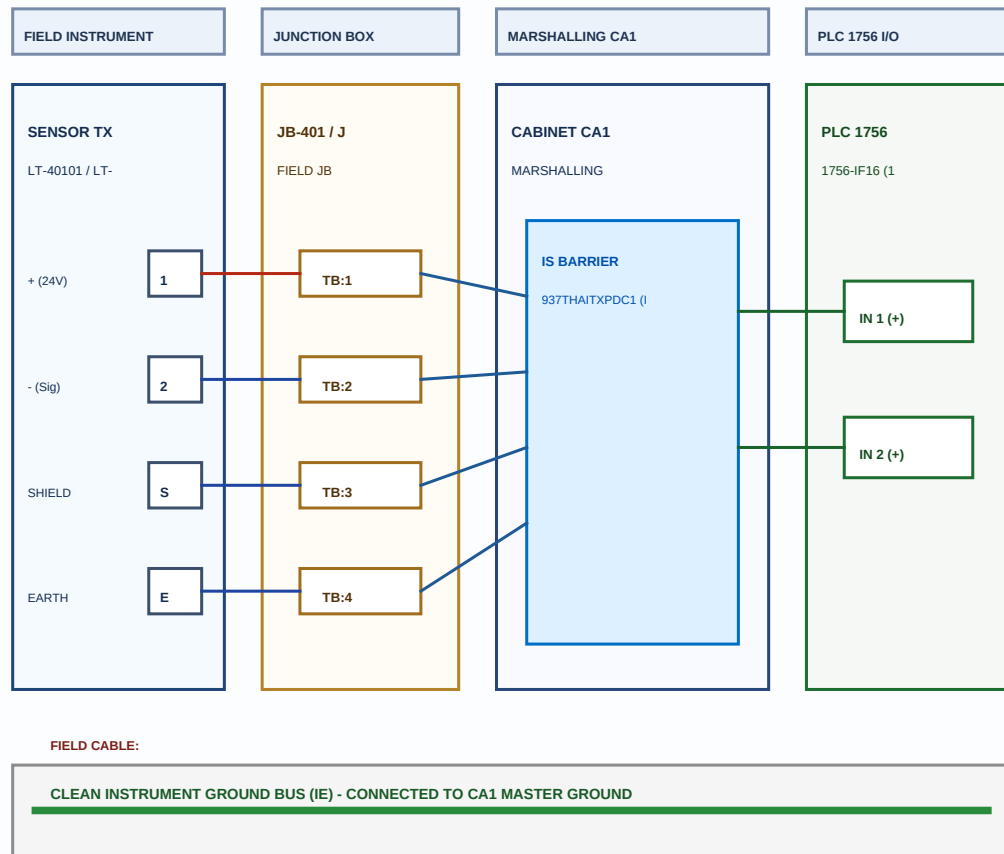
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN80 (3") ASME Cl.150 RF Fla..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Level (0-100% / 0-6.0m)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Pair Shielded (Belde..
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Micropilot FMR67B (80 GHz High-Frequency Radar)

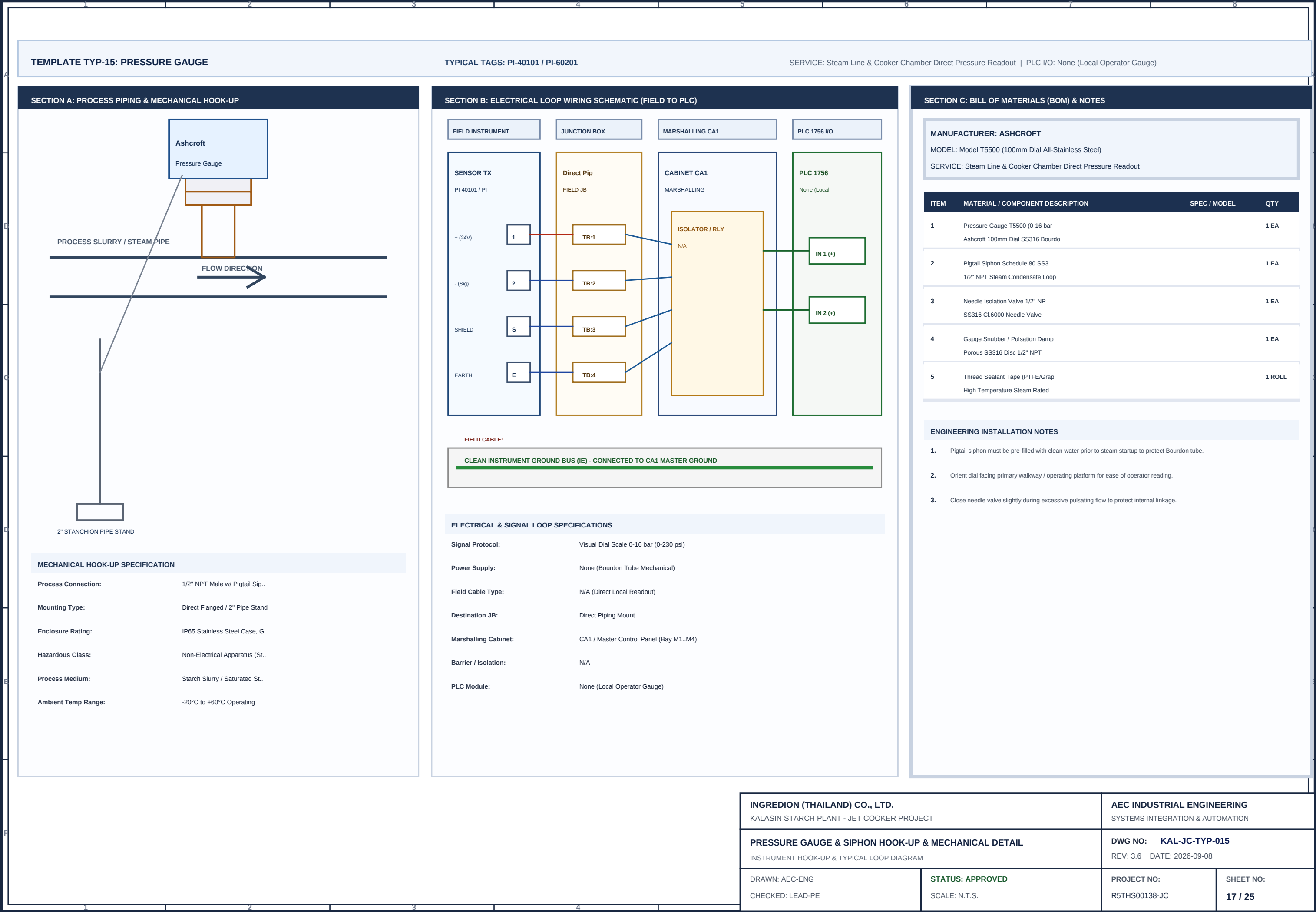
SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Micropilot FMR67B Radar Transm E+H 80GHz w/ Drip-off PTFE Anten		1 EA
2	Tank Nozzle Flange DN80 Cl.150 SS316 Tank Flange Neck		1 EA
3	PTFE Envelope Gasket DN80 Cl.1 Chemical Resistant Gasket		1 EA
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Gland		1 EA
5	Air Purge Connection Kit 1/4" Purge Connection for Condensate		1 SET

ENGINEERING INSTALLATION NOTES

- Mount nozzle height must not exceed antenna length so horn protrudes clear into vessel interior.
- Distance to vessel wall should be min. 1/6 of vessel diameter to prevent false sidewall echoes.
- Activate Heartbeat Technology and mapping curves to mask agitator blade echoes during commissioning.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONRADAR LEVEL TRANSMITTER (80 GHz) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-014
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
16 / 25

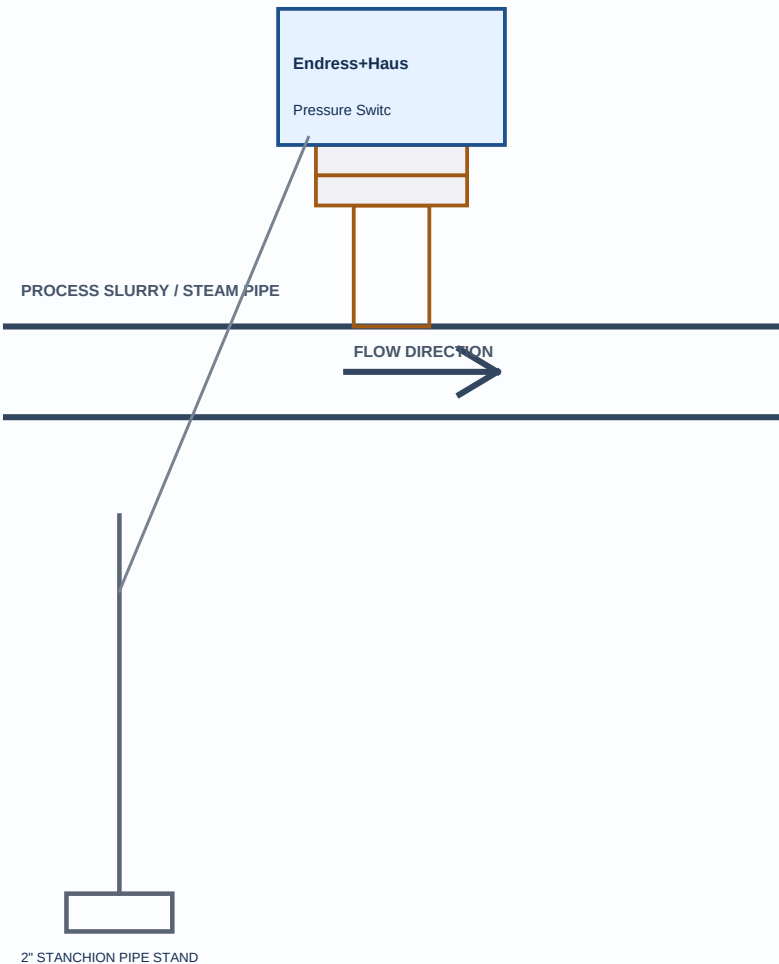


TEMPLATE TYP-16: PRESSURE SWITCH

TYPICAL TAGS: PSH-60201 / PSL-60201

SERVICE: High / Low Pressure Safety Trip Interlock | PLC I/O: 1756-IB32 (24V DC Digital Input)

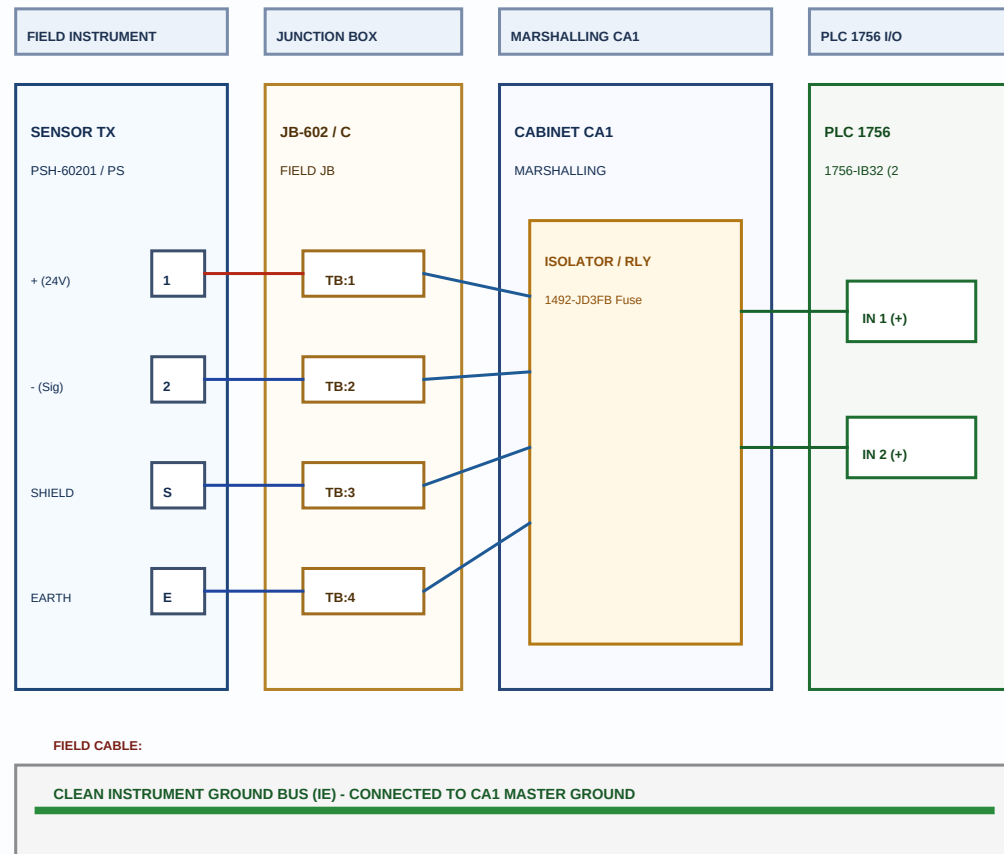
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Threaded / Tri..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 / IP67 Stainless Steel ..
Hazardous Class:	General Industrial Zone (Opt..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: Trip Contact (Open on Alarm)
Power Supply:	24V DC (10-30V DC, 20mA)
Field Cable Type:	3Cx1.5mm² Shielded Instrumentation Cable
Destination JB:	JB-602 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Protection Terminal
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Ceraphant PTC31B (Ceramic Sensor Diaphragm)

SERVICE: High / Low Pressure Safety Trip Interlock

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Ceraphant PTC31B Pressure Swit E+H Ceramic Cell, LED Display		1 EA
2	2-Valve Isolation Manifold 1/2 SS316 Block & Bleed Valve		1 EA
3	M12 4-Pin Field Wireable Conne IP67 Female M12 Connector		1 EA
4	Cable Gland M16x1.5 Brass Nickel-plated Cable Gland		1 EA
5	Impulse Adapter 1/2" NPT x 1/2 Swagelok SS316 Tube Fitting		1 EA

ENGINEERING INSTALLATION NOTES

- Program switch point and reset hysteresis via front push-buttons and LED display.
- Configure output as Normally Closed (NC) fail-safe: circuit de-energizes on overpressure or power loss.
- Provide block-and-bleed valve for routine verification with field calibrator.

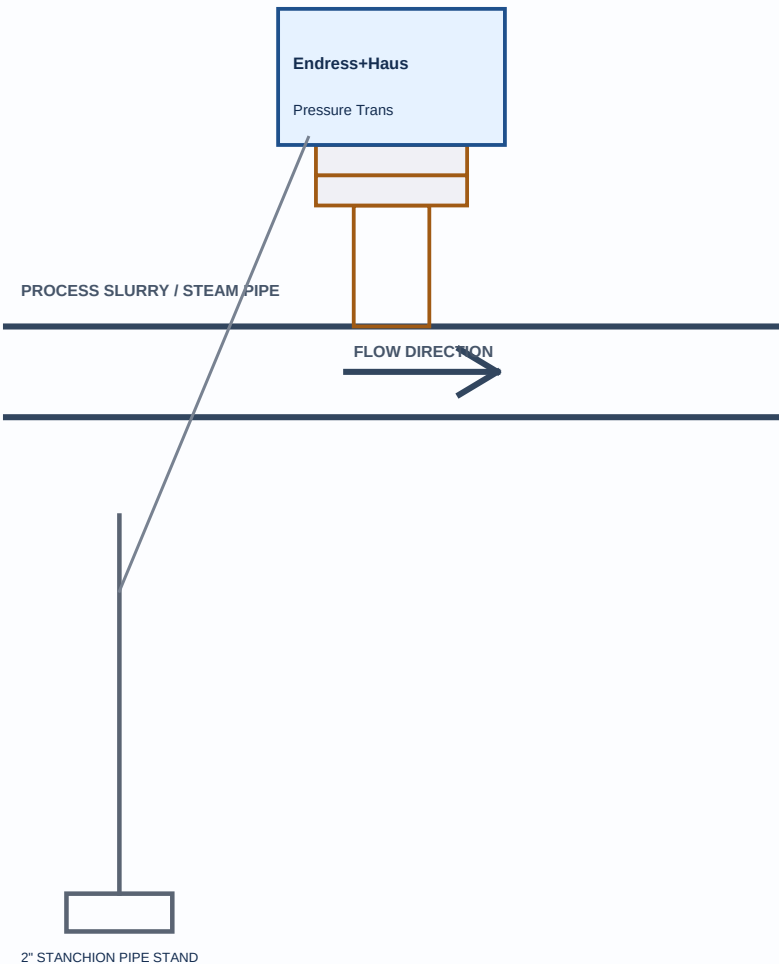
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTRONIC PRESSURE SWITCH HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-016
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
18 / 25

TEMPLATE TYP-17: PRESSURE TRANSMITTER

TYPICAL TAGS: PT-40101 / PT-60202

SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank | PLC I/O: 1756-IF16 (16-Point Differential AI)

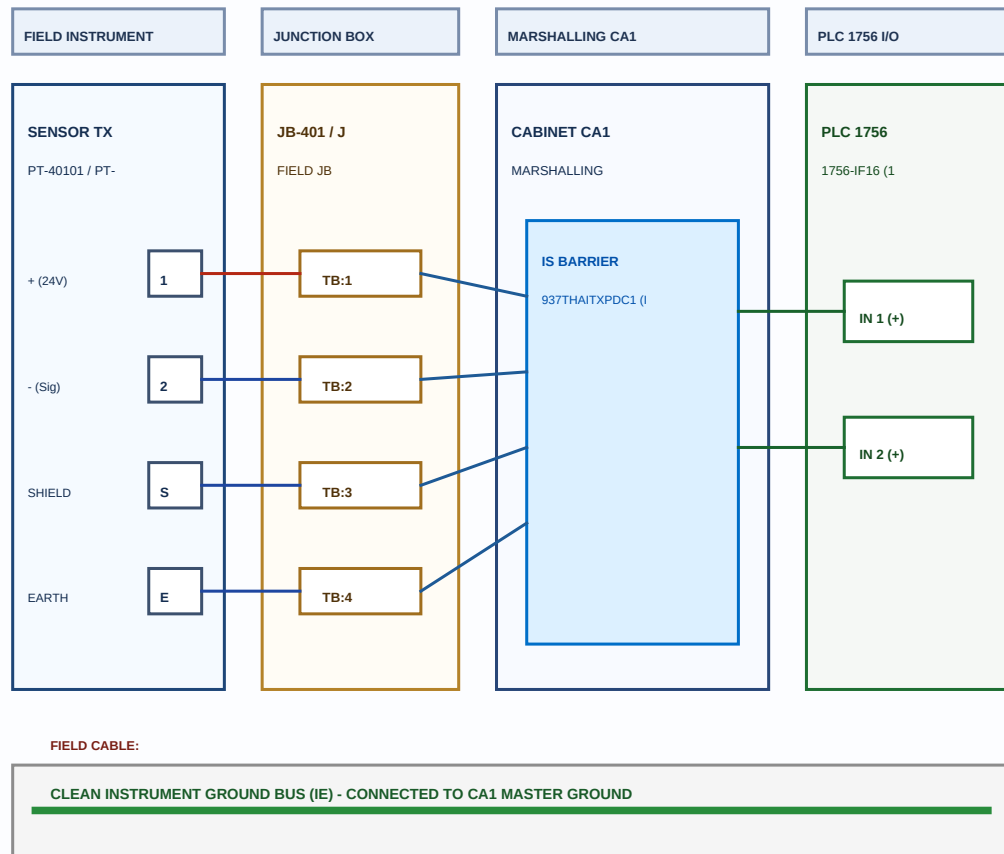
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Thread / 2" Tr..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-10 bar / 0-16 bar gauge)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Shielded (Belden 8761)
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Cerabar PMP51B / PMC51B (Metallic / Ceramic)

SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Cerabar PMP51B Pressure Transm E+H Piezoresistive SS316L Cell		1 EA
2	2-Valve Block & Bleed Manifold SS316 1/2" NPT Male x Female		1 EA
3	2" Pipe Mounting Bracket Kit SS304 Universal Bracket & U-Bolt		1 SET
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Armored Gland		1 EA
5	Siphon / Cooling Neck (Steam D SS316 Pigtail Loop 1/2" NPT		1 EA

ENGINEERING INSTALLATION NOTES

- Use flush diaphragm seal (PMC51B) for thick starch slurry to eliminate clogging in impulse ports.
- Ground transmitter body to local pipe ground; shield wire grounded only at CA1 cabinet earth bar.
- HART configuration: Set damping to 1.0s to filter process pump pulsations.

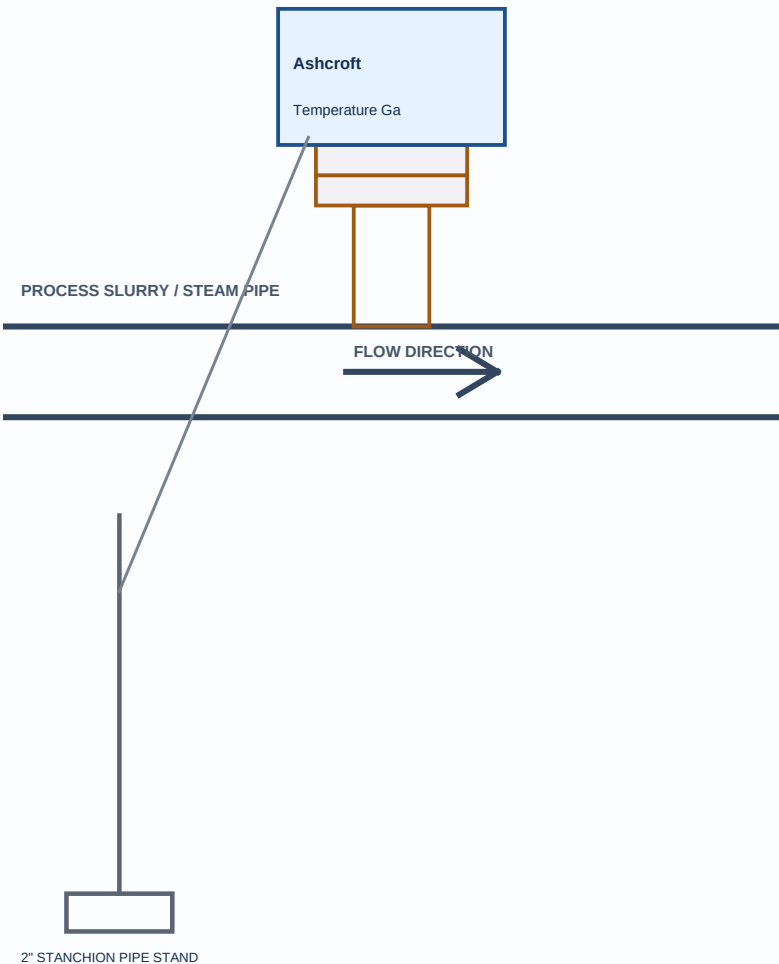
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONPRESSURE TRANSMITTER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-017
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
19 / 25

TEMPLATE TYP-18: TEMPERATURE GAUGE

TYPICAL TAGS: TI-40101 / TI-60201

SERVICE: Direct Pipe & Tank Slurry Temperature Indicator | PLC I/O: None (Local Operator Readout)

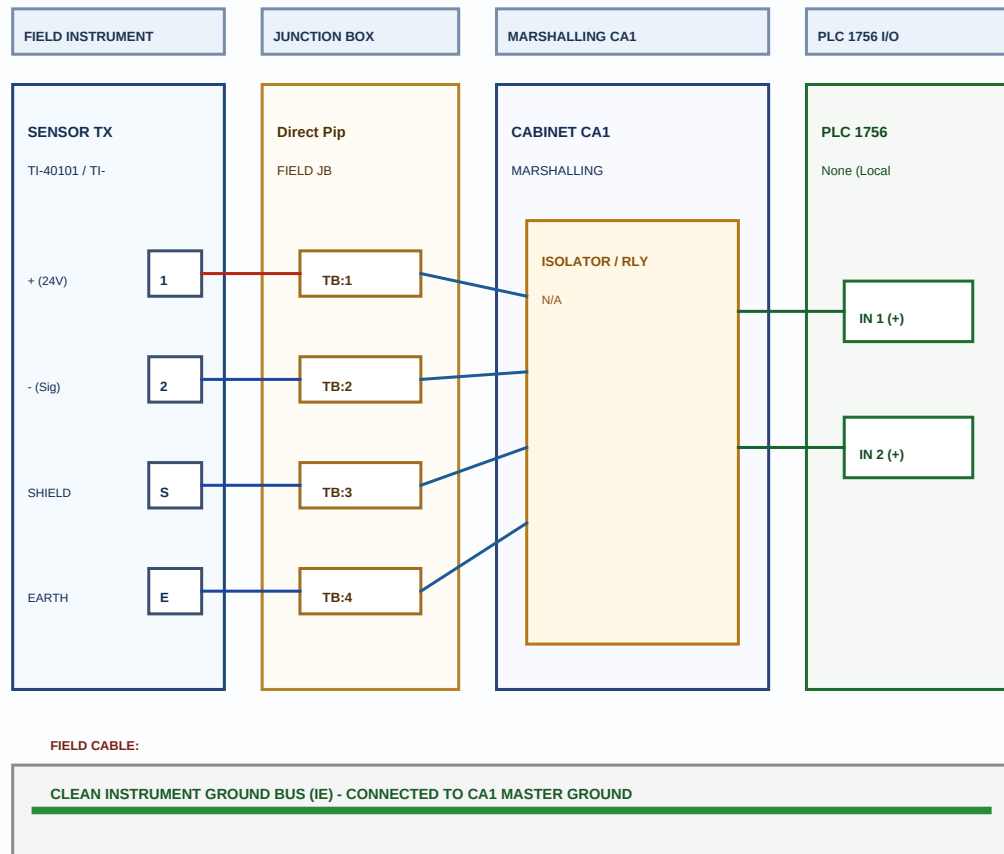
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Thread into Ba..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 Hermetically Sealed Sta..
Hazardous Class:	Non-Electrical Apparatus (St..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	Visual Scale 0-150°C (32-300°F)
Power Supply:	None (Bimetal Coil Driven)
Field Cable Type:	N/A (Local Dial Readout)
Destination JB:	Direct Piping Well Mount
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	N/A
PLC Module:	None (Local Operator Readout)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ASHCROFT

MODEL: Model FI (100mm Dial Everyangle Bimetal) + Thermowell

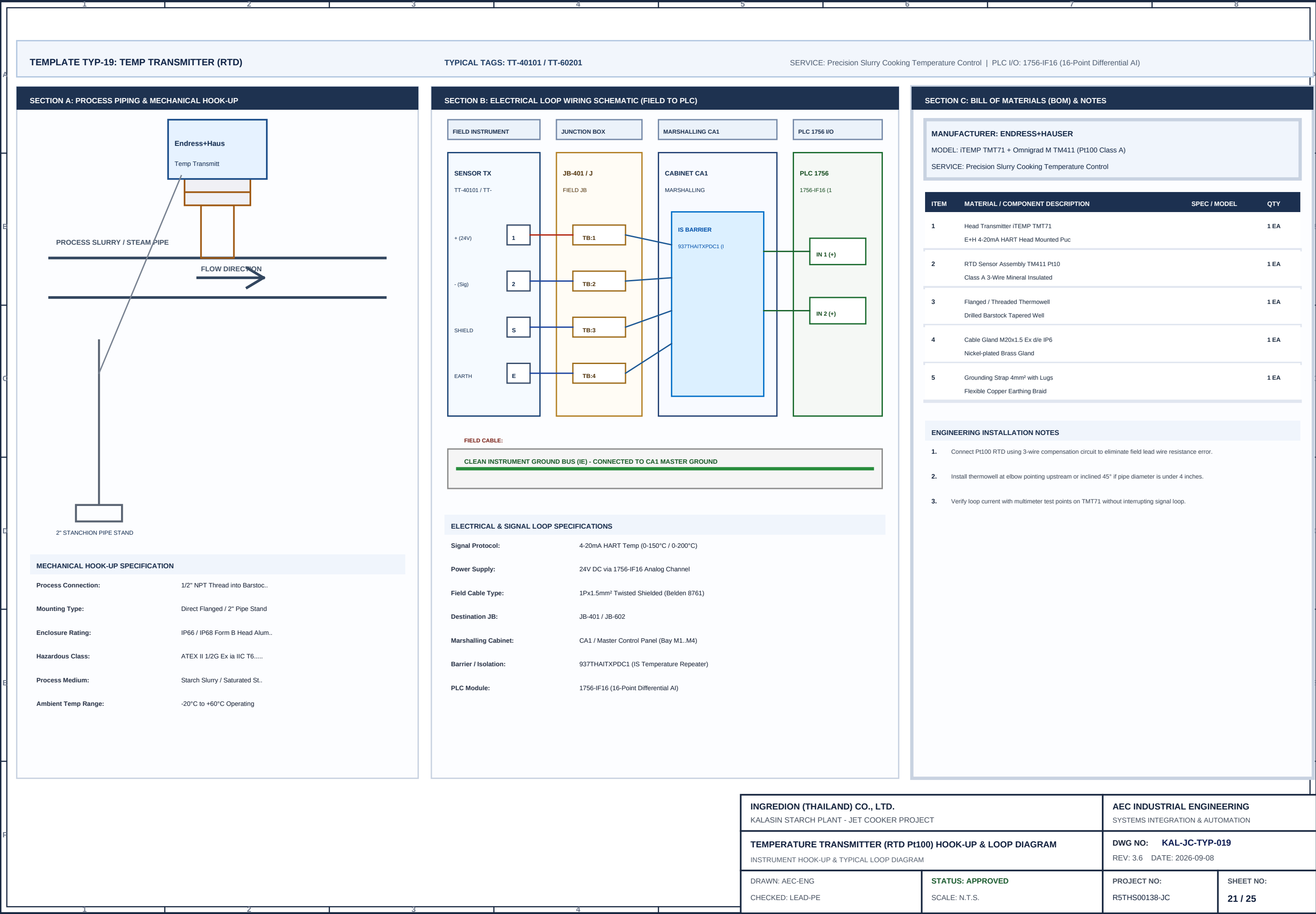
SERVICE: Direct Pipe & Tank Slurry Temperature Indicator

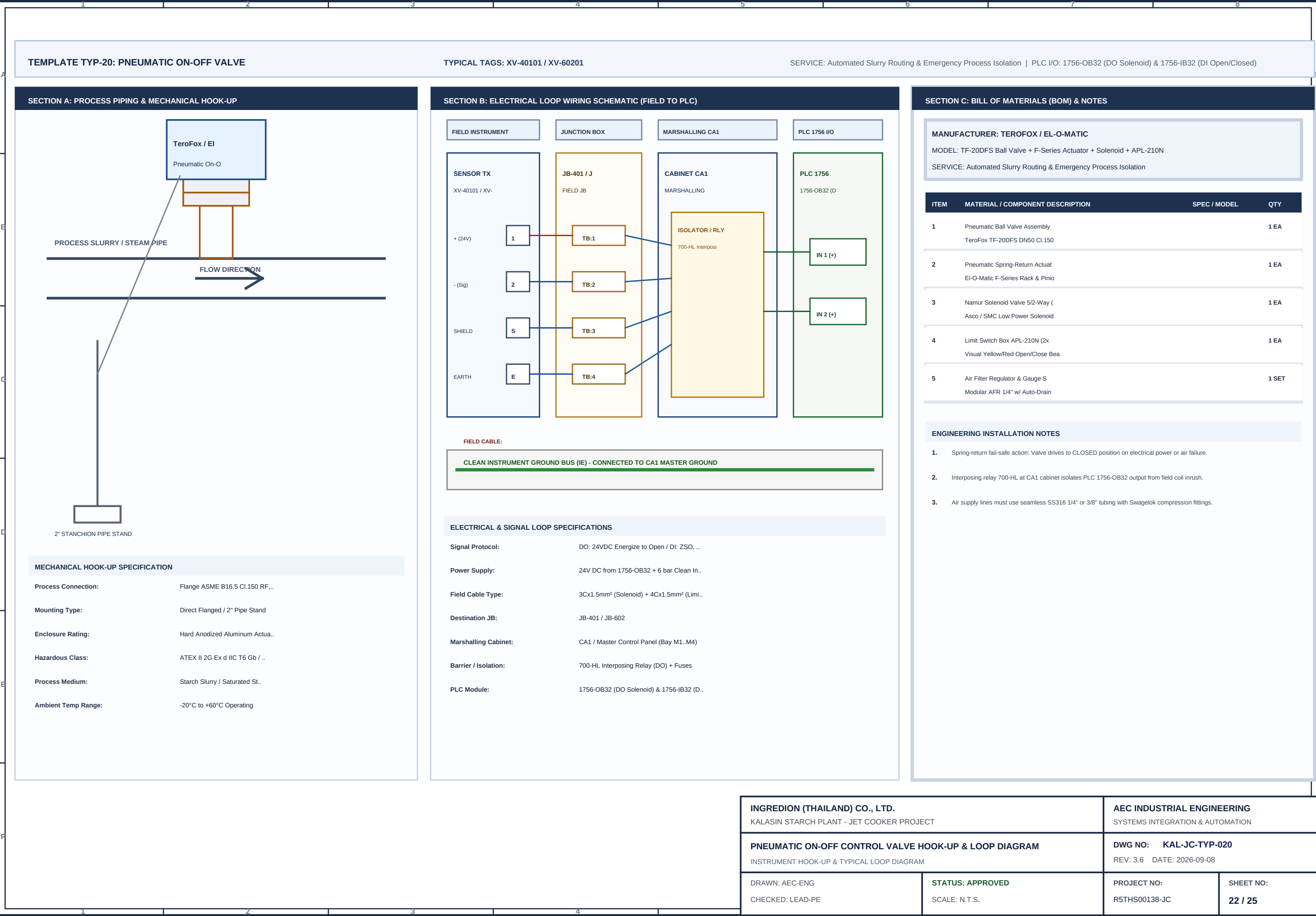
ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Bimetal Thermometer Model FI (Ashcroft 100mm Dial Everyangle S		1 EA
2	Barstock Thermowell SS316 (U=1 1/2" NPT x 1" NPT Process Thread		1 EA
3	Weldolet / Thredolet 1" NPT ASTM A182 F316 Pipe Fitting		1 EA
4	Thermal Conductive Paste High Temperature Aluminum Oxide		1 TUBE
5	Safety Wire & Tag Plate SS316 Laser Engraved Tag Plate		1 EA

ENGINEERING INSTALLATION NOTES

- Thermowell tip must be immersed into center third of process pipe for representative measurement.
- Apply thermal conductive paste inside thermowell bore before inserting gauge stem to improve heat transfer.
- Thermowell allows removal of thermometer for calibration without process shutdown.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONBIMETAL TEMPERATURE GAUGE & THERMOWELL HOOK-UP DETAIL
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-018
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
20 / 25



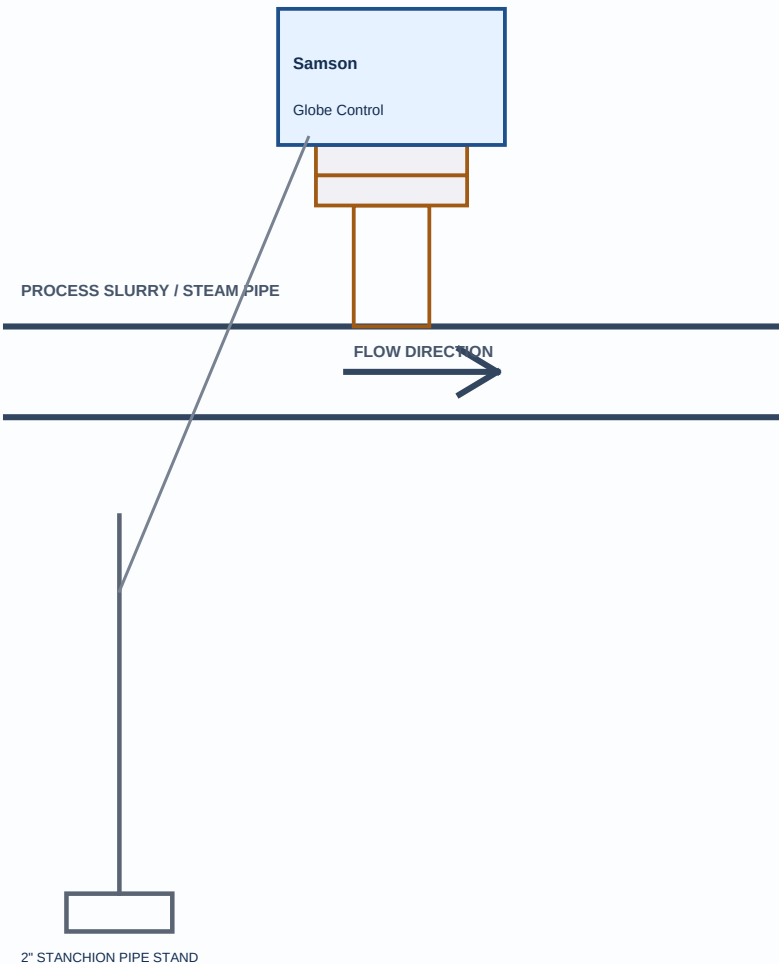


TEMPLATE TYP-21: GLOBE CONTROL VALVE

TYPICAL TAGS: TCV-40222 / TV-60202

SERVICE: High-Precision Steam & Slurry Flow Modulation | PLC I/O: 1756-OF8 (AO Setpoint) & 1756-IF16 (AI Feedback)

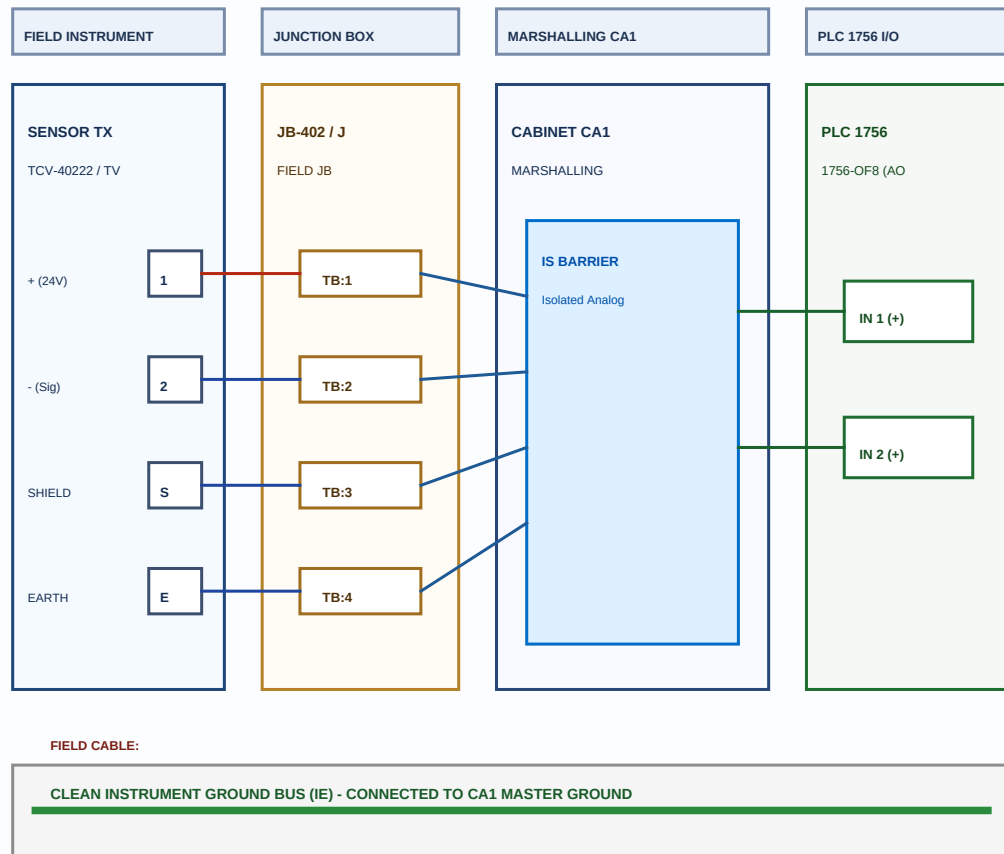
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	Flange ASME B16.5 Cl.150/300..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP66 Die-cast Aluminum Posit..
Hazardous Class:	ATEX II 2G Ex ia IIC T6 / Ex..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



FIELD CABLE:

CLEAN INSTRUMENT GROUND BUS (IE) - CONNECTED TO CA1 MASTER GROUND

ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AO: 4-20mA Valve Cmd (0-100%) / AI: Po..
Power Supply:	2-Wire 4-20mA Loop + 6.0 bar Instrumen..
Field Cable Type:	2Px1.5mm ² Twisted Shielded (Belden 8777)
Destination JB:	JB-402 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Isolated Analog Output / IS Repeater
PLC Module:	1756-OF8 (AO Setpoint) & 1756-IF16 (AI..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: SAMSON

MODEL: Type 3241 Globe Valve + Trovis 3730-1 Smart Positioner

SERVICE: High-Precision Steam & Slurry Flow Modulation

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Globe Control Valve Type 3241 Samson Cast Steel / SS316 Body		1 EA
2	Pneumatic Diaphragm Actuator T Samson Multi-Spring Actuator		1 EA
3	Smart Positioner Trovis 3730-1 HART Communication, Auto-Tuning		1 EA
4	SMC Air Filter Regulator AW20 Clean Air 1/4" w/ Pressure Gauge		1 SET
5	Cable Glands M20x1.5 Ex d/e IP Nickel-plated Brass Glands		2 EA

ENGINEERING INSTALLATION NOTES

- Perform Trovis 3730-1 automatic initialization routine after mechanical mounting is completed.
- Connect 4-20mA setpoint from PLC 1756-OF8 to Positioner Terminals 11(+) and 12(-).
- Ensure air supply pressure to positioner is regulated strictly between 4.0 and 6.0 bar.

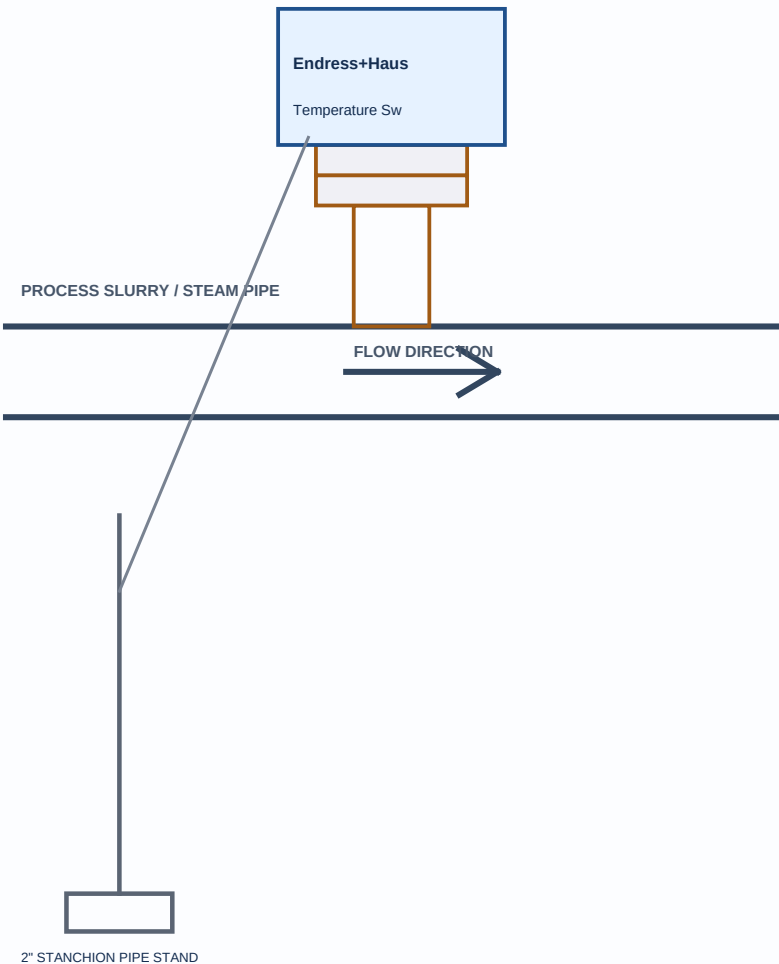
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONMODULATING GLOBE CONTROL VALVE HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-021
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
23 / 25

TEMPLATE TYP-22: TEMPERATURE SWITCH

TYPICAL TAGS: TSH-40101 / TSH-60201

SERVICE: High Temperature Safety Trip & Alarm | PLC I/O: 1756-IB32 (24V DC Digital Input)

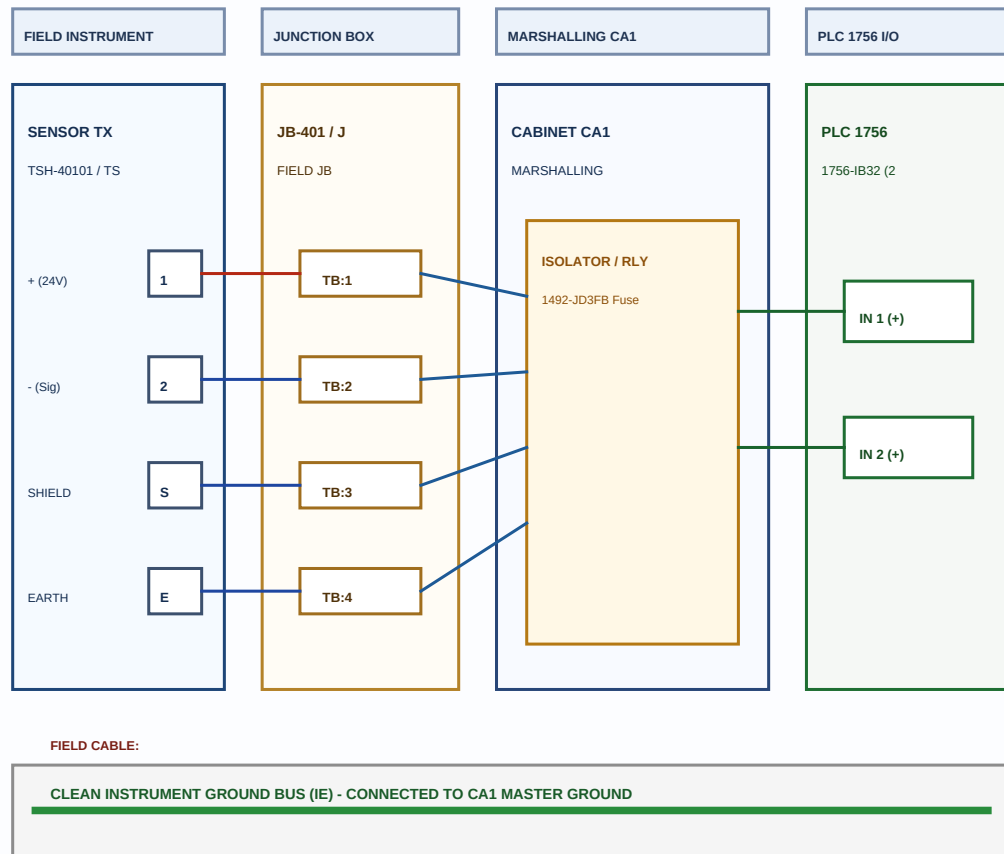
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Threaded / Hyg..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 / IP67 Stainless Steel ..
Hazardous Class:	General Industrial / Hygieni..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Temperature Alarm Trip
Power Supply:	24V DC (10-30V DC, 35mA)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Protection Terminal
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Thermophant TTR35 (Compact Pt100 Switch)

SERVICE: High Temperature Safety Trip & Alarm

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Thermophant TTR35 Temp Switch E+H Pt100 Class A, Digital Displ		1 EA
2	Barstock Thermowell SS316L (U= 1/2" NPT x 1/2" NPT Process Well		1 EA
3	M12 4-Pin Female Cable Assembl M12 Right Angle Molded Connector		1 EA
4	Thredolet / Boss 1/2" NPT SS3 Pipe Welding Fitting		1 EA
5	Grounding Clamp & Lug Copper Earth Wire		1 SET

ENGINEERING INSTALLATION NOTES

- Configure switch point via optical touch keys on front cover without opening enclosure.
- Normally closed (NC) contact output configuration guarantees fail-safe alarm on wire break.
- Thermowell allows switch replacement without draining the process line.

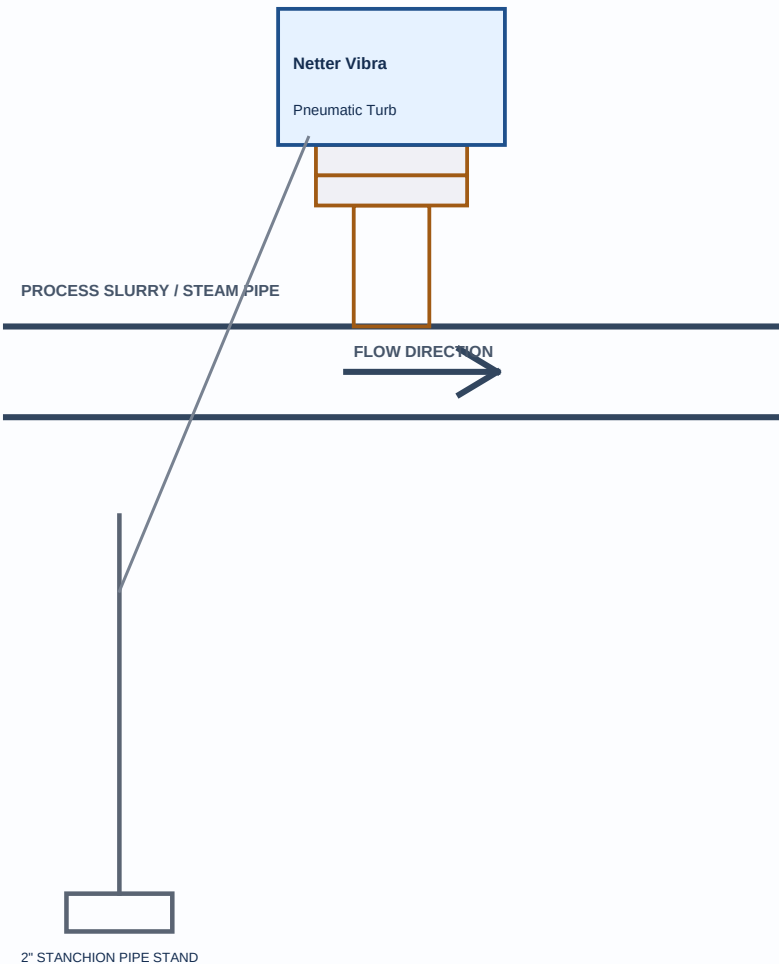
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTRONIC TEMPERATURE SWITCH HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-022
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
24 / 25

TEMPLATE TYP-23: PNEUMATIC TURBINE VIBRATOR

TYPICAL TAGS: XV-60901 Series

SERVICE: Powder Silo & Hopper Wall Discharge Agitation | PLC I/O: 1756-OB32 (24V DC Digital Output)

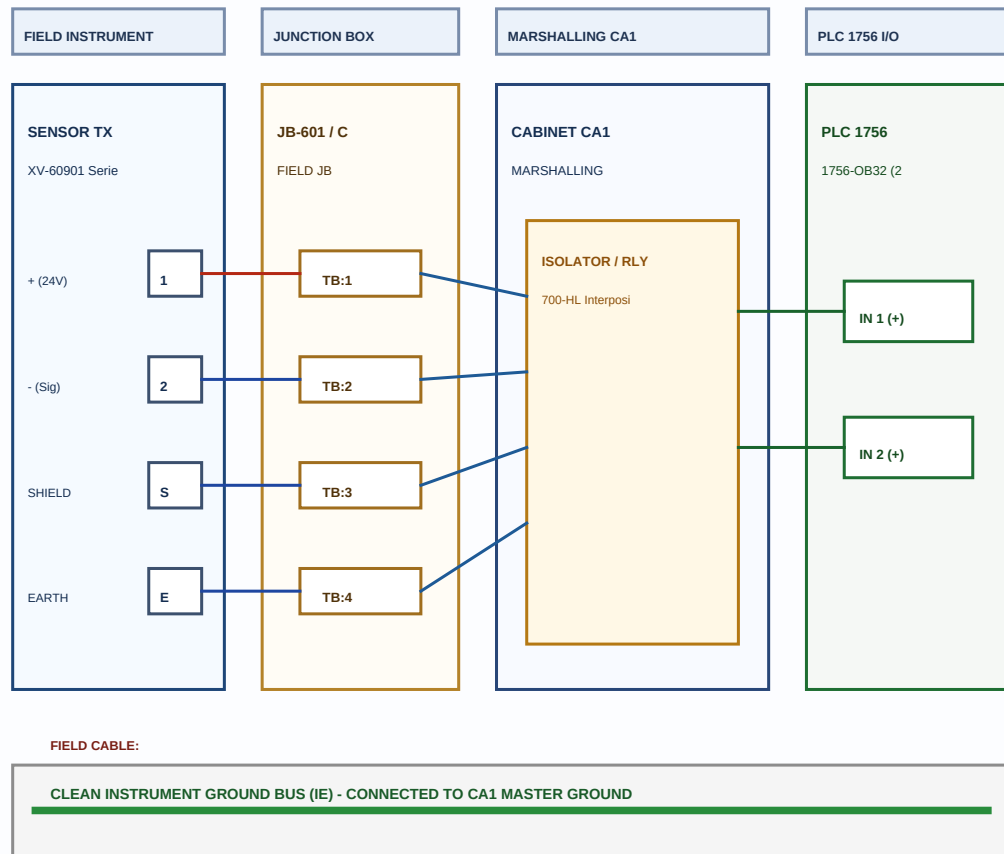
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	G 1/8" BSP Air Port / Rigid ..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	Extruded Aluminum Hard Anodi..
Hazardous Class:	ATEX II 2GD c T6 85°C (Mecha..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DO: Pulsed Vibration Burst (Timer Cont..
Power Supply:	24V DC from 1756-OB32 + 2 to 6 bar Lub..
Field Cable Type:	3Cx1.5mm² Shielded Cable (Belden 9533)
Destination JB:	JB-601 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	700-HL Interposing Relay with Diode
PLC Module:	1756-OB32 (24V DC Digital Output)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: NETTER VIBRATION

MODEL: NCT 5 Pneumatic Turbine Vibrator

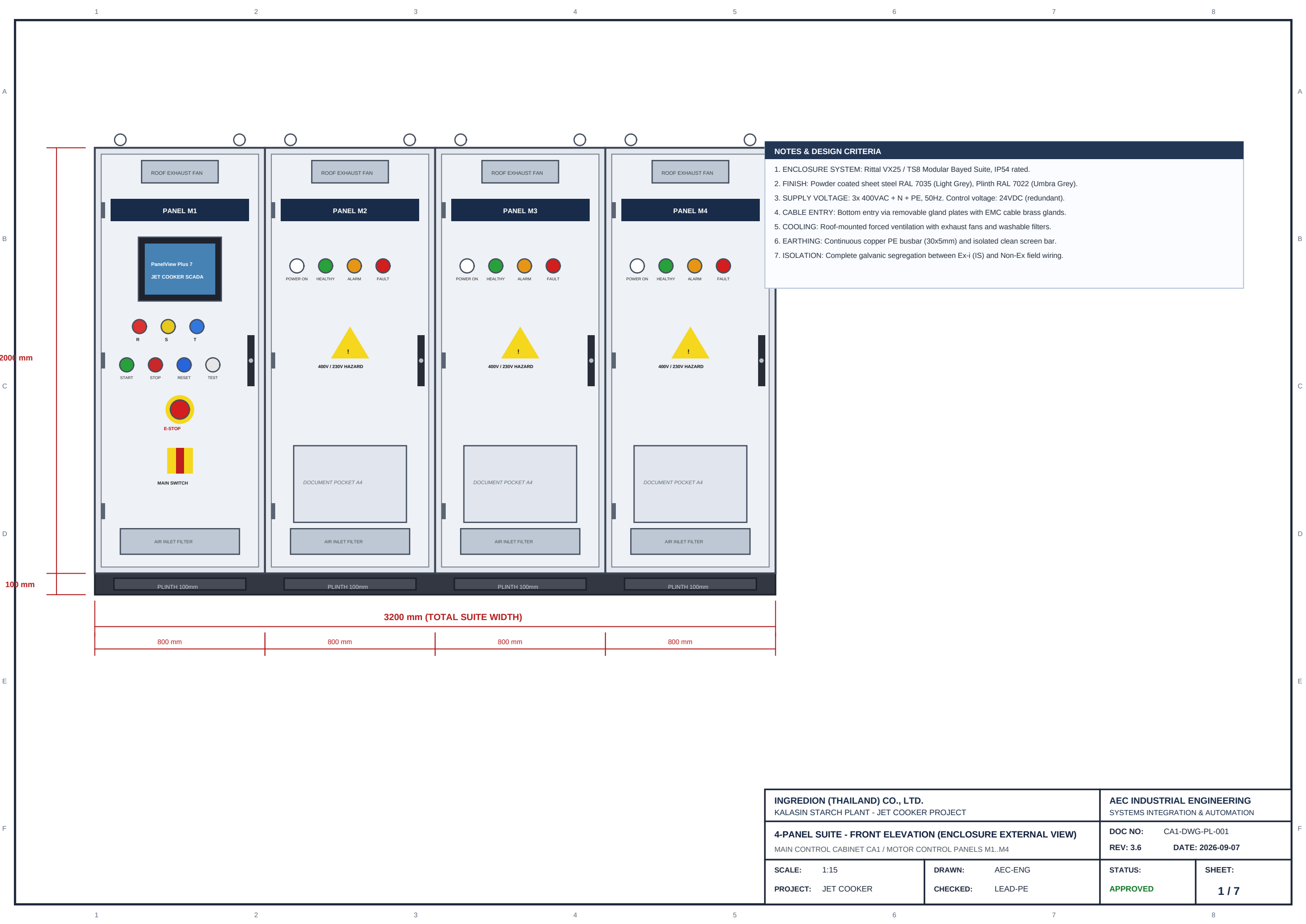
SERVICE: Powder Silo & Hopper Wall Discharge Agitation

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Pneumatic Turbine Vibrator NCT Netter Vibration High-Speed Unit		1 EA
2	Solenoid Valve 3/2-Way 24VDC (High Flow Fast-Cycling Solenoid		1 EA
3	Silencer / Exhaust Air Filter Sintered Bronze Sinter Filter		1 EA
4	SMC Air Filter Regulator Kit Modular AFR w/ Pressure Gauge		1 SET
5	Weld-on Hopper Mounting Channe Reinforced Channel Profile Steel		1 EA

ENGINEERING INSTALLATION NOTES

- Mount vibrator on stiffened channel beam welded to hopper cone to transmit vibration into bulk powder.
- Operate in intermittent pulse mode (e.g. 5 seconds ON, 30 seconds OFF) via PLC timer logic.
- Interposing relay 700-HL at CA1 panel protects PLC output module from inductive flyback.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTPNEUMATIC TURBINE VIBRATOR HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-023
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
25 / 25



A

B

C

D

E

F

A

B

C

D

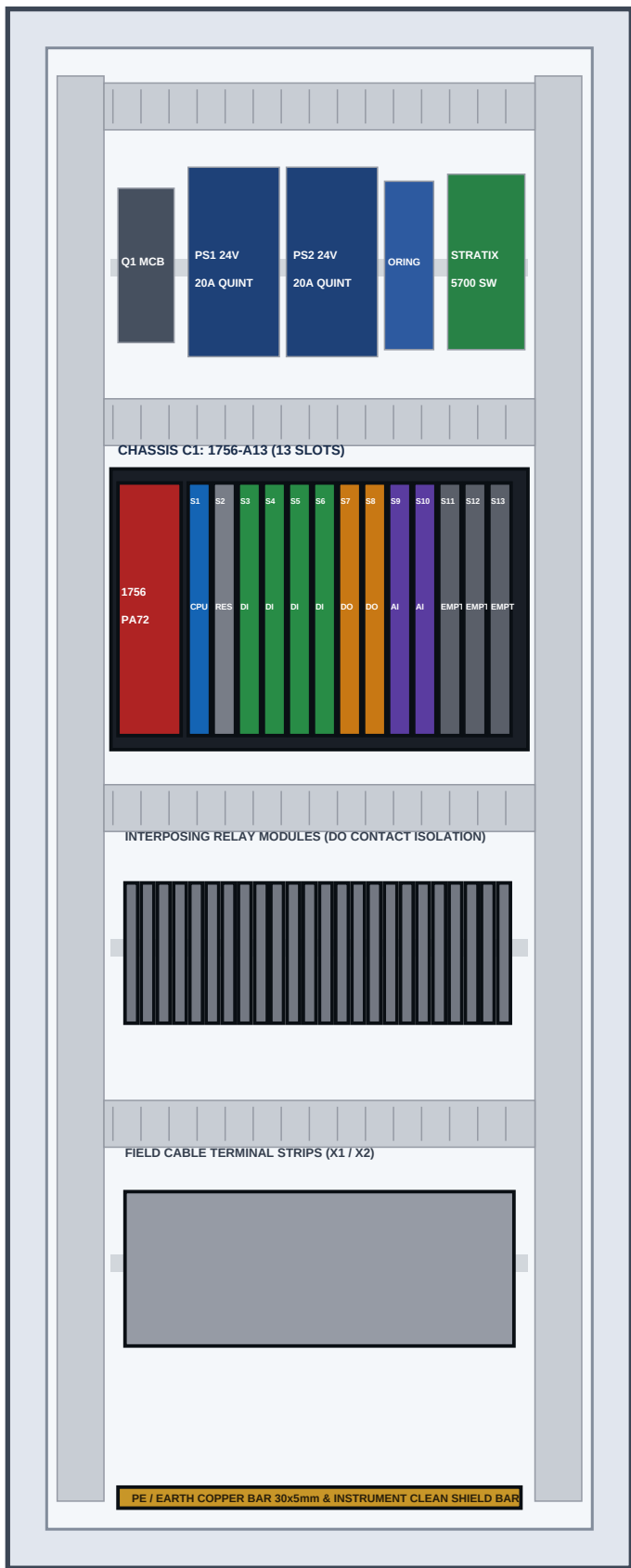
E

F



NOTES & DESIGN CRITERIA			
1. CHASSIS SYSTEM: Rockwell Automation Allen-Bradley ControlLogix 1756-A13 (13-slot racks).			
2. WIRING SEGREGATION: 50mm min. clearance between 230VAC, 24VDC, and analog signal paths.			
3. INTRINSIC SAFETY: Panels M3 & M4 feature dedicated blue IS terminal strips and 50mm physical air gap partition plates complying with IEC 60079-11 / IEC 60079-14 Ex-i standards.			
4. SPARE CAPACITY: All 4 panels maintain >25% spare DIN rail space and spare card slots.			
5. WIRE DUCT: Halogen-free slotted PVC wire duct with snap-on covers (60x80mm & 80x100mm).			

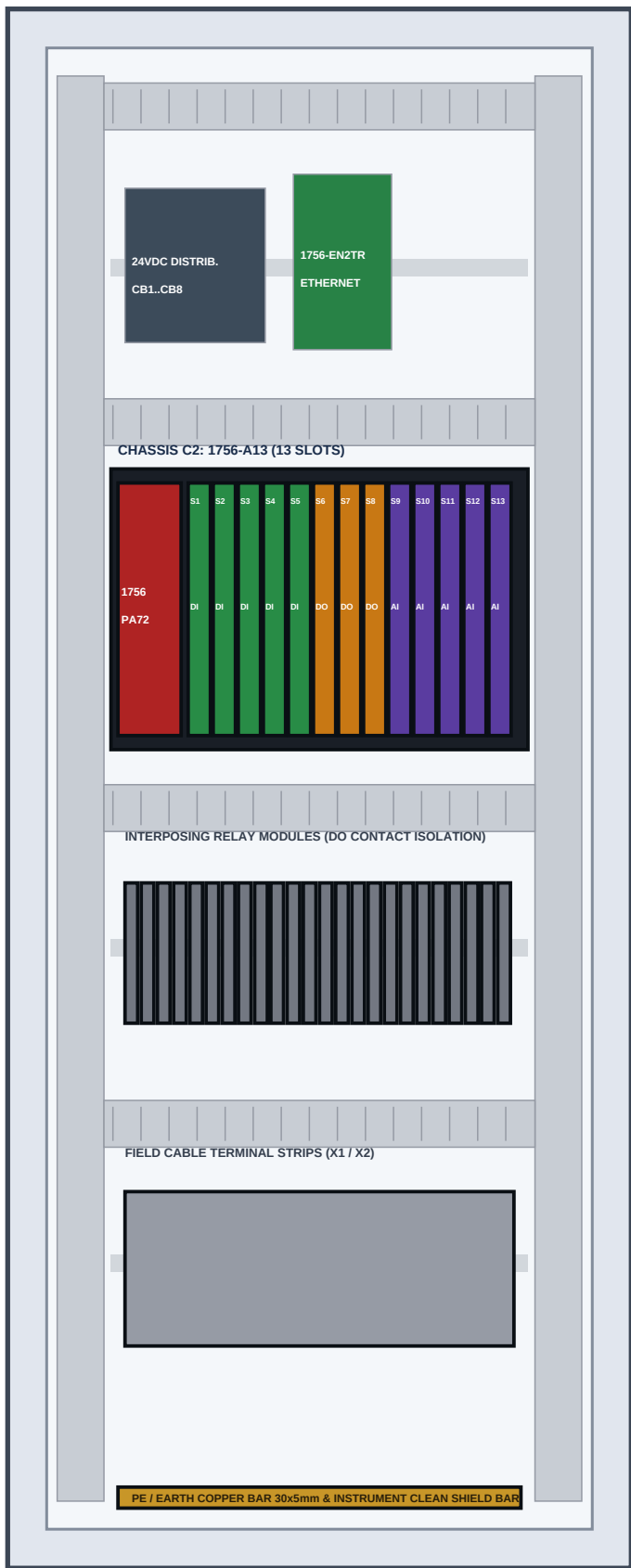
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
4-PANEL SUITE - INTERNAL GENERAL ARRANGEMENT (DOORS OPEN) MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4		DOC NO: CA1-DWG-PL-002 REV: 3.6	DATE: 2026-09-07
SCALE: 1:15 PROJECT: JET COOKER	DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED	SHEET: 2 / 7



Panel M1 - Rack C1 Hardware Allocation & Field Terminal Schedule							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-L950TPSXT	ControlLogix 5580 Controller (50MB Memory)	CPU	1	ETH1/2	Plant Master Controller / SCADA	ACTIVE (MASTER)
02	Reserve	Reserved for 1756-EN2TR / Redundancy	RES	-	-	Reserved for Controller Redundancy	SPARE (RESERVE)
03	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S03-X1,X2	JB-401 Field Signals & Spares	ACTIVE
04	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S04-X1,X2	JB-401 Field Signals	ACTIVE
05	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S05-X1,X2	JB-601 Field Signals	ACTIVE
06	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S06-X1,X2	JB-601 Field Signals & CA1 Spares	ACTIVE
07	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C1S07-X1,X2	JB-401 Solenoid Valves & DO	ACTIVE
08	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C1S08-X1,X2	JB-601 Solenoids & CA1 Spares	ACTIVE
09	1756-IF16	16-Point High Resolution Analog Input	AI	16	C1S09-X1	JB-401 Analog Transmitters (4-20mA)	ACTIVE
10	1756-IF16	16-Point High Resolution Analog Input	AI	16	C1S10-X1	JB-601 Analog Transmitters (4-20mA)	ACTIVE
11	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
12	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
13	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)

PANEL M1 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.	- FIELD DESTINATIONS: Connects to field junction boxes: JB-401, JB-601, CA1.
	- POWER INFEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.
	- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.
	- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).
	- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M1 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1..M4		DOC NO: CA1-DWG-PL-003 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8 PROJECT: JET COOKER	DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED	SHEET: 3 / 7



Panel M2 - Rack C2 Hardware Allocation & Field Terminal Schedule							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S01-X1,X2	JB-402 & JB-602 Flow & Limit Switches	ACTIVE
02	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S02-X1,X2	JB-602 & JB-607 Field Inputs	ACTIVE
03	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S03-X1,X2	JB-607 Spray Dryer 7th Flr DI	ACTIVE
04	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S04-X1,X2	JB-607 Spray Dryer 7th Flr DI	ACTIVE
05	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S05-X1,X2	JB-607 Spray Dryer 7th Flr DI	ACTIVE
06	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C2S06-X1,X2	JB-402 & JB-602 On/Off Valves	ACTIVE
07	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C2S07-X1,X2	JB-607 Solenoid Valves (DO)	ACTIVE
08	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C2S08-X1,X2	JB-607 Solenoid Valves (DO)	ACTIVE
09	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S09-X1	JB-402 Jet Cooker Flow/Press AI	ACTIVE
10	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S10-X1	JB-602 Spray Dryer 3rd Flr AI	ACTIVE
11	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S11-X1	JB-602 Spray Dryer 3rd Flr AI	ACTIVE
12	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S12-X1	JB-607 Spray Dryer 7th Flr AI	ACTIVE
13	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S13-X1	JB-607 Spray Dryer 7th Flr AI	ACTIVE

PANEL M2 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
Panel M2 Installation Specifications & Field Connections	- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.
	- FIELD DESTINATIONS: Connects to field junction boxes: JB-602, JB-402, JB-607.
	- POWER INFEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.
	- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.
	- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).
- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M2 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1..M4		DOC NO: CA1-DWG-PL-004 REV: 3.6 DATE: 2026-09-07	
SCALE: 1:8	DRAWN: AEC-ENG	STATUS: APPROVED	SHEET: 4 / 7
PROJECT: JET COOKER	CHECKED: LEAD-PE		

A

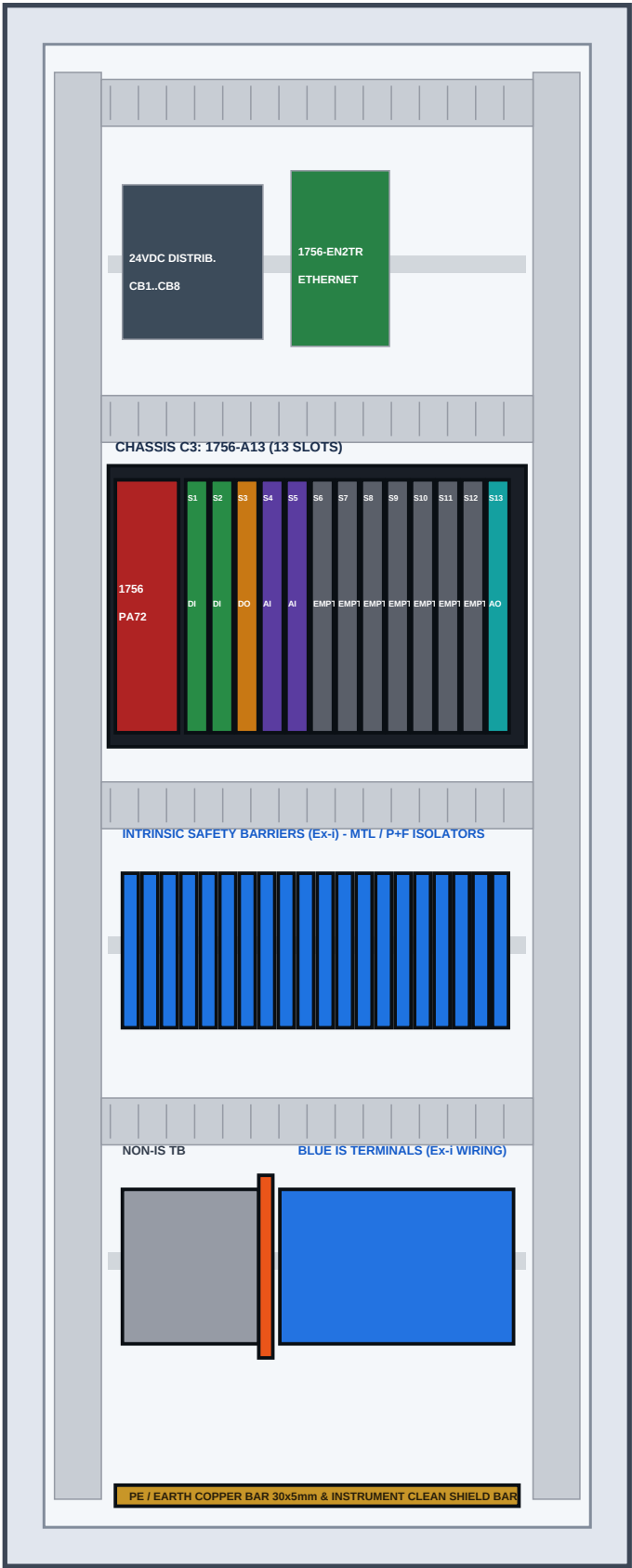
B

C

D

E

F



PANEL M3 - RACK C3 HARDWARE ALLOCATION & FIELD TERMINAL SCHEDULE							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C3S01-X1,X2	IS-JB-603 & IS-JB-618 Ex-i DI	ACTIVE (IS)
02	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C3S02-X1,X2	IS-JB-618 & JB-618 Field DI	ACTIVE
03	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C3S03-X1,X2	JB-618 Packing Tower 8th Flr DO	ACTIVE
04	1756-IF16	16-Point High Resolution Analog Input	AI	16	C3S04-X1	IS-JB-603 Ex-i Transmitters (AI)	ACTIVE (IS)
05	1756-IF16	16-Point High Resolution Analog Input	AI	16	C3S05-X1	IS-JB-618 & JB-618 Transmitters	ACTIVE (IS)
06	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
07	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
08	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
09	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
10	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
11	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
12	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
13	1756-OF8	8-Point High Resolution Analog Output	AO	8	C3S13-X1	Control Valves / Speed References	ACTIVE

PANEL M3 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.	
- FIELD DESTINATIONS: Connects to field junction boxes: IS-JB-603, IS-JB-618, JB-618.	
- POWER INFEEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.	
- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.	
- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).	
- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M3 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4				DOC NO: CA1-DWG-PL-005 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8	DRAWN: AEC-ENG		STATUS: APPROVED		SHEET: 5 / 7
PROJECT: JET COOKER	CHECKED: LEAD-PE				

A

B

C

D

E

F

A

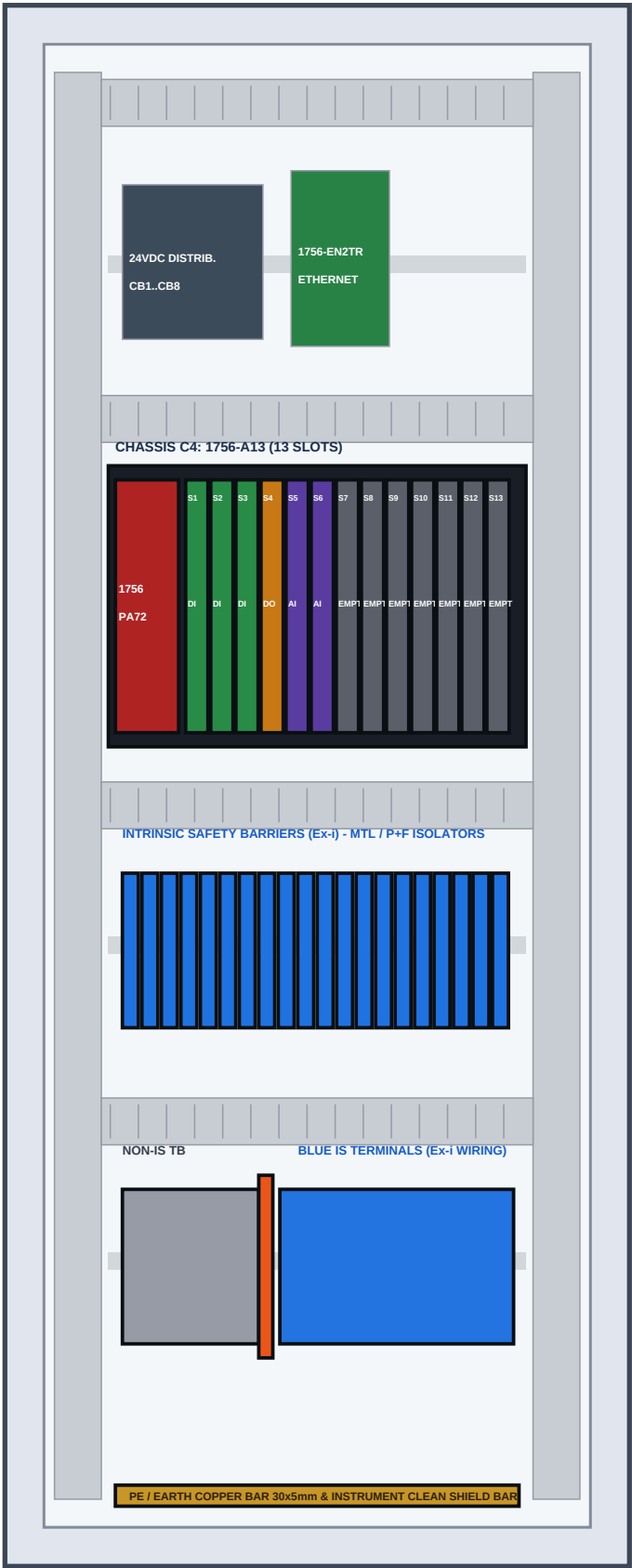
B

C

D

E

F



PANEL M4 - RACK C4 HARDWARE ALLOCATION & FIELD TERMINAL SCHEDULE							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C4S01-X1,X2	JB-606 & IS-JB-612 Ex-i DI	ACTIVE
02	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C4S02-X1,X2	IS-JB-612 & JB-612 Field DI	ACTIVE (IS)
03	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C4S03-X1,X2	IS-JB-608 & JB-608 Ex-i DI	ACTIVE (IS)
04	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C4S04-X1,X2	JB-606, JB-612, IS-JB-608 DO	ACTIVE
05	1756-IF16	16-Point High Resolution Analog Input	AI	16	C4S05-X1	JB-606 & IS-JB-612 Ex-i AI	ACTIVE (IS)
06	1756-IF16	16-Point High Resolution Analog Input	AI	16	C4S06-X1	JB-612, IS-JB-608, JB-608 AI	ACTIVE (IS)
07	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
08	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
09	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
10	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
11	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
12	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
13	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)

PANEL M4 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.	
- FIELD DESTINATIONS: Connects to field junction boxes: JB-606, IS-JB-612, JB-612, IS-JB-608, JB-608.	
- POWER INFEEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.	
- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.	
- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).	
- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M4 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4				DOC NO: CA1-DWG-PL-006 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8	DRAWN: AEC-ENG		STATUS: APPROVED		SHEET: 6 / 7
PROJECT: JET COOKER	CHECKED: LEAD-PE				

A

B

C

D

E

F

A

B

C

D

E

F

MAJOR HARDWARE BILL OF MATERIALS (BOM) - ENCLOSURES, PLC HARDWARE & ACCESSORIES								
Item	Part Number / Catalog	Manufacturer	Component Description	Installed Location	Qty	Spare Qty	Technical Specifications / Ratings	Remarks
1	VX 8806.500	Rittal	Industrial Bayed Enclosure Suite (4 bays)	M1, M2, M3, M4	4	-	800x2000x800 mm, IP54, Sheet Steel RAL 7035	Complete with side panels &
2	1756-A13	Rockwell Automation	ControlLogix 13-Slot Chassis	M1, M2, M3, M4	4	-	Standard backplane, horizontal mounting	Chassis C1, C2, C3, C4
3	1756-PA72	Rockwell Automation	ControlLogix Power Supply	M1, M2, M3, M4	4	1	85-265V AC, 72W output @ 60 deg C	One supply per chassis
4	1756-L950TPSXT	Rockwell Automation	ControlLogix 5580 Controller	Panel M1 (C1S01)	1	1	Extreme environment, 50MB memory, 1Gbps ETH	Master Plant Controller
5	1756-EN2TR	Rockwell Automation	EtherNet/IP Dual-Port DLR Comm Module	M2, M3, M4	3	1	10/100 Mbps, 128 TCP connections, DLR ring	I/O Remote Adapter modules
6	1756-IB32	Rockwell Automation	32-Point 24VDC Digital Input Module	M1, M2, M3, M4	14	2	24VDC Sink/Source, 10-31.2VDC, Isolated groups	544 total DI channels availa
7	1756-OB32	Rockwell Automation	32-Point 24VDC Digital Output Module	M1, M2, M3, M4	7	1	24VDC Source, 10-31.2VDC, 0.5A per point	224 total DO channels availa
8	1756-IF16	Rockwell Automation	16-Point Analog Input Module	M1, M2, M3, M4	11	1	Current / Voltage (4-20mA, 0-10V), 16-bit resolution	176 total AI channels availa
9	1756-OF8	Rockwell Automation	8-Point Analog Output Module	Panel M3 (C3S13)	1	1	Current / Voltage (4-20mA, 0-10V), 15-bit resolution	8 total AO channels availabl
10	1756-N2	Rockwell Automation	ControlLogix Slot Filler Plate	M1, M3, M4	17	-	Chassis slot cover plate for dust/cooling protection	Covers unpopulated slots
11	QUINT4-PS/1AC/24DC/20	Phoenix Contact	Primary Switched Redundant 24VDC Power Supply	Panel M1	2	1	Input 100-240VAC, Output 24VDC 20A, SFB Technology	Primary/Secondary Redundant
12	QUINT4-DIODE/40	Phoenix Contact	Diode Redundancy Module	Panel M1	1	-	Redundancy decoupling module 2x 20A / 1x 40A	Decouples PS1 & PS2
13	Stratix 5700	Rockwell Automation	Industrial Managed Ethernet Switch	Panel M1	1	-	16 Ports 10/100 Mbps, DLR, CIP Sync, VLAN support	Panel Network Backbone
14	KFD2-STC4-Ex1	Pepperl+Fuchs	Intrinsically Safe SMART Transmitter Isolator	Panel M3, M4	38	5	Ex-ia IIC, 4-20mA current repeater with HART support	For IS-JB-603, 608, 612, 618
15	KFD2-SL2-Ex2	Pepperl+Fuchs	Intrinsically Safe Digital Switch Isolator	Panel M3, M4	46	6	Ex-ia IIC, Dual-channel dry contact / NAMUR sensor	For IS Level & Limit Switche
16	Finder 38.51	Finder	Interposing Relay Module (DO Isolation)	M1, M2, M3, M4	120	15	24VDC Coil, 1 CO 6A contact, LED indicator	DIN Rail mount relay interfa
17	WDU 2.5 / WDU 4	Weidmuller	Feed-Through Terminal Blocks (Gray)	M1, M2, M3, M4	950	100	800V, 24A, Screw connection, multi-tier	Standard Field Wiring
18	WDU 2.5 BL	Weidmuller	Intrinsically Safe Terminal Blocks (Blue)	Panel M3, M4	320	50	Ex-i Blue colored terminal blocks for hazardous area	IS Field Wiring Only
19	SK 3328.540	Rittal	Roof Mounted Cooling Unit / Fans	M1, M2, M3, M4	4	-	230V, 50/60Hz, 1500W cooling capacity / high flow	Temperature maintenance < 35

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
4-PANEL SUITE - COMPLETE BILL OF MATERIALS (BOM) & SPECIFICATION MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4				DOC NO: CA1-DWG-PL-007 REV: 3.6	DATE: 2026-09-07
SCALE: N/A	DRAWN: AEC-ENG		STATUS:		SHEET:
PROJECT: JET COOKER	CHECKED: LEAD-PE		APPROVED		7 / 7

A

B

C

D

E

F